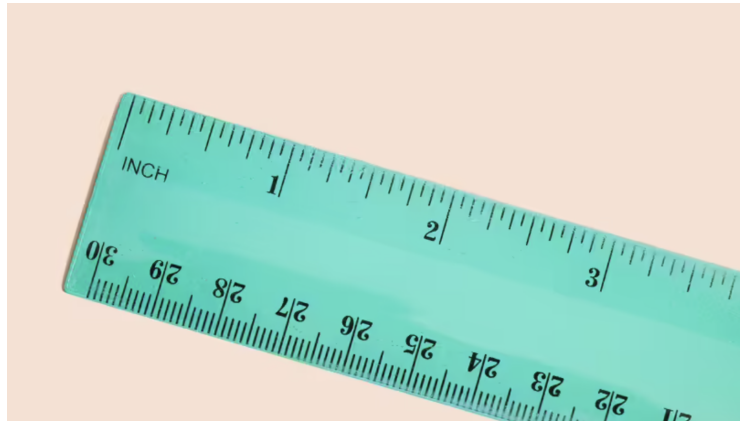


McGILL UNIVERSITY

MATH 454: HONOURS ANALYSIS 3

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Notes on Measure Theory

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Contents

1	Measure	2
1.1	Construction of the Lebesgue measure (on $\mathbb{R}$)	2
1.2	Lebesgue Measure and its properties	7
1.3	Relation Between $\mathcal{B}_{\mathbb{R}}$ and $\mathcal{M}$	13
1.4	Some Special Sets	14
2	Integration Theory	20
2.1	Measurable Functions	20
2.2	Approximation by simple functions	26
2.3	Convergence a.e vs Convergence in Measure	30
2.4	The Theorems of Egorov and Lusin	32
2.5	Construction of Integrals	34
2.6	Convergence Theorems of Integrals	41
2.7	Riemann vs Lebesgue integral	47
2.8	L^p spaces	49
2.9	Convergence modes, uniform integrability	51
3	Product Spaces	55
3.1	Product σ -algebra	55
3.2	Product Measure	58
3.3	Fubini's Theorem	63
4	Differentiation	66
4.1	Hardy-Littlewood maximal function	66
4.2	Lebesgue Differentiation Theorem	69
4.3	Monotonic (Increasing) functions	71
4.4	Functions of Bounded Variation	72
4.5	Absolutely Continuous Functions	73
4.6	A Sprinkling of Probability	76

Chapter 1: Measure

1.1 Construction of the Lebesgue measure (on $\mathbb{R}$)

We want a measure on $\mathbb{R}$ such that:

- a) it assigns to an interval the length of that interval
- b) it is translation invariant (i.e. A and $A + x$ have the same measure)

Definition 1.1.1: Outer Measure

For all $A \subseteq \mathbb{R}$ define:

$$m^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \ell(I_n) \mid A \subseteq \bigcup_{n=1}^{\infty} I_n \right\}$$

Where I_n s are open intervals.

Then m^* is called a (Lebesgue) outer measure on $\mathbb{R}$.

Proposition 1.1.2

Simple properties of m^* :

- 1) $\forall A \subseteq \mathbb{R}, m^*(A) \geq 0$, and $m^*(\emptyset) = 0$.
- 2) Monotonicity: if $A, B \subseteq \mathbb{R}$ with $A \subseteq B$, then $m^*(A) \leq m^*(B)$ - i.e., any open-interval covering of B is automatically one of A).

Proposition 1.1.3

3) m^* has countable subadditivity : if $\{A_n; n \geq 1\}$ is a sequence of subsets of $\mathbb{R}$, then $m^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} m^*(A_n)$

Proof. If $m^*(A_n) = \infty$ for some $n \geq 1$ then the inequality holds trivially. Therefore, wlog. we assume that $\forall n \geq 1, m^*(A_n) < \infty$.

Then, for all $\varepsilon > 0$ one can choose a squence of open intervals $\{I_{n,i}; i \geq 1\}$ s.t:

$$A_n \subseteq \bigcup_{i=1}^{\infty} I_{n,i} \text{ and } m^*(A_n) \leq \sum_{i=1}^{\infty} \ell I_{n,i} \leq m^*(A_n) + \frac{\varepsilon}{2^n} \quad [\Delta]$$

Therefore $\bigcup_{n=1}^{\infty} A_n \subseteq \bigcup_{n=1}^{\infty} \bigcup_{i=1}^{\infty} I_{n,i}$. (i.e. RHS is an open-interval covering of $\bigcup_{n=1}^{\infty} A_n$). Then by the definition of m^* :

$$\begin{aligned} m^*\left(\bigcup_{n=1}^{\infty} A_n\right) &\leq \sum_{n,i} \ell(I_{n,i}) = \sum_{n=1}^{\infty} \sum_{i=1}^{\infty} \ell(I_{n,i}) \quad [\Delta] \\ &\leq \sum_{n=1}^{\infty} \left(m^*(A_n) + \frac{\varepsilon}{2^n}\right) = \sum_{n=1}^{\infty} m^*(A_n) + \varepsilon \end{aligned}$$

$$\text{So } \forall \varepsilon > 0, m^*\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} m^*(A_n) + \varepsilon$$

Since ε is arbitrarily small, $m^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} m^*(A_n)$. □

Proposition 1.1.4

4) If $I \subseteq \mathbb{R}$ is an interval then $m^*(I) = \ell(I)$.

Proof. We first prove the result for $I = [a, b]$ for $a, b \in \mathbb{R}, a < b$.

For all $\varepsilon > 0$, set $I_1 = (a - \varepsilon, b + \varepsilon)$. Then, $I \subseteq I_1$

$$\implies m^*(I) \leq \ell(I_1) = (b - a) + 2\varepsilon \implies m^*(I) \leq b - a = \ell(I)$$

We only need to prove $m^*(I) \geq \ell(I)$. It's sufficient to show that $\forall$ open-interval coverings $\{I_n; n \geq 1\}$ of I , $\sum_{n=1}^{\infty} \ell(I_n) \geq \ell(I)$.

Let $\{I_n; n \geq 1\}$ be one such covering. Since $I = [a, b]$ is compact, I can be covered by finitely many I_n s. In particular, we can extract a finite collection $I_1, I_2, \dots, I_N$ with $I_n = (a_n, b_n) \forall n \geq 1$ with $a_1 < a$ and $b_N > b$ s.t $a_n < b_{n-1} \forall 2 \leq n \leq N$.

$$\begin{aligned} \text{Then } \sum_{n=1}^{\infty} \ell(I_n) &\geq \sum_{n=1}^N \ell(I_n) = b_1 - a_1 + \sum_{n=2}^N (b_n - a_n) \\ &\geq b_1 - a_1 + \sum_{n=2}^N (b_n - b_{n-1}) = b_N - a_1 \geq b - a = \ell(I) \end{aligned}$$

Next, assume I is a finite interval with end points $a, b; a < b$.

Then $\forall \varepsilon > 0$, $[a + \varepsilon, b - \varepsilon] \subseteq I \subseteq [a - \varepsilon, b + \varepsilon]$ ($\varepsilon < \frac{b-a}{2}$).

By monotonicity, $b - a - 2\varepsilon \leq m^*(I) \leq b - a + 2\varepsilon \implies m^*(I) = b - a = \ell(I)$.

Finally, take I to be an infinite interval $\forall M > 0$. Then $\exists$ a closed finite interval I_M s.t $I_M \subseteq I$ and $\ell(I_M) \geq M$. $\implies m^*(I) \geq m^*(I_M) \geq M$.

Since M is arbitrarily large, $m^*(I) = \infty$. □

Proposition 1.1.5

5) m^* is translation invariant, i.e., $\forall A \subseteq \mathbb{R}, x \in \mathbb{R}, m^*(A + x) = m^*(A)$

Proof sketch (key points): $\{I_n; n \geq 1\}$ is an open-interval covering of $A \iff \{I_n + x; n \geq 1\}$ is an open-interval covering of $A + x$, and $\ell(I_n) = \ell(I_n + x)$.

Proposition 1.1.6

6) "Regularity": $\forall A \subseteq \mathbb{R}, m^*(A) = \inf\{m^*(B); B \subseteq \mathbb{R} \text{ is open and } A \subseteq B\}$.

Proof. $\forall B$ open with $A \subseteq B$, $m^*(A) \leq m^*(B) \implies m^*(A) \leq \text{RHS}$,

We need to prove that $m^*(A) \geq \text{RHS}$. Wlog. assume that $m^*(A) < \infty$. $\forall \varepsilon > 0$, $\exists$ open intervals $\{I_n; n \geq 1\}$ s.t. $A \subseteq \bigcup_{n=1}^{\infty} I_n$ and $\sum_{n=1}^{\infty} \ell(I_n) \leq m^*(A) + \varepsilon$.

Set $B = \bigcup_{n=1}^{\infty} I_n$. B is open, $A \subseteq B$ and:

$$m^*(B) = m^*\left(\bigcup_{n=1}^{\infty} I_n\right) \leq (\text{by count. subadd.}) \sum_{n=1}^{\infty} \ell(I_n) \leq m^*(A) + \varepsilon$$

Then it is done, using the earlier properties. $\square$

Proposition 1.1.7

7) If $A = A_1 \cup A_2 \subseteq \mathbb{R}$ with $d(A_1, A_2) > 0$, then $m^*(A) = m^*(A_1) + m^*(A_2)$.

Proof. $m^*(A) \leq m^*(A_1) + m^*(A_2)$ (finite subadd.)

Then we just need to prove $m^*(A) \geq m^*(A_1) + m^*(A_2)$. Wlog. assume $m^*(A) < \infty$. Set $\delta = d(A_1, A_2) > 0$.

Let $\varepsilon > 0$. By definition of outer measure, $\exists$ a sequence of open intervals $\{I_n\}$ such that $A \subseteq \bigcup_{n=1}^{\infty} I_n$ and $\sum_{n=1}^{\infty} \ell(I_n) \leq m^*(A) + \varepsilon$.

For each n , either $I_n \cap A_1 \neq \emptyset$ or $I_n \cap A_2 \neq \emptyset$ (or both). However, if I_n intersects both A_1 and A_2 , then $\ell(I_n) \geq \delta$ since $d(A_1, A_2) = \delta$. By shrinking such intervals if necessary (keeping them open), we may assume that each I_n intersects at most one of A_1, A_2 .

Let $\mathcal{I}_1 = \{I_n : I_n \cap A_1 \neq \emptyset\}$ and $\mathcal{I}_2 = \{I_n : I_n \cap A_2 \neq \emptyset\}$. Then $A_1 \subseteq \bigcup_{I \in \mathcal{I}_1} I$ and $A_2 \subseteq \bigcup_{I \in \mathcal{I}_2} I$.

Therefore,

$$m^*(A_1) + m^*(A_2) \leq \sum_{I \in \mathcal{I}_1} \ell(I) + \sum_{I \in \mathcal{I}_2} \ell(I) = \sum_{n=1}^{\infty} \ell(I_n) \leq m^*(A) + \varepsilon$$

Since $\varepsilon > 0$ was arbitrary, $m^*(A_1) + m^*(A_2) \leq m^*(A)$.

Proposition 1.1.8

8) If $A = \cup_{k=1}^{\infty} J_k$ where J_k s are almost disjoint intervals (at most sharing end points, e.g. $(0, 1], [1, 2)$), then:

$$m^*(A) = \sum_{k=1}^{\infty} \ell(J_k)$$

[This is still not additivity!]

Proof. Clearly, $m^*(A) \leq \sum_{k=1}^{\infty} \ell(J_k)$ by countable subadd.

If $\ell(J_k) = \infty$ for some k , then since $J_k \subseteq A$ monotonicity $\implies m^*(A) = \infty$. In this case both sides of the relation above $= \infty$.

So now assume $\ell(J_k) < \infty \forall k \geq 1$. Fix an arbitrary $\varepsilon > 0 \forall k \geq 1$, choose an open interval $I_k \subseteq J_k$ s.t $\ell(J_k) \leq \ell(I_k) + \frac{\varepsilon}{2^k}$.

Furthermore, for any fixed $N \geq 1$ we can make the choice of $I_1, I_2, \dots, I_N$ s.t they are disjoint with a positive distance from one another.

$$\text{Then } \bigcup_{k=1}^N I_k \subseteq \bigcup_{k=1}^N J_k \subseteq A \implies m^*(A) \geq m^*\left(\bigcup_{k=1}^N J_k\right)$$

$$\text{Then by 7) combined with induction: } = \sum_{k=1}^N \ell(J_k) \geq \sum_{k=1}^N \left(\ell(J_k) - \frac{\varepsilon}{2^k}\right) \geq \sum_{k=1}^N \ell(J_k) - \varepsilon$$

Since ε arbitrarily small, N arbitrarily large, we find that $m^*(A) \geq \sum_{k=1}^{\infty} \ell(J_k)$. $\square$

Note: in general, A, B disjoint does NOT imply that $m^*(A \cup B) = m^*(A) + m^*(B)$. m^* is **NOT** a measure!

Definition 1.1.9: m^* -measurable

$A \subseteq \mathbb{R}$ is m^* -measurable if $\forall B \subseteq \mathbb{R}, m^*(B) = m^*(B \cap A) + m^*(B \cap A^c)$.

Otherwise, A is non-measurable. [Shocking!]

Remark.

By subadd., $\forall A, B \subseteq \mathbb{R}, m^*(B) \leq m^*(B \cap A) + m^*(B \cap A^c)$. Always!

1) Therefore, to show that A is m^* -measurable we only need to prove that $\forall B \subseteq \mathbb{R}, m^*(B) \geq m^*(B \cap A) + m^*(B \cap A^c)$.

2) Whether or not A is m^* -measurable depends on whether or not " $<$ " can occur for some B

3) A non measurable set exists (relies on the Axiom of Choice)

1.2 Lebesgue Measure and its properties

Theorem 1.2.1: Caratheodory's Theorem

Let $\mathcal{M} = \{A \subseteq \mathbb{R}; A \text{ is } m^*\text{-measurable}\}$. Then, $\mathcal{M}$ is a σ -algebra of subsets of $\mathbb{R}$. Define $m : \mathcal{M} \rightarrow [0, \infty]$ by $\forall A \in \mathcal{M}, m(A) = m^*(A)$.

Then, m is a measurable called the Lebesgue measure on $\mathbb{R}$. If $A \in \mathcal{M}$, then A is (Lebesgue) measurable.

Proof. First, we show that $\mathcal{M}$ is a σ -algebra.

1) $\mathbb{R} \in \mathcal{M} \checkmark$

2) $A \in \mathcal{M} \implies A^c \in \mathcal{M} \checkmark$

3) if $\{A_n, n \geq 1\} \subseteq \mathcal{M} \implies \bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$

We first verify that $\mathcal{M}$ is closed under finite union, i.e., $A_1, \dots, A_N \in \mathcal{M} \implies \bigcup_{n=1}^N A_n \in \mathcal{M}$.

It's sufficient to treat $N = 2$, because after that, induction! (i.e. want to show $A_1 \cup A_2 \in \mathcal{M}$)

$$\forall B \subseteq \mathbb{R}, m^*(B) = m^*(B \cap A_1) + m^*(B \cap A_1^c) = m^*(B \cap A_1) + m^*(B \cap A_1^c \cap A_2) + m^*(B \cap A_1^c \cap A_2^c)$$

Note: $m^*(B \cap A_1^c \cap A_2^c) = m^*(B \cap (A_1 \cup A_2)^c)$, and note: $(B \cap A_1) \cup (B \cap A_1^c \cap A_2) = B \cap (A_1 \cup A_2)$.

Then by subadd:

$$\geq m^*(B \cap (A_1 \cup A_2)) + m^*(B \cap (A_1 \cup A_2)^c)$$

$$\implies A_1 \cup A_2 \in \mathcal{M} \implies A_1 \cup \dots \cup A_N \in \mathcal{M} \implies A_1 \cap \dots \cap A_N \in \mathcal{M}$$

Consider $\{A_n; n \geq 1\} \subseteq \mathcal{M}$. Set $B_1 = A_1$, $B_n = A_n \setminus (\bigcup_{i=1}^{n-1} A_i)$ for $n \geq 2$.

B_n s are disjoint. $\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n$, $B_n \in \mathcal{M} \forall n \geq 1$. To show $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$, we only need to show $\bigcup_{n=1}^{\infty} B_n \in \mathcal{M}$.

For every $n \geq 1$, set $E_n = \bigcup_{i=1}^n B_i$, $\implies E_n \in \mathcal{M} \forall n \geq 1$.

$$\forall B \subseteq \mathbb{R} : m^*(B) = m^*(B \cap E_n) + m^*(B \cap E_n^c); \quad E_n \subseteq \bigcup_{n=1}^{\infty} B_n; \quad E_n^c \supseteq \left(\bigcup_{n=1}^{\infty} B_n \right)^c$$

$$\geq m^*(B \cap B_n) + m^*(B \cap E_{n-1}) + m^*(B \cap \left(\bigcup_{n=1}^{\infty} B_n \right)^c)$$

$$\geq \sum_{i=1}^n m^*(B \cap B_i) + m^*(B \cap (\bigcup_{i=1}^{\infty} B_i)^c) \xrightarrow{n \rightarrow \infty} m^*(B) \geq \sum_{n=1}^{\infty} m^*(B \cap B_n) + m^*(B \cap (\bigcup_{n=1}^{\infty} B_n)^c)$$

$$m^*(B) \geq \sum_{n=1}^{\infty} m^*(B \cap B_n) + m^*(B \cap (\cup_{n=1}^{\infty} B_n)^c)$$

$$\text{Subadd.} \implies \geq m^*(B \cap (\cup_{n=1}^{\infty} B_n)) + m^*(B \cap (\cup_{n=1}^{\infty} B_n)^c) \implies \bigcup_{n=1}^{\infty} B_n \in \mathcal{M}$$

Therefore if $\{A_n; n \geq 1\} \subseteq \mathcal{M}, \cup_{n=1}^{\infty} A_n \in \mathcal{M} \checkmark$

So $\mathcal{M}$ is a σ -algebra. □

" B_1 is the only one that survives this intersection!" - Linan Chen

Proof. Next, we check that $m = m^*|_{\mathcal{M}}$ is a measure.

1) $m(\emptyset) = 0 \checkmark$

2) Given $\{A_n; n \geq 1\} \subseteq \mathcal{M}$ disjoint, following exactly the same argument as above, we can get $\forall n \geq 1, m(\cup_{i=1}^n A_i) = \sum_{i=1}^n m(A_i)$

$$\text{Therefore } m(\cup_{i=1}^{\infty} A_i) \geq m(\cup_{i=1}^n A_i) = \sum_{i=1}^n m(A_i) \implies m(\cup_{i=1}^{\infty} A_i) \geq \sum_{i=1}^{\infty} m(A_i)$$

And subadd. implies the reverse ineq. Thus m is countably additive and a measure. □

Proposition 1.2.2

$\mathcal{M}$ and m are translation invariant, i.e., $\forall A \in \mathcal{M}, x \in \mathbb{R}$:

$A + x = \{y + x; y \in A\} \in \mathcal{M}$ and $m(A + x) = m(A)$.

Proof. $\forall B \subseteq \mathbb{R} : m^*(B) = m^*(B - x) = m^*((B - x) \cap A) + m^*((B - x) \cap A^c)$
 $= m^*(B \cap (A + x)) + m^*(B \cap (A + x)^c)$ [recall that $A^c + x = (A + x)^c$] $\implies A + x \in \mathcal{M}$,
 and $m(A + x) = m(A)$ because $m^*(A + x) = m^*(A)$.

Proposition 1.2.3

$\forall a, b \in \mathbb{R}, a < b, (a, b) \in \mathcal{M}$ and $m((a, b)) = b - a$.

Proof. Given $a, b \in \mathbb{R}, a < b$, it's enough to show that $\forall B \subseteq \mathbb{R}, m^*(B) \geq m^*(B \cap (a, b)) + m^*(B \cap (a, b)^c)$ (the other direction is given by subadditivity.)

Wlog. assume that $m^*(B) < \infty$. $\forall \varepsilon > 0$, choose $\{I_n : n \geq 1\}$ open intervals s.t $B \subseteq \cup_{n=1}^{\infty} I_n$ and $m^*(B) \geq \sum_{n=1}^{\infty} \ell(I_n) - \varepsilon$.

For each $n \geq 1$ set $J_n = I_n \cap (a, b)$, $K_n = I_n \cap (-\infty, a]$, and $L_n = I_n \cap [b, \infty)$. Then J_n, K_n, L_n are almost disjoint intervals, and $m^*(I_n) = m^*(J_n) + m^*(K_n) + m^*(L_n)$ (and so equivalently $\ell(I_n) = \ell(J_n) + \ell(K_n) + \ell(L_n)$).

$$m^*(B) \geq \sum_{n=1}^{\infty} \ell(I_n) - \varepsilon = \sum_{n=1}^{\infty} \ell(J_n) + \sum_{n=1}^{\infty} (\ell(K_n) + \ell(L_n)) - \varepsilon \quad [d(K_n, L_n) > 0]$$

$$= \sum_{n=1}^{\infty} m^*(I_n \cap (a, b)) + \sum_{n=1}^{\infty} m^*(I_n \cap (a, b)^c) - \varepsilon$$

$$\begin{aligned} \text{By countable subadd.} \quad &\geq m^*\left(\left(\bigcup_{n=1}^{\infty} I_n\right) \cap (a, b)\right) + m^*\left(\left(\bigcup_{n=1}^{\infty} I_n\right) \cap (a, b)^c\right) - \varepsilon \\ &\geq m^*(B \cap (a, b)) + m^*(B \cap (a, b)^c) - \varepsilon \end{aligned}$$

Since ε arbitrarily small, we have the statement, masha'allah. $\square$

Corollary 1.2.4

$\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}$; $\forall$ intervals $I, I \in \mathcal{M}, m(I) = \ell(I)$.

Proposition 1.2.5

Regularity Properties

- 1) $\forall A \in \mathcal{M}, \varepsilon > 0, \exists$ open G s.t $A \subseteq G, m(G \setminus A) \leq \varepsilon$.
- 2) $\forall A \in \mathcal{M}, \varepsilon > 0, \exists$ closed F s.t $F \subseteq A, m(A \setminus F) \leq \varepsilon$.
- 3) $\forall A \in \mathcal{M}, m(A) = \inf\{m(G) : G \text{ open and } A \subseteq G\}$ (outer regularity)
- 4) $\forall A \in \mathcal{M}, m(A) = \sup\{m(K) : K \text{ compact and } K \subseteq A\}$ (inner regularity)
- 5) $\forall A \in \mathcal{M}$ where $m(A) < \infty$ (finite), $\varepsilon > 0 \exists$ a compact set $K \subseteq A$ s.t. $m(A \setminus K) \leq \varepsilon$.
- 6) $\forall A \in \mathcal{M}$ where $m(A) < \infty, \varepsilon > 0 \exists$ a finite collection of (open) intervals $I_1, \dots, I_N$ s.t $m(A \Delta (\bigcup_{n=1}^N I_n)) \leq \varepsilon$ (recall $A \Delta B = (A \setminus B) \cup (B \setminus A)$).

Proof. of 5).

$\forall \varepsilon > 0$, by 2) $\exists$ closed F s.t $F \subseteq A$ and $m(A \setminus F) \leq \varepsilon$. We must first bound F . For each $n \geq 1$ set $F_n = [-n, n] \cap F$; F_n is compact. Hence $A \setminus F_n$ is a decreasing sequence of sets, and we have $\bigcap_{n=1}^{\infty} (A \setminus F_n) = A \setminus F$.

By continuity (from above) [and also the fact that $m(A) < \infty$] of m :

$$\frac{\varepsilon}{2} \geq m(A \setminus F) = \lim_{n \rightarrow \infty} m(A \setminus F_n); \exists \text{ large enough } N \text{ s.t } m(A \setminus F_N) \leq \varepsilon.$$

So F_N is compact; $F_N \subseteq A$ and $m(A \setminus F_N) \leq \varepsilon$. $\square$

Proof. of 6). $\forall \varepsilon > 0$ choose $\mathcal{I}_n : n \geq 1\}$ open intervals s.t $A \subseteq \bigcup_{n=1}^{\infty} I_n$ nad $m(A) \geq \sum_{n=1}^{\infty} \ell(I_n) - \frac{\varepsilon}{2}$. Since $m(A) < \infty, \sum_{n=1}^{\infty} \ell(I_n)$ is convergent.

$\exists$ large enough $N \geq 1$ s.t $\sum_{n=N+1}^{\infty} \ell(I_n) \leq \frac{\varepsilon}{2}$, so:

$$m(A \Delta (\bigcup_{n=1}^N I_n)) = m(A \setminus \bigcup_{n=1}^N I_n) + m(\bigcup_{n=1}^N I_n \setminus A) \leq m(\bigcup_{n=N+1}^{\infty} I_n) + m(\bigcup_{n=1}^{\infty} I_n \setminus A)$$

$$\text{By countable subadd.} \leq \sum_{n=N+1}^{\infty} \ell(I_n) + m(\bigcup_{n=1}^{\infty} I_n) - m(A) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

□

Fact 1.2.6

Reminder: for some sigma algebra $\mathcal{F}$, for $A \in \mathcal{F}$, in general, $B \subseteq A \not\Rightarrow B \in \mathcal{F}$

Proposition 1.2.7

Completeness (of Lebesgue Measure)

$(\mathbb{R}, \mathcal{M}, m)$ is a complete measure space in the sense that $\forall A \subseteq \mathbb{R}$, if $\exists B \in \mathcal{M}$ s.t $A \subseteq B$ and $m(B) = 0$ then $A \in \mathcal{M}$ and $m(A) = 0$.

(any subset of a null set is again a null set).

Proof. Assume $A \subseteq \mathbb{R}$ s.t $A \subseteq B$ for some $B \in \mathcal{M}$ with $m(B) = 0$. We need to show that $\forall E \in \mathbb{R}$, $m^*(E) \geq m^*(E \cap A) + m^*(E \cap A^c)$.

Since $A \subseteq B$, $m^*(E \cap A) \leq m^*(E \cap B)$. Since $B \in \mathcal{M}$ and $m(B) = 0$:

$$m^*(E \cap A^c) = m^*(E \cap A^c \cap B) + m^*(E \cap A^c \cap B^c) \leq m^*(B) + m^*(E \cap B^c) = m^*(E \cap B^c)$$

$$m^*(E \cap A) + m^*(E \cap A^c) \leq m^*(E \cap B) + m^*(E \cap B^c) = m^*(E) \implies A \in \mathcal{M} \therefore m(A) \leq m(B) = 0$$

□

Remark.

There is an equivalent statement of completeness for $(\mathbb{R}, \mathcal{M}, m)$:

$\forall E \subseteq \mathbb{R}$, if $\exists F, G \in \mathcal{M}$ s.t $F \subseteq E \subseteq G$ and $m(G \setminus F) = 0$ then:

$$E \in \mathcal{M} \text{ and } m(E) = m(G) = m(F)$$

(Basically the squeeze theorem for measures - a very hot statement)

Proposition 1.2.8

(Uniqueness of m) (Now, let's consider $m|_{\mathcal{B}_{\mathbb{R}}}(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, m)$ is a measure space)

Up to rescaling, m is the unique (non-trivial) measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ that is finite on compact sets on translation invariant, i.e., if μ is another such measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$, then $\mu = c \cdot m$ for some $c \geq 0$ [and $c = \mu((0, 1))$].

Check: if $\forall A \in \mathcal{B}_{\mathbb{R}}, cm(A) = c \cdot (m(A))$, then cm is a measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$.

The uniqueness of m can be derived from Dynkin's $\pi - d$ theorem.

Theorem 1.2.9: Dynkin's $\pi - d$ Theorem

Given a space x , let $\mathcal{C}$ be a collection of subsets of X . $\mathcal{C}$ is called a π -system if $A, B \in \mathcal{C} \implies A \cap B \in \mathcal{C}$ (π -system is closed under finite intersction.)

Assume $\mathcal{F} = \sigma(\mathcal{C})$ and μ_1, μ_2 are two FINITE measures on $(X, \mathcal{F})$ s.t $\mu_1(X) = \mu_2(X)$ and $\mu_1 = \mu_2$ on $\mathcal{C}$, i.e. $\forall A \in \mathcal{C}, \mu_1(A) = \mu_2(A)$.

Then $\mu_1 = \mu_2$ on $\mathcal{F}$, i.e., $\forall B \in \mathcal{F}, \mu_1(B) = \mu_2(B)$.

Remark.

Back to $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$:

$$\mathcal{B}_{\mathbb{R}} = \sigma(\{(a, b) : a, b \in \mathbb{R}, a < b\}) = \sigma(\emptyset \cup \mathcal{C}) \quad [\text{a } \pi\text{-system}]$$

For each $n \geq 1$, $\mathcal{C}_{\cap[-n, n]} = \{(a, b) \cap [-n, n] : a, b \in \mathbb{R}\}$, a π -system - then: $\sigma(\mathcal{C}_{\cap[-n, n]}) = \sigma(\mathcal{C})_{\cap[-n, n]} \implies \mathcal{B}_{[-n, n]} = \sigma(\mathcal{C}_{\cap[-n, n]})$

Proposition 1.2.10

If μ is a measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ s.t t $\forall$ intervals I , $\mu(I) = \ell(I)$ then $\mu = m$ (μ must be the Lebesgue measure).

Proof. $\forall n \geq 1$, consider μ restricted on $\mathcal{B}_{[-n, n]}$. Clearly $\mu([-n, n]) = \ell([-n, n]) = 2n < \infty$, so μ is a finite measure on $[-n, n]$.

Let $\mathcal{I} = \{(a, b) : a < b\}$ be the collection of all open intervals. Then $\mathcal{I}$ is a π -system (closed under finite intersections) and $\sigma(\mathcal{I}) = \mathcal{B}_{\mathbb{R}}$.

By assumption, $\forall (a, b) \in \mathcal{I}, \mu((a, b)) = \ell((a, b)) = b - a = m((a, b))$.

Therefore μ and m agree on the π -system $\mathcal{I}$. By the π - λ theorem (or uniqueness of measures on generated σ -algebras), since both μ and m are finite measures on $\mathcal{B}_{[-n, n]}$ that agree on

$\mathcal{I} \cap \mathcal{B}_{[-n,n]}$, we have

$$\mu(E) = m(E) \quad \forall E \in \mathcal{B}_{[-n,n]}$$

Finally, for any $E \in \mathcal{B}_{\mathbb{R}}$, write $E = \bigcup_{n=1}^{\infty} (E \cap [-n, n])$. Then

$$\mu(E) = \lim_{n \rightarrow \infty} \mu(E \cap [-n, n]) = \lim_{n \rightarrow \infty} m(E \cap [-n, n]) = m(E)$$

Thus $\mu = m$ on $\mathcal{B}_{\mathbb{R}}$. □

Proposition 1.2.11

If μ is a measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ assigning finite values to compact sets, and μ is translation invariant, then $\mu = c \cdot m$ for some $c > 0$.

Remark.

We need to first check that $\mathcal{B}_{\mathbb{R}}$ is translation invariant, i.e. $\forall B \in \mathcal{B}_{\mathbb{R}}, \forall x \in \mathbb{R}, B+x \in \mathcal{B}_{\mathbb{R}}$. Proof sketch: Use the "good set strategy": define a collection of the "good sets" $\Sigma = \{B \in \mathcal{B}_{\mathbb{R}} : B+x \in \mathcal{B}_{\mathbb{R}}\}$. Check that Σ is a σ -algebra, then show that $\{(a, b) : a, b \in \mathbb{R}\} \subseteq \Sigma \subseteq \mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{R}} = \sigma(\{(a, b) : a, b \in \mathbb{R}\}) \subseteq \Sigma$ and so $\Sigma = \mathcal{B}_{\mathbb{R}}$.

Proof. Set $c = \mu((0, 1])$. We take $c > 0$:

$$\left(c = 0 \implies \forall n \in \mathbb{Z}, \mu((n, n+1]) = 0 \implies \mathbb{R} = \bigcup_{n \in \mathbb{Z}} (n, n+1] \implies \mu(\mathbb{R}) = \sum_n \mu((n, n+1]) = 0 \right)$$

$$c > 0 \implies \forall n \geq 1, \mu((0, \frac{1}{n}]) = \frac{c}{n}$$

$$\implies \forall m = 1, 2, \dots, n-1 : \mu((0, \frac{m}{n}]) = \frac{m}{n}c \implies \forall q \in \mathbb{Q} \cap (0, 1], \mu((0, q]) = qc$$

$$\implies \forall \text{ intervals } I, \mu(I) = c\ell(I) \text{ [by continuity]} \implies \forall q \in \mathbb{Q}^+, \mu((0, q]) = qc$$

$$\therefore \forall n \geq 1, \forall a, b \in \mathbb{R}, \mu((a, b) \cap [-n, n]) = c\ell((a, b) \cap [-n, n]) = cm((a, b) \cap [-n, n])$$

$$\implies \mu = cm \text{ on } \mathcal{B}_{[-n,n]} \implies \mu = cm \text{ on } \mathcal{B}_{\mathbb{R}}$$

□

Proposition 1.2.12

(Scaling Property)

m has the scaling property that $\forall A \in \mathcal{M}, \forall c \in \mathbb{R}, cA = \{cx : x \in A\} \in \mathcal{M}$ and $m(cA) = |c|m(A)$. In particular, m is reflection symmetric, i.e. $\forall A \in \mathcal{M}, -A \in \mathcal{M}$ and $m(-A) = m(A)$.

Proof. Method 1: direct. A sketch.

Assume $c \neq 0$. Given any $A \subseteq \mathbb{R}$, observe that $\{I_n : n \geq 1\}$ is an open interval covering of A iff. $\{cI_n\}$ is an open interval covering of cA , and observe that $\ell(cI_n) = |c|\ell(I_n) \implies m^*(cA) = |c|m^*(A)$.

Now assume that $A \in \mathcal{M}$. Argue that $\forall B \subseteq \mathbb{R} : m^*(B) = m^*(B \cap (cA)) + m^*(B \cap (cA)^c)$. [If A is a 'good cookie cutter', argue that cA is also a 'good cookie cutter'.]

(Observe that $B \cap (cA) = c(\frac{1}{c}B \cap A)$). Then $cA \in \mu$. $\square$

Proof. Method 2: alternative via Borel. Requires some further facts.

Assume $c \neq 0$. First prove the scaling property on $\mathcal{B}_{\mathbb{R}}$. $\forall B \in \mathcal{B}_{\mathbb{R}}$ define $\mu_c(B) = m(cB)$. Check that μ_c is indeed a measure and that μ_c is translation invariant $\implies \mu_c = |c|m$ on $\mathcal{B}_{\mathbb{R}}$. $\implies \forall B \in \mathcal{B}_{\mathbb{R}} \forall c \in \mathbb{R} \setminus \{0\}, m(cB) = |c|m(B)$.

But how do we go from $\mathcal{B}_{\mathbb{R}}$ to $\mathcal{M}$? We need some more information on the relation. $\square$

1.3 Relation Between $\mathcal{B}_{\mathbb{R}}$ and $\mathcal{M}$

Recall that $(\mathbb{R}, \mu, m)$ is complete, i.e., $\forall B \subseteq \mathbb{R}$, if $\exists A \in \mathcal{M}$ with $m(A) = 0$ s.t $B \subseteq A$, then $B \in \mathcal{M}$ and $m(B) = 0$.

Definition 1.3.1: Completion

Given a measure space $(X, \mathcal{F}, \mu)$, consider the following collection of subsets of X : $\mathcal{N} = \{B \subseteq X : \exists A \in \mathcal{F} \text{ with } \mu(A) = 0 \text{ s.t } B \subseteq A\}$.

Then, $\overline{\mathcal{F}} = \sigma(\mathcal{F} \cup \mathcal{N})$ is called the completion of $\mathcal{F}$ with respect to μ . If $\mathcal{N} \subseteq \mathcal{F}$, then $\overline{\mathcal{F}} = \mathcal{F}$, in which case $\mathcal{F}$ is already complete.

Proposition 1.3.2

$$\overline{\mathcal{F}} = \{F \subseteq X : \exists E, G \in \mathcal{F} \text{ s.t } E \subseteq F \subseteq G \text{ and } \mu(G \setminus E) = 0\}$$

Proof. (Mostly a sketch)

Denote by $\mathcal{G}$ the collection on the RHS. First verify that $\mathcal{G}$ is a σ -algebra.

Since $\mathcal{F} \subseteq \mathcal{G}$ and $\mathcal{N} \subseteq \mathcal{G} \implies \overline{\mathcal{F}} = \sigma(\mathcal{F} \cup \mathcal{N}) \subseteq \mathcal{G}$. Meanwhile, for any $F \in \mathcal{G} \exists E, G \in \mathcal{F}$ s.t $E \subseteq F \subseteq G$ with $\mu(G \setminus E) = 0$.

Write $F = E \cup (F \setminus E) \implies F \in \sigma(\mathcal{F} \cup \mathcal{N}) = \overline{\mathcal{F}}$. (Since $F \setminus E \subseteq G \setminus E$ with $\mu(G \setminus E) = 0 \implies F \setminus E \in \mathcal{N}$). $\implies \mathcal{G} \subseteq \overline{\mathcal{F}} \implies \overline{\mathcal{F}} = \mathcal{G}$. $\square$

Definition 1.3.3: Extension to Completeness

Given a measure space $(X, \mathcal{F}, \mu)$, μ can be extended to $\overline{\mathcal{F}}$ as follows: if $F \in \mathcal{F}$, assume $E, G \in \mathcal{F}$ s.t $E \subseteq F \subseteq G$ with $\mu(G \setminus E) = 0$, then $\mu(F) = \mu(E) = \mu(G)$. Then $(X, \overline{\mathcal{F}}, \mu)$ is a complete measure space.

Example.

Exercise: verify μ is well-defined on $\overline{\mathcal{F}}$ given $F \in \overline{\mathcal{F}}$, i.e if $E, G \in \mathcal{F}$ s.t $E \subseteq F \subseteq G$ with $\mu(G \setminus E) = 0$ and $E', G' \in \mathcal{F}$ s.t $E' \subseteq F' \subseteq G'$ with $\mu(G' \setminus E') = 0$, then $\mu(F) = \mu(E) = \mu(G) = \mu(F') = \mu(E') = \mu(G')$.

Theorem 1.3.4: Borel Completion

$(\mathbb{R}, \mathcal{M}, m)$ is the completion of $(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, m)$.

Proof. Recall the regularity properties (1) and (2) of m . Given $A \in \mathcal{M}$, $\forall n \geq 1 \exists$ open set G_n s.t $A \subseteq G_n$ and $m^*(G_n \setminus A) \leq \frac{1}{n}$ and $\exists$ closed set F_n s.t $F_n \subseteq A$ and $m^*(A \setminus F_n) \leq \frac{1}{n}$. Set $C = \bigcap_{n=1}^{\infty} G_n, B = \bigcup_{n=1}^{\infty} F_n \implies C, B \in \mathcal{B}_{\mathbb{R}}, B \subseteq A \subseteq C$. Moreover, $m(C \setminus A) \leq m(G_n \setminus A) \leq \frac{1}{n} \forall n \geq 1$ and $m(A \setminus B) \leq m(A \setminus F_n) \leq \frac{1}{n}$.

$$\implies m(C \setminus B) = m(C \setminus A) + m(A \setminus B) \leq \frac{2}{n}$$

$$\implies \text{as } n \rightarrow \infty, m(C \setminus B) = 0 \implies A \in \overline{\mathcal{B}_{\mathbb{R}}} \implies \mathcal{M} = \overline{\mathcal{B}_{\mathbb{R}}}$$

And so any measurable set is different from a Borel set by at most a null set: $\implies$ if $m(cB) = |c|m(B) \forall B \in \mathcal{B}_{\mathbb{R}}$, then $m(cA) = |c|m(A) \forall A \in \mathcal{M}$. (i.e., the scaling method #2 proof from earlier is now doable.) $\square$

1.4 Some Special Sets

Q1: is there $A \in \mathcal{M}$ s.t $m(A) = 0$ and A is uncountable? (One direction: if A is countable, then $m(A) = 0$.) But there are uncountable sets with zero measure, say, the Cantor set.)

Theorem 1.4.1: Cantor's Uncountable Set of Zero Measure

- 1) C is closed, since C_n is the union of 2^n disjoint closed intervals. So C_n is closed as it involves finitely many closed intervals so C is closed. And so $C \in \mathcal{B}_{\mathbb{R}}$ - it is Borel.
- 2) $m(C) = 0$. Since C_n is just a bunch of disjoint intervals, $m(C_n)$ will just be the sums of their lengths, i.e:

$$m(C_n) = 2^n \frac{1}{3^n} = \left(\frac{2}{3}\right)^n \text{ so by continuity of measure } m(C) = \lim_{n \rightarrow \infty} m(C_n) = 0$$

- 3) C is uncountable. Observe that $\forall x \in [0, 1], \exists (a_1, a_2, a_3, \dots)$ s.t $\forall n \geq 1, a_n \in \{0, 1, 2\}$ s.t $x = \sum_{n=1}^{\infty} a_n \frac{1}{3^n}$ (base-3 expansion of x). [Typical proof that Cantor is uncountable.]

Definition 1.4.2: The Cantor-Lebesgue Function

(The Devil's Staircase) ("Night at the Museum" metaphor).

Define a function $f : [0, 1] \rightarrow [0, 1]$:

$$f(x) = \begin{cases} \sum_{n=1}^{\infty} \frac{a_n}{2} \frac{1}{2^n} & \text{if } x \in C \text{ and } x = \sum_{n=1}^{\infty} a_n \frac{1}{3^n} \\ f(a) & \text{if } x \notin C \text{ then } x \in (a, b) \text{ s.t } (a, b) \text{ gets removed} \end{cases}$$

Its properties:

- 1) $f(0) = 0, f(1) = 1, f(\frac{1}{3}) = f(\frac{2}{3}) = \frac{1}{2}, f(\frac{1}{9}) = f(\frac{2}{9}) = \frac{1}{4}$; $f = \frac{1}{2}$ on $(\frac{1}{3}, \frac{2}{3})$, and so on.

Note: Since $m(C) = 0$, then $m([0, 1] \setminus C) = 1$.

- 2) $f : [0, 1] \rightarrow [0, 1]$ is surjective.

- 3) f is nondecreasing.

Proof: if $x_1, x_2 \in C$ with $x_1 < x_2$, say in base-3 expansion, $x_1 \rightarrow (a_1, a_2, \dots)$ and $x_2 \rightarrow (b_1, b_2, \dots)$, then it must be that $a_1 = b_1, a_2 = b_2, \dots, a_i = b_i \forall i \leq n$ for some n and then $a_{n+1} \leq b_{n+1} \implies f(x_1) \leq f(x_2)$.

$\begin{matrix} \parallel & \parallel \\ 0 & 2 \end{matrix}$

- 4) f is continuous. Proof: f is continuous on $[0, 1] \setminus C$ (f is piecewise constant on $[0, 1] \setminus C$) and f is one-sided continuous at $\frac{1}{3}, \frac{2}{3}, \frac{1}{9}, \frac{2}{9}$, etc. If $x \in C$ then $\forall n \geq 1$ there must be x_n, x'_n s.t $x_n < x < x'_n$ (if $x = 0$, then we only need $x < x'_n$; if $x = 1$, we only need $x_n < x$) s.t $f(x'_n) - f(x_n) \leq \frac{1}{2^n} \implies f$ is continuous at x by the monotonicity of f . (Since $\forall y, x \in (x_n, x'_n), |f(x) - f(y)| \leq \frac{1}{2^n}$.)

Q2: IS there $A \subseteq \mathbb{R}$ that is non measurable? Yes.

Definition 1.4.3: Axiom of Choice

If Σ is a collection of non-empty sets, then $\exists$ a function $S : \Sigma \rightarrow \bigcup_{A \in \Sigma} A$ s.t. $\forall A \in \Sigma, S(A) \in A$ (S is called a selection function). We refer to $S(A)$ as a representative of A .

Lemma 1.4.4: Construction of a Non-Measurable Set

Define an equivalence relation " $\sim$ " on $[0, 1]$:

$$\text{for } a, b \in [0, 1] \quad a \sim b \iff a - b \in \mathbb{Q}$$

(Recall equiv. relation properties: $a \sim a, a \sim b \iff b \sim a, a \sim b \& b \sim c \implies a \sim c$.)

Denote by E_a the equivalence class that contains a . Set $\Sigma = \{\text{collection of all the equivalence classes}\}$. $\forall E_a \in \Sigma, E_a \neq \emptyset$.

By the axiom of choice, we can select exactly one element S_a from E_a for each $E_a \in \Sigma$. Set $N = \{S_a : S_a \text{ is a representative of } E_a, E_a \in \Sigma\}$.

(You can see that this depends on the axiom of choice in order to make the choice in each set.)

Proposition 1.4.5

N is non-measurable.

Proof. Assume otherwise, i.e., assume $N \in \mathcal{M}$. Consider $[1, 1] \cap \mathbb{Q}$. Assume $\{q_k : k \geq 1\}$ is an enumeration of $[1, 1] \cap \mathbb{Q}$. For each $k \geq 1$, set $N_k = N + q_k$. By the assumption, $N_k \in \mathcal{M}$ and $m(N_k) = m(N)$.

Claim: N_k s are disjoint (i.e. $N_k \cap N_\ell = \emptyset \forall k \neq \ell$).

Assume not: $\exists q_k \neq q_\ell, S_a, S_b \in N$ s.t $S_a + q_k = S_b + q_\ell$

$\implies S_a - S_b = q_\ell - q_k \in \mathbb{Q} \implies S_a, S_b$ are the same equivalence class $S_a \sim S_b \implies S_a = S_b \implies q_k = q_\ell$; contradiction. Then N_k s are disjoint.

Claim: $[0, 1] \subseteq \bigcup_{k=1}^{\infty} N_k$.

To see it, take $x \in [0, 1]$. Then $x \sim S_a$ for some $S_a \in N \implies x - S_a \in \mathbb{Q}$. Since $x - S_a \in [-1, 1]$, $\exists k$ s.t $x - S_a = q_k \implies x \in N_k$.

Main proof continues. On the other hand, by default, $\bigcup_{n=1}^{\infty} N_k \subseteq [-1, 2]$
 $\implies N_k$ s are disjoint and $[0, 1] \subseteq \bigcup_{k=1}^{\infty} N_k \subseteq [-1, 2] \quad (*)$:

$$\implies m\left(\bigcup_{n=1}^{\infty} N_k\right) = \sum_{k=1}^{\infty} m(N_k) \text{ and } 1 \leq m\left(\bigcup_{n=1}^{\infty} N_k\right) \leq 3$$

Impossible! $\implies N \notin \mathcal{M}$.

Then N is called a **Vitali** set. But note at $(*)$ that this directly implies that $m^*(N) > 0$ since all subsets of the real line are outer-measurable.

Corollary 1.4.6

Vitali sets are ubiquitous, i.e., $\forall A \in \mathcal{M}$ with $m(A) > 0$, $\exists B \subseteq A$ s.t B is non-measurable.

Proof. Assume otherwise, i.e. $\exists A \in \mathcal{M}$ with $m(A) > 0$ s.t $\forall B \subseteq A, B \text{ in } \mathcal{M}$.

$$A \subseteq \bigcup_{n \in \mathbb{Z}} (A \cap [n, n+1]) \implies \exists n \text{ s.t } m(A \cap [n, n+1]) > 0 \implies m(A \cap [n, n+1] - n) > 0$$

Define $A' = A \cap [n, n+1] - n$, so we know $A' \subseteq [0, 1], m(A') > 0$. Then $\forall B' \subseteq A', B' + n \subseteq A$. By hypothesis, $B' + n \in \mathcal{M} \implies B' \in \mathcal{M}$.

Now we have $A' \subseteq [0, 1]$ with $m(A') > 0$ s.t $\forall B' \subseteq A', B' \in \mathcal{M}$.

Now let N, q_k , and N_k be as in the Vitali set. Set $A'_k = A' \cap N_k \forall k \geq 1$. We know (for the same reason as earlier) that A'_k s are disjoint, and:

$$A' = [0, 1] \cap A' \subseteq \bigcup_{k=1}^{\infty} (N_k \cap A') = \bigcup_{k=1}^{\infty} A'_k$$

Since $m(A') > 0$, then there must $\exists k$ such that $m(A'_k) > 0$.

Fix k . Set $L = \{\ell \geq 1 : q_\ell + \text{equals to } q_k \text{ fixed} \in [-1, 1]\}$, which is countably infinite.

Now we translate. Consider $\{q_\ell + A'_k : \ell \in L\}$. Since $q_\ell + q_k \in [-1, 1] \cap \mathbb{Q}$, $\implies q_\ell + q_k = q_m$ for some unique m .

$$\implies q_\ell + A'_k = q_\ell + A' \cap (N + q_k) \subseteq N_m$$

$\implies \{q_\ell + A'_k : \ell \in L\}$ is a family of disjoint sets.

$$\text{Again } \bigcup_{\ell \in L} (q_\ell + A'_k) \subseteq [-1, 2] \implies \sum_{\ell \in L} m(q_\ell + A'_k) \leq 3$$

$$\implies m(A'_k) = 0. \text{ Contradiction.}$$

Q3: Is there a set A that is measurable ($A \in \mathcal{M}$) but $A \notin \mathcal{B}_{\mathbb{R}}$? Yes.

Lemma 1.4.7: Construction of a Measurable Non-Borel Set

Let $f : [0, 1] \rightarrow [0, 1]$ be the Cantor-Lebesgue function, and write $g(x) = f(x) + x$. This implies that g is continuous and strictly increasing ($\implies$ injective).

Then $g : [0, 1] \rightarrow [0, 2]$ is a bijection, and so $g^{-1} : [0, 2] \rightarrow [0, 1]$ exists and is also continuous, so g is a homeomorphism, and maps open to open, and closed to closed.

$\implies$ if $A \in \mathcal{B}_{\mathbb{R}}$, then $g(A) \in \mathcal{B}_{\mathbb{R}}$.

Observe that if (a, b) is an open interval that gets removed from the construction of C then f is constant on $(a, b) \implies g$ will map (a, b) to an interval of length $b - a$

$$\implies m(g([0, 1] \setminus C)) = m([0, 1] \setminus C) = 1 \implies m(g(C)) = 2 - 1 = 1 > 0$$

Since $g(C) \subseteq [0, 2]$, $\implies \exists B \subseteq g(C)$ s.t $B \notin \mathcal{M}$. Set $A = g^{-1}(B) \implies A \subseteq g^{-1}(g(C)) = C$. Since $m(C) = 0$, $A \in \mathcal{M}$ and $m(A) = 0$.

But $A \notin \mathcal{B}_{\mathbb{R}}$ (because if $A \in \mathcal{B}_{\mathbb{R}}$ then $B = g(A) \in \mathcal{B}_{\mathbb{R}}$ (which would contradict $B \notin \mathcal{M}$)).

Chapter 2: Integration Theory

2.1 Measurable Functions

We consider a function f defined on $\mathbb{R}$ (or some $A \subseteq \mathbb{R}$) that could take $\pm\infty$ as its value, i.e. $f : \mathbb{R} \rightarrow \overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$ - f is extended-real-valued.

Definition 2.1.1: Finite-Valued

If $-\infty < f(x) < \infty \forall x \in \mathbb{R}$ then f is finite-valued (or real-valued).

Note that f^{-1} is not the inverse function of f - it is its inverse image. Now, $\forall a \in \mathbb{R}$:

$$f^{-1}([-\infty, a)) = \{x \in \mathbb{R} : f(x) \in [-\infty, a)\} = \{f < a\}$$

$$\text{And: } f^{-1}((a, \infty]) = \{x \in \mathbb{R} : f(x) \in (a, \infty]\} = \{f > a\}$$

i.e., we include infinity (or negative infinity) when we write $f > a$ (or vice-versa). Generally:

$$\forall B \in \mathbb{R}, f^{-1}(B) = \{x \in \mathbb{R} : f(x) \in B\} = \{f \in B\}$$

Fact 2.1.2

These are all followings from the fact that:

- 1) $\forall B \in \mathbb{R}, f^{-1}(B^c) = (f^{-1}(B))^c$
- 2) $\forall A, B \subseteq \mathbb{R} f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$
- 3) $\forall A, B \subseteq \mathbb{R} f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$

And same for countable union and intersection; in general, inverse image plays very nicely with set relations.

Definition 2.1.3: Measurable Function

$f : \mathbb{R} \rightarrow \overline{\mathbb{R}}$ is measurable if $\forall a \in \mathbb{R}, f^{-1}([-\infty, a)) \in \mathcal{M}$ (i.e. $\{f < a\} \in \mathcal{M}$).

But there are many equivalent definitions of measurability of f :

f measurable $\iff \forall a \in \mathbb{R}, f^{-1}([a, \infty]) \in \mathcal{M}$

$\iff \forall a \in \mathbb{R}, f^{-1}((a, \infty]) \in \mathcal{M}$

$\iff \forall a \in \mathbb{R}, f^{-1}([-\infty, a]) \in \mathcal{M}$

If f is finite-valued, replace ∞ with b to have intervals of the form (a, b) , $[a, b]$, etc.

Definition 2.1.4: Extended Borel Algebra

Consider the Borel algebra $\mathcal{B}_{\mathbb{R}}$ of subsets of $\overline{\mathbb{R}}$:

$$\mathcal{B}_{\overline{\mathbb{R}}} = \sigma(\mathcal{B}_{\mathbb{R}} \cup \{\{\infty\}, \{-\infty\}\})$$

Fact 2.1.5

It is true that $\mathcal{B}_{\overline{\mathbb{R}}} = \sigma(\{[-\infty, a) : a \in \mathbb{R}\})$.

Proof. $\forall a \in \mathbb{R}$, write $[-\infty, a) = (-\infty, a) \cup \{-\infty\}$; then clearly this $\in \sigma(\mathcal{B}_{\mathbb{R}} \cup \{\{\infty\}, \{-\infty\}\})$
 $\implies \sigma(\{[-\infty, a) : a \in \mathbb{R}\}) \subseteq \mathcal{B}_{\overline{\mathbb{R}}}$

For the other direction:

Note that $\{-\infty\} = \bigcup_{n=1}^{\infty} [-\infty, -n)$ and $\{\infty\} = \overline{\mathbb{R}} \setminus (\bigcup_{n=1}^{\infty} [-\infty, n]) \implies \{-\infty\}\{\infty\} \in \sigma(\{[-\infty, a) : a \in \mathbb{R}\})$

$\forall a \in \mathbb{R}, (-\infty, a) = [-\infty, a) \setminus \{-\infty\} \in \sigma(\{[-\infty, a) : a \in \mathbb{R}\}) \implies \mathcal{B}_{\mathbb{R}} \subseteq \sigma(\{[-\infty, a) : a \in \mathbb{R}\})$
 All together: $\implies \mathcal{B}_{\overline{\mathbb{R}}} \subseteq \sigma(\{[-\infty, a) : a \in \mathbb{R}\})$. $\square$

Proposition 2.1.6

$f : \mathbb{R} \rightarrow \overline{\mathbb{R}}$ is measurable $\iff \forall B \in \mathcal{B}_{\overline{\mathbb{R}}}, f^{-1}(B) \in \mathcal{M}$

Similarly, if f is finite-valued, then f measurable $\iff \forall B \in \mathcal{B}_{\mathbb{R}}, f^{-1}(B) \in \mathcal{M}$.

Proof. $\Leftarrow$ is immediate because $[-\infty, a) \in \mathcal{B}_{\overline{\mathbb{R}}} \forall a \in \mathbb{R}$.

$\implies$: We'll use the result from an assignment problem:
 Given a collection $\mathcal{C}$ of subsets of $\overline{\mathbb{R}}$:

$$f^{-1}(\sigma(\mathcal{C})) = \{f^{-1}(B) : B \in \sigma(\mathcal{C})\} \implies f^{-1}(\sigma(\mathcal{C})) = \sigma(f^{-1}(\mathcal{C}))$$

(by the "good set principle".)

Take $\mathcal{C} = \{[-\infty, a) : a \in \mathbb{R}\}$. Then:

$$f^{-1}(\mathcal{B}_{\overline{\mathbb{R}}}) = f^{-1}(\sigma(\{[-\infty, a) : a \in \mathbb{R}\}) = \sigma(\{f^{-1}([-\infty, a)) : a \in \mathbb{R}\}) \subseteq \mathcal{M}$$

(Since the interior is by definition of a measurable function $\in \mathcal{M}$.)

$\implies$ the title result. □

Proposition 2.1.7

Given $f : \mathbb{R} \rightarrow \overline{\mathbb{R}}$, define:

$$f_{\mathbb{R}}(x) = \begin{cases} f(x) & \text{if } -\infty < f(x) < \infty \\ 0 & \text{otherwise} \end{cases}$$

Then, f measurable $\iff \forall B \in \mathcal{B}_{\mathbb{R}}, f_{\mathbb{R}}^{-1}(B) \in \mathcal{M}$ AND $\{f = -\infty\}, \{f = \infty\} \in \mathcal{M}$.

Definition 2.1.8: Almost Everywhere

If a statement is true for every $x \in A$ where $A \in \mathcal{M}$ s.t $m(A^c) = 0$ then we say that the statement is true a.e ("almost everywhere") or for a.e x .

Proposition 2.1.9

If $f : \mathbb{R} \rightarrow \overline{\mathbb{R}}$ is measurable and $f = g$ a.e, then g is measurable. [Ass. 2]

Corollary 2.1.10

If f is finite-valued a.e then f is measurable $\iff f_{\mathbb{R}}$ is measurable $\iff \forall a, b \in \mathbb{R}, a < b, f^{-1}((a, b)) \in \mathcal{M}$.

Proposition 2.1.11

If $f = c$, i.e., f is the constant function, then f is measurable.

Proof. Assume $f = c$.

$$\forall a \in \mathbb{R}, f^{-1}([-\infty, a)) = \begin{cases} \mathbb{R} & \text{if } c < a \\ \emptyset & \text{if } c \geq a \end{cases} \in \mathcal{M}$$

□

Proposition 2.1.12

If $f = 1_A$ for some $A \subseteq \mathbb{R}$ (f is the characteristic function of A) then f is measurable $\iff A \in \mathcal{M}$.

Proof. Assume $f = 1_A$.

$$\forall a \in \mathbb{R}, f^{-1}([-\infty, a)) = \begin{cases} \mathbb{R} & \text{if } a > 1 \\ A^c & \text{if } 0 < a \leq 1 \\ \emptyset & \text{if } a \leq 0 \end{cases} \in \mathcal{M} \iff A \in \mathcal{M}$$

□

Proposition 2.1.13

If f is a (finite-valued) continuous (continuous necessarily implies FV) on $\mathbb{R}$, then f is measurable.

Proof. $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous $\iff \forall$ open $G \subseteq \mathbb{R}, f^{-1}(G)$ is open $\implies \forall a, b \in \mathbb{R}, f^{-1}((a, b))$ is open $\in \mathcal{M}$.

In fact, if $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then $\forall B \in \mathcal{B}_{\mathbb{R}}, f^{-1}(B) \in \mathcal{B}_{\mathbb{R}}$, so:

$$f^{-1}(\mathcal{B}_{\mathbb{R}}) = f^{-1}(\sigma(\{\text{open sets}\})) = \sigma(f^{-1}(\{\text{open sets}\})) \subseteq \mathcal{B}_{\mathbb{R}} \subseteq \text{open sets}$$

Furthermore, if f^{-1} (the INVERSE function of f) exists and is continuous then $\forall B \in \mathcal{B}_{\mathbb{R}}, f(B) \in \mathcal{B}_{\mathbb{R}}$.

$$(f(B)) = (f^{-1})^{-1}(B).$$

□

Proposition 2.1.14

If $f : \mathbb{R} \rightarrow \mathbb{R}$ (finite-valued) is measurable and $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then $g \circ f : \mathbb{R} \rightarrow \mathbb{R}$ is measurable [order matters! other order not necessarily true.]

Proof. $\forall a \in \mathbb{R}, (g \circ f)^{-1}((-\infty, a)) = \{x \in \mathbb{R} : g(f(x)) < a\} = \{x \in \mathbb{R} : f(x) \in g^{-1}((-\infty, a))\} = f^{-1}(g^{-1}((-\infty, a))) \in \mathcal{M}$.

□

Proposition 2.1.15

If $f : \mathbb{R} \rightarrow \overline{\mathbb{R}}$ is measurable, then:

- 1) $\forall c \in \mathbb{R}, cf$ is measurable (in particular, $-f$!)
- 2) $|f|$ is measurable
- 3) $\forall k \in \mathbb{N}, f^k$ is measurable

Proof. (of 3).

$$\forall a \in \mathbb{R}, (f^k)^{-1}([-\infty, a)) = \begin{cases} f^{-1}([-\infty, a^{\frac{1}{k}})) & \text{if } k \text{ is odd} \\ \emptyset & \text{if } k \text{ is even and } a \leq 0 \\ f^{-1}([-a^{\frac{1}{k}}, a^{\frac{1}{k}})) & \text{if } k \text{ is even and } a > 0 \end{cases} \in \mathcal{M}$$

(If $k = 0, f^k = 1$ is trivial.) □

Proposition 2.1.16

If f, g are two finite-valued measurable functions, then $f+g, f \cdot g, f \vee g = \max\{f, g\}, f \wedge g = \min\{f, g\}$ are measurable too.

Proof. $\forall a \in \mathbb{R}, (f+g)^{-1}([-\infty, a)) = \{x \in \mathbb{R} : f(x) + g(x) < a\} = \{x \in \mathbb{R} : f(x) < a - g(x)\}$
 Since this inequality is STRICT, by density, we can always insert a rational number.

$$= \bigcup_{q \in \mathbb{Q}} \{x \in \mathbb{R} : f(x) < q < a - g(x)\} = \bigcup_{q \in \mathbb{Q}} \{x \in \mathbb{R} : f(x) < q\} \cap \{x \in \mathbb{R} : g(x) < a - q\}$$

But both these components are $\in \mathcal{M}$ so their intersection is $\in \mathcal{M}$ as well.

$$\implies f - g \text{ measurable}$$

$$\implies (f+g)^2 \text{ and } (f-g)^2 \text{ measurable} \implies f \cdot g = \frac{1}{4}[(f+g)^2 - (f-g)^2] \text{ measurable}$$

$$\implies f \vee g = \frac{1}{2}(|f-g| + (f+g)) \text{ measurable, } \implies f \wedge g = -\max\{-f, -g\} \text{ measurable.}$$

□

Corollary 2.1.17

If f is measurable, then $f^+ = f \vee 0 = \max\{f, 0\}$ is measurable, and likewise:
 $f^- = -(f \wedge 0) = \max\{-f, 0\}$ is measurable.

Remark.

- $f = f^+ - f^-$ for all f , and this is always well-defined as $\infty - \infty$ will not occur.
- Moreover, $|f| = f^+ + f^-$.

Remark.

- $\forall k \in \mathbb{R}, f \wedge k$ is measurable.

Now we discuss sequences of measurable functions.

Proposition 2.1.18

Let $\{f_n : n \geq 1\}$ be a sequence of measurable functions.

Then $\sup_n f_n, \inf_n f_n, \limsup_{n \rightarrow \infty} f_n, \liminf_{n \rightarrow \infty} f_n$ are measurable functions.

Proof. To show $\sup_n f_n$ measurable, we will show that $\forall a \in \mathbb{R}, \{\sup_n f_n \leq a\} \in \mathcal{M}$. Now:

$$x \in \{\sup_n f_n \leq a\} \iff \sup_n f_n(x) \leq a \iff f_n(x) \leq a \forall n \iff x \in \bigcap_{n=1}^{\infty} \{f_n \leq a\}$$

So what we have established is that $\{\sup_n f_n \leq a\} = \bigcap_{n=1}^{\infty} \{f_n \leq a\} \in \mathcal{M}$ (since each of the intersection is measurable.) Then $\sup_n f_n$ is measurable.

$\implies \inf_n f_n = -\sup_n(-f_n)$ is measurable.

"Supremum is no bigger than something $\iff$ every element of the sequence is no bigger than that something."

Note: if we study $\{\sup_n f_n < a\}$ we find that $\{\sup_n f_n < a\} \neq \bigcap_{n=1}^{\infty} \{f_n < a\}$; the strict inequality does not hold.

(As an example: if $f_n = a - \frac{1}{n}$, then $\sup_n f_n = a$.)

(To solve, pursue $\{\sup_n f_n < a\} = \bigcup_{k=1}^{\infty} \{\sup_n f_n \leq a - \frac{1}{k}\} = \bigcup_{k=1}^{\infty} \bigcap_{n=1}^{\infty} \{f_n \leq a - \frac{1}{k}\} \in \mathcal{M}.$)

□

Proof. We now want to show that $\limsup_n f_n = \inf_{m \geq 1} (\sup_{n \geq m} f_n) = \inf_{m \geq 1} g_m$.

We know g_m is measurable $\forall m \geq 1 \implies \inf_m g_m$ is measurable, which implies that $\limsup_n f_n$ is measurable.

Then clearly by similar logic, $\liminf_n f_n = \sup_{n \geq 1} (\inf_{m \geq n} f_m)$ is measurable as well.

□

Proposition 2.1.19

Let $\{f_n : n \geq 1\}$ be a sequence of measurable functions. Then, all of the following sets are measurable:

$$\{x \in \mathbb{R} : \lim_{n \rightarrow \infty} f_n \text{ exists in } \mathbb{R}\} \quad \{\lim_n f_n = \infty\} \quad \{\lim_n f_n = -\infty\} \quad \{\lim_n f_n = c \ (c \in \mathbb{R})\}$$

Moreover (hwk), if $\lim f_n$ exists a.e, and you set $f = \lim_n f_n$ a.e, then f is measurable.

Proof.

$$\begin{aligned} \{\lim_n \text{ exists in } \mathbb{R}\} &= \{\limsup_n f_n = \liminf_n f_n : |\limsup_n f_n| < \infty\} \\ &= \{-\infty < \liminf_n f_n < \infty\} \cap \{-\infty < \limsup_n f_n < \infty\} \cap \{\liminf_n f_n - \limsup_n f_n = 0\} \in \mathcal{M} \end{aligned}$$

$$\{\lim_n f_n = c\} = \left\{x \in \mathbb{R} : \forall k \geq 1 \exists n \geq 1 \text{ s.t. } \forall m \geq n, |f_m(x) - c| \leq \frac{1}{k}\right\}$$

$$= \bigcap_{k=1}^{\infty} \bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} \left\{|f_m - c| \leq \frac{1}{k}\right\}$$

”In Linan Chen language... $\forall = \bigcap, \exists = \bigcup$.” □

Example.

Rewrite $\lim_n f_n$ exists in $\mathbb{R}$ using def. of Cauchy sequences.

2.2 Approximation by simple functions

Given $f : \mathbb{R} \rightarrow \overline{\mathbb{R}}$ measurable, in some steps:

- 1) $f = f^+ - f^-$, for f^+, f^- non-negative and measurable
- 2) $\forall n \geq 1 f_n^+ = (f^+ \wedge n)1_{[-n, n]}$ - we combine truncation and cut off. Non-negative, measurable, bounded (by n), zero outside of a bounded set, and moreover monotonic and converging, as $f_n^+ \uparrow; \lim_{n \rightarrow \infty} f_n^+ = f^+$.
Similarly, $f_n^- = (f^- \wedge n)1_{[-n, n]}$; same properties, $f_n^- \uparrow \lim_n f_n^- = f^-$.
- 3) Fix n ; consider f_n^+ . For $k = 0, 1, 2, \dots, n2^n$, define:

$$A_{n,k} = \left\{x \in [-n, n] : \frac{k}{2^n} \leq f_n^+(x) < \frac{k+1}{2^n}\right\} \in \mathcal{M}$$

For $A_{n,k} \cap A_{n,\ell} = \emptyset$ if $k \neq \ell$. Now set:

$$\varphi_n = \sum_{k=0}^{n2^n} 1_{A_{n,k}} \frac{k}{2^n}$$

(if $x \in A_{n,k}$ then $\varphi_n(x) = \frac{k}{2^n}$.)

Definition 2.2.1: Simple Function

We call φ_n s simple functions. φ is a simple function if:

$$\varphi = \sum_{k=1}^L 1_{E_k} \cdot a_k$$

Where L is a positive integer, a_k s are constant, $E_k \in \mathcal{M}$ and $m(E_k) < \infty$.

And (**important!**) moreover... $\varphi_n \uparrow$, so $\lim_{n \rightarrow \infty} \varphi_n = f^+$.

"Cut up your space into lots of tiny little pieces, and on each piece, take the lower bound of your function."

"We cut it in a way such that there is no overlapping between the slices."

Proof. (Of $\lim_{n \rightarrow \infty} \varphi_n(x) = f^+(x) \forall x \in \mathbb{R}$).

$\forall x \in \mathbb{R}$, when n is large enough, $x \in [-n, n]$, which implies that $f^+(x) = f^+(x)1_{[-n,n]}(x)$.

Assume $f^+(x) < \infty$. Then, by making n even larger (if necessary), we can make $f^+(x) < n$.

Then $f^+(x) = ((f \wedge n)1_{[-n,n]})(x) = f_n^+(x)$.

Further, we observe that:

$$0 \leq f_n^+(x) - \varphi_n(x) < 2^{-n} \implies 0 \leq f^+(x) - \varphi_n(x) \leq 2^{-n} \implies \lim_{n \rightarrow \infty} \varphi_n(x) = f^+(x)$$

(i.e. we can take a mesh fine enough so that they are equivalent).

Now: $f^+(x) = \infty$. Then $\varphi_n(x) = n$ for all large n , so $\lim_{n \rightarrow \infty} \varphi_n(x) = \infty = f^+(x)$. $\square$

Theorem 2.2.2: Simple Functions Converge Pointwise

If g is measurable and non-negative, then $\exists$ a sequence of simple functions $\{\varphi_n : n \geq 1\}$ such that $\varphi_n \uparrow$ and $\lim_{n \rightarrow \infty} \varphi_n(x) = g(x) \forall x \in \mathbb{R}$. (Pointwise!!!!!!)

Theorem 2.2.3: Simple Functions Approximate

If f is measurable, then $\exists$ a sequence of simple functions $\{\Psi_n : n \geq 1\}$ s.t. $|\Psi_n| \uparrow$ ($|\Psi_n| \leq |\Psi_{n+1}|$) and $|\Psi_n| \leq |f|$, and $\forall x \in \mathbb{R}$, $\lim_{n \rightarrow \infty} \Psi_n(x) = f(x)$.
Pointwise!

Proof. Take $\Psi_n = \varphi_n - \tilde{\varphi}_n$ ($\forall x \in \mathbb{R}$ at least one of $\varphi_n(x)$ and $\tilde{\varphi}_n(x)$ must be 0).

Then $|\Psi_n| = \varphi_n + \tilde{\varphi}_n < f^+ + f^- = |f|$ and $\lim \Psi_n(x) = \lim \varphi_n(x) - \lim \tilde{\varphi}_n(x)$

(Since $f^+ + f^- = |f|$, and $f^+ - f^- = f$.) □

Now, we would like to "approximate" f by step functions.

Definition 2.2.4: Step Function

θ is called a step function if:

$$\theta(x) = \sum_{k=1}^L a_k 1_{I_k}(x)$$

Where $L \in \mathbb{N}$, a_k s are constant, and I_k is a finite open interval.

Theorem 2.2.5: Approximation by Step Functions

If f is measurable, then $\exists$ a sequence of step functions $\{\theta_n : n \geq 1\}$ s.t. $\lim_{n \rightarrow \infty} \theta_n(x) = f(x)$ for **almost every** $x \in \mathbb{R}$.

Proof. Wlog. we assume f non-negative.

Given $A \in \mathcal{M}$ with $m(A) < \infty$, recall that $\forall \varepsilon > 0$, $\exists I_1, I_2, \dots, I_N$ open finite intervals s.t. $m(A \Delta (\bigcup_{i=1}^N I_i)) < \varepsilon$.

By renaming or re-arranging I_i s if necessary, we may assume that the I_i s are disjoint (and hence that $1_{\bigcup_{i=1}^N I_i} = \sum_{i=1}^N 1_{I_i}$).

Define a step function $\theta_A = \sum_{i=1}^N 1_{I_i}$. Then:

$$\begin{aligned} m(\{x \in \mathbb{R} : 1_A(x) \neq \theta_A(x)\}) &\leq \varepsilon \\ &\parallel \\ &A \Delta (\bigcup_{i=1}^N I_i) \end{aligned}$$

Since f is measurable and non-negative, $\exists \{\varphi_n : n \geq 1\}$, a sequence of simple functions such that $\lim_{n \rightarrow \infty} \varphi_n = f$. In particular, $\varphi_n = \sum_{k=0}^{n2^n} \frac{k}{2^n} 1_{A_{n,k}}$.

Then we can find a step function $\theta_{n,k}$ s.t. $m(\{x \in \mathbb{R} : 1_{A_{n,k}}(x) \neq \theta_{n,k}(x)\}) < \frac{1}{2^n(n2^n+1)}$.

Set $\theta_n = \sum_{k=0}^{n2^n} \frac{k}{2^n} 1_{\theta_{n,k}}$; then θ_n is a step function $\forall n$.

$$\text{Set } E_n = \{x \in \mathbb{R} : \theta_n(x) \neq \varphi_n(x)\} \implies m(E_n) \leq m\left(\bigcup_{k=0}^{n2^n} \{\theta_{n,k} \neq 1_{A_{n,k}}\}\right)$$

$$\leq \sum_{k=0}^{n2^n} m(\{\theta_{n,k} \neq 1_{A_{n,k}}\}) \leq (n2^n + 1) \frac{1}{2^n(n2^n + 1)} = 2^{-n}$$

Then $m(E_n) \leq 2^{-n}$. We choose φ_n s s.t. $\forall x \in \mathbb{R}, |\varphi_n(x) - f_n(x)| \leq \frac{1}{2^n}$.

If $F_n = \{x \in \mathbb{R} : |\theta_n(x) - f_n(x)| > 2^{-n}\}$ then $F_n \subset E_n$ which implies $m(F_n) \leq 2^{-n}$.

"If F_n happens, then E_n has surely happened..."

Claim. For almost every $x \in \mathbb{R}$, $\exists m \geq 1$ s.t. $\forall n \geq m, |\theta_n(X) - f_n(x)| \leq 2^{-n}$.

To prove the claim let's consider to the complement. We want to show:

$$m\left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{x \in \mathbb{R} : |\theta_n(x) - f_n(x)| > 2^{-n}\}\right) = 0$$

i.e. $m(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} F_n) = 0$.

To this end, call $B_m = \bigcup_{n=m}^{\infty} F_n$. Then $B_m \downarrow$.

$$\implies m\left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} F_n\right) = \lim_{m \rightarrow \infty} m(B_m) \leq \lim_{m \rightarrow \infty} \sum_{n=m}^{\infty} 2^{-n} = 0$$

Claim proven. $\implies$ a.e. $\exists m \geq 1$ s.t. $\forall n \geq m, |\theta_n - f_n| \leq 2^{-n} \implies$ a.e. $\lim_{n \rightarrow \infty} (\theta_n - f_n) = 0$.

Therefore, a.e. $\theta_n = (\theta_n - f_n) + f_n \rightarrow f$ as $n \rightarrow \infty$. In fact, we have proven that if $\{F_n : n \geq 1\} \subseteq \mathcal{M}$ such that $\sum_{n=1}^{\infty} m(F_n) < \infty$, then $m(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} F_n) = 0$.

(i.e. the **Borel-Cantelli Lemma**.)

□

2.3 Convergence a.e vs Convergence in Measure

Definition 2.3.1: Converges a.e

If $\{f_n\}, f$ are measurable functions, we say f_n converges to f a.e and write ' $f_n \rightarrow f$ a.e' if for a.e $x \in \mathbb{R}$, $\lim_{n \rightarrow \infty} f_n(x) = f(x)$.

Definition 2.3.2: Converges in measure

If $\{f_n\}, f$ are measurable finite-valued functions, we say f_n converges to f in measure and write ' $f_n \rightarrow f$ in measure' if $\forall \delta > 0, \lim_{n \rightarrow \infty} m(\{x \in \mathbb{R} : |f_n(x) - f(x)| \geq \delta\}) = 0$.

Similarly, we have $f_n \rightarrow f$ a.e on A if $\exists B \subseteq A$ with $m(B) = 0$ s.t. $\forall x \in A \setminus B$, $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ (for convergence in measure, sub A for R).

Proposition 2.3.3

Given finite-valued measurable functions $\{f_n\}$ and f and $A \in \mathcal{M}$ with $m(A) < \infty$, if $f_n \rightarrow f$ a.e on A , then $f_n \rightarrow f$ in measure on A .

Proof. $\forall \delta > 0, \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{x \in A : |f_n(x) - f(x)| > \delta\} \subseteq \{x \in A : \lim_{n \rightarrow \infty} f_n(x) \neq f(x)\}$

$$\begin{aligned} \implies m\left(\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{|f_n - f| > \delta\}\right) &= 0 \implies \lim_{m \rightarrow \infty} m(B_m) = 0 \implies \{|f_m - f| > \delta\} \subseteq \bigcup_{n=m}^{\infty} \{|f_n - f| > \delta\} \\ &\quad \parallel \\ &\quad B_m \\ \implies m(\{|f_m - f| > \delta\}) &\leq m(B_m) \rightarrow 0 \text{ (as } m \rightarrow \infty\text{)}. \end{aligned}$$

□

Example.

As to why the assumption $m(A)$ is necessary. Let $f_n = 1_{[n, \infty)}, f \equiv 0$. Then $\lim_{n \rightarrow \infty} f_n(x) = f(x) = 0 \forall x \in \mathbb{R}$.

$$\text{But } m(\{x \in \mathbb{R} : |f_n(x) - f(x)| = 1\}) = m([n, \infty)) = \infty$$

Remark.

In general, $f_n \rightarrow f$ in measure $\not\Rightarrow f_n \rightarrow f$, even on sets of finite measure.

Proposition 2.3.4

Given $\{f_n\}, f$ measurable finite-valued functions; if $f_n \rightarrow f$ in measure then $\exists$ a subsequence $\{n_k : k \geq 1\} \subseteq \mathbb{N}$ s.t $f_{n_k} \rightarrow f$ almost everywhere (as $k \rightarrow \infty$).

(Not as strong as having the sequence go there, but at least you can extract a subsequence.)

Proof. Assume $f_n \rightarrow f$ in measure. $\forall \delta > 0, m(\{|f_n - f| > \delta\}) \rightarrow 0$ as $n \rightarrow \infty$

$$\implies \forall k \geq 1, \exists n_k \text{ s.t } m\left(\left\{|f_{n_k} - f| > \frac{1}{k}\right\}\right) \leq \frac{1}{k^2} \implies \sum_{k=1}^{\infty} m(A_k) < \infty$$

(By Borel-Cantelli Lemma)

$$\implies m\left(\bigcap_{\ell=1}^{\infty} \bigcup_{k=\ell}^{\infty} A_k\right) = \lim_{\ell \rightarrow \infty} m\left(\bigcup_{k=\ell}^{\infty} A_k\right) \leq \lim_{\ell \rightarrow \infty} \sum_{k=\ell}^{\infty} m(A_k) = 0$$

So now since that is a null set, the COMPLEMENT is a 'full set' almost everywhere.

$$\implies \text{a.e } \exists \ell \text{ s.t } \forall k \geq \ell, |f_{n_k} - f| \leq \frac{1}{k} \implies \text{a.e } \lim_{k \rightarrow \infty} |f_{n_k} - f| = 0$$

And so $f_{n_k} \rightarrow f$ a.e (as k to ∞). □

Proposition 2.3.5

(Subsequence Test). Given $\{f_n\}, f$ measurable finite-valued functions: $f_n \rightarrow f$ in measure iff. $\forall$ subsequence $\{n_k\}$, $\exists$ a subsubsequence $\{n_{k_\ell}\} \subseteq \{n_k\} \subseteq \mathbb{N}$ such that $f_{n_{k_\ell}} \rightarrow f$ in measure as $\ell \rightarrow \infty$.

Proof. $\implies$) is obvious.

$\Leftarrow$) Assume $f_n \not\rightarrow f$ in measure for contradiction.

Then $\exists \delta > 0, \exists \{n_k\}$ such that $m(\{|f_{n_k} - f| > \delta\}) > \delta \forall k \geq 1$.

But, by the hypothesis, $\exists \{n_{k_\ell}\} \subseteq \{n_k\}$ s.t $f_{n_{k_\ell}} \rightarrow f$ in measure. Contradiction. □

2.4 The Theorems of Egorov and Lusin

Recall from previously: if f is measurable then $\exists \{\theta_n\}$ a sequence of 'step functions' such that $\theta_n \rightarrow f$ almost everywhere.

Theorem 2.4.1: Egorov's Theorem

Given $\{f_n\}$, f measurable functions and $A \in \mathcal{M}$ with $m(A) < \infty$; if $f_n \rightarrow f$ a.e on A , then $\forall \varepsilon > 0 \exists$ closed $A_\varepsilon \subseteq A$ s.t $m(A \setminus A_\varepsilon) < \varepsilon$ and s.t $f_n \rightarrow f$ uniformly on A_ε .

"You need to sacrifice a little bit of measure to achieve uniform convergence."

Proof. We assume that f is finite-valued on A (otherwise replace A by $A \cap \{-\infty < f < \infty\}$). We want to show that $\forall \varepsilon > 0, \exists$ closed $A_\varepsilon \subseteq A$ s.t $m(A \setminus A_\varepsilon) < \varepsilon$ AND $\sup_{x \in A_\varepsilon} |f_n(x) - f(x)| \rightarrow 0$ (as $n \rightarrow \infty$).

For each $k \geq 1$ (for now, fix k) and $\forall n \geq 1$, set $E_n^k = \{x \in A : |f_j(x) - f(x)| \leq \frac{1}{k} \forall j \geq n\}$. Then $E_n^k \subseteq E_{n+1}^k$ (if it behaves well starting from n , it behaves well past $n...$). Then:

$$\bigcup_{n=1}^{\infty} E_n^k = \left\{ x \in A : \exists n \geq 1 \text{ s.t } \forall j \geq n, |f_j(x) - f(x)| \leq \frac{1}{k} \right\} \supseteq \{x \in A : \lim_{n \rightarrow \infty} f_n(x) = f(x)\}$$

$$\implies \text{ if } A' = \{x \in A : \lim_{n \rightarrow \infty} f_n(x) = f(x)\} \text{ then (by hypothesis) } m(A^k) = m(A)$$

$$\implies m(A) = m(A') \leq m\left(\bigcup_{n=1}^{\infty} E_n^k\right) = \lim_{n \rightarrow \infty} m\left(\bigcup_{n=1}^{\infty} E_n^k\right) \leq m(A) \implies \lim_{n \rightarrow \infty} m(E_n^k) = m(A) \forall k$$

Given any $\varepsilon > 0$, $\exists n_k \geq 1$ s.t $m(A \setminus E_{n_k}^k) = m(A) - m(E_{n_k}^k) < \frac{1}{2^k} \frac{\varepsilon}{2}$

Set B to be "the worst scenario", i.e. $B = A \setminus \left(\bigcap_{k=1}^{\infty} E_{n_k}^k\right)$. Then $m(B) = m\left(\bigcup_{k=1}^{\infty} (A \setminus E_{n_k}^k)\right) \leq \sum_{k=1}^{\infty} m(A \setminus E_{n_k}^k) < \frac{\varepsilon}{2}$. (Complement of intersection is union of complement.)

Set $\tilde{A}$ to be the set "with perfect behaviour", i.e. $\tilde{A} = A \setminus B = \bigcap_{k=1}^{\infty} E_{n_k}^k$. Then if $x \in \tilde{A}$ then $x \in E_{n_k}^k$ for every k . Then $\forall k \geq 1$ and $\forall j \geq n_k$, $|f_j(x) - f(x)| \leq \frac{1}{k}$.

$\implies f_n \rightarrow f$ uniformly on $\tilde{A}$. By regularity of m , $\exists$ closed $A_\varepsilon \subseteq \tilde{A}$ s.t $m(\tilde{A} \setminus A_\varepsilon) < \frac{\varepsilon}{2}$

$\implies f_n \rightarrow f$ uniformly on A_ε and $m(A \setminus A_\varepsilon) = m(A \setminus \tilde{A}) + m(\tilde{A} \setminus A_\varepsilon) = m(B) + m(\tilde{A} \setminus A_\varepsilon) < \varepsilon$

□

Remark.

It is very necessary that $m(A) < \infty$. You can cook up some example: $f_n = 1_{[n, \infty)}$; then $f_n \rightarrow 0$ pointwise but $\forall a \in \mathbb{R}$ f_n does not converge to 0 uniformly on (a, ∞) .

Fact 2.4.2

In general, Egorov's Theorem $\not\Rightarrow f_n \rightarrow f$ uniformly a.e. For example, on $[0, 1]$, $f_n(x) = x^n$; $f(x) \equiv 0 \forall x \in [0, 1)$ and then $f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$; then $f_n \rightarrow f$ a.e on $[0, 1]$.

If $A \subseteq [0, 1]$ is closed s.t $1 \in A$ then $f_n \not\rightarrow f$ uniformly on A . To see this, take $\{x_m\} \subseteq A$ s.t $x_m \uparrow$ and $\lim_{m \rightarrow \infty} x_m = 1$. Then:

$$\sup_{x \in A} |f_n(x) - f(x)| \geq \sup_{m \geq 1} |f_n(x_m) - f(x_m)| = \sup_m x_m^n = 1$$

(The biggest gap is always 1). $\Rightarrow f_n \not\rightarrow f$ uniformly on A .

(In fact, $\forall \varepsilon > 0$, $f_n \rightarrow f$ uniformly on $[0, 1 - \varepsilon]$).

"If you want uniform convergence, you need to stay away from 1."

Theorem 2.4.3: Lusin's Theorem

An application of Egorov's Theorem.

Given f measurable, finite-valued, and $A \in \mathcal{M}$ with $m(A) < \infty$; $\forall \varepsilon > 0$, $\exists$ closed $A_\varepsilon \subseteq A$ s.t $m(A \setminus A_\varepsilon) < \varepsilon$ and $f|_{A_\varepsilon}$ is continuous.

"Every measurable function is nearly continuous."

Remark.

Lusin's Theorem says that $f|_{A_\varepsilon}$, as a function on A_ε , is continuous. Which is NOT the same as f as a function on A is continuous at points in A_ε (NO!).

e.g. $f = 1_{\mathbb{Q} \cap [0, 1]}$ is not continuous anywhere on $[0, 1]$. However, $f|_{\mathbb{Q} \cap [0, 1]}$ IS constant and hence continuous on $\mathbb{Q} \cap [0, 1]$.

Proof. $\exists$ a sequence $\{\theta_n\}$ of step functions s.t $\theta_n \rightarrow f$ a.e on A .

(θ_n is piecewise constant and hence piecewise continuous.) Given any $\varepsilon > 0$, $\forall n \geq 1$ it is possible to find an open set E_n such that $\theta_n|_{E_n^c}$ is continuous and $m(E_n) \leq \frac{\varepsilon}{2^{2n}}$.

Meanwhile, Egorov's Theorem implies that $\exists$ closed $B \subseteq A$ s.t $m(A \setminus B) \leq \frac{\varepsilon}{2}$ s.t $\theta_n \rightarrow f$ uniformly on B .

Set $A_\varepsilon = B \setminus (\bigcup_{n=1}^{\infty} E_n)$. $A_\varepsilon \subseteq A$ and A_ε is closed, then:

$$m(A \setminus A_\varepsilon) \leq m(A \setminus B) + m\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \frac{\varepsilon}{2} + \sum_{n=1}^{\infty} m(E_n) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Finally, on A_ε , $\theta_n \rightarrow f$ uniformly and $\theta_n|_{A_\varepsilon}$ is continuous; $\Rightarrow f|_{A_\varepsilon}$ is continuous (limit of uniform convergence of continuous functions is continuous). $\square$

Remark.

Lusin's Theorem does NOT imply that f is continuous a.e. on A . The sets A_k depend on k and there may be no point that belongs to all A_k simultaneously. Even though $m(A \setminus A_k) \rightarrow 0$, the intersection $\bigcap_{k=1}^{\infty} A_k$ could be empty or have measure zero, so we cannot conclude continuity at a.e. point of A .

Remark.

Lusin's Theorem $\implies \forall k$ large $\exists$ closed $A_k \subseteq A$ s.t. $m(A \setminus A_k) \leq \frac{1}{k}$ and $f|_{A_k}$ is continuous. In fact, we can take $A_k \uparrow$ (otherwise $\tilde{A}_k = \bigcup_{i=1}^k A_i$ replaces A_k).

2.5 Construction of Integrals

Step 1.

Definition 2.5.1: Integral of a Simple Function

Given a simple function $\varphi = \sum_{k=1}^L a_k 1_{E_k}$, then the Lebesgue integral of φ is defined as:

$$\int_{\mathbb{R}} \varphi(x) dx = \sum_{k=1}^L a_k m(E_k)$$

Remark.

$\forall A \in \mathcal{M}$ with $m(A) < \infty$,

$$\int_{\mathbb{R}} 1_A = m(A)$$

What are the properties of $\int_{\mathbb{R}} \varphi$ for a simple function φ ?

Proposition 2.5.2

Well-definedness

If $\varphi = \sum_{k=1}^L a_k 1_{E_k} = \sum_{\ell=1}^M b_{\ell} 1_{F_{\ell}}$, for $E_k, F_{\ell} \in \mathcal{M}$, $a_k, b_{\ell} \in \mathbb{R}$, then:

$$\sum_{k=1}^L a_k m(E_k) = \sum_{\ell=1}^M b_{\ell} m(F_{\ell})$$

Proof. Wlog. we can assume that the E_k s are disjoint, and the same for the F_{ℓ} s. Set $a_0 = b_0 = 0$ and $E_0 = (\bigcup_{k=1}^L E_k)^c$, $F_0 = (\bigcup_{\ell=1}^M F_{\ell})^c$, i.e., $\{E_0, E_1, \dots, E_L\}$ is a partition of

$\mathbb{R}$ (and likewise for the F s.) Then:

$$\varphi = \sum_{k=1}^L a_k 1_{E_k} = \sum_{k=0}^L a_k 1_{E_k} = \sum_{k=0}^L \sum_{\ell=0}^M a_k 1_{E_k \cap F_\ell}$$

$$\left(1_{E_k} = \sum_{\ell=0}^M 1_{E_k \cap F_\ell} \text{ because } E_k = \bigcup_{\ell=0}^L (E_k \cap F_\ell) \text{ a disjoint union} \right)$$

Similarly:

$$\varphi = \sum_{\ell=1}^M b_\ell 1_{F_\ell} = \sum_{\ell=0}^M \sum_{k=0}^L b_\ell 1_{E_k \cap F_\ell}$$

If $E_k \cap F_\ell \neq \emptyset$ then $a_k = b_\ell$.

$$\Rightarrow \int_{\mathbb{R}} \varphi = \sum_{k=1}^L a_k m(E_k) = \sum_{k=0}^L \sum_{\ell=0}^M a_k m(E_k \cap F_\ell); \quad \text{and} \quad \int_{\mathbb{R}} \varphi = \sum_{\ell=1}^M b_\ell m(F_\ell) = \sum_{\ell=0}^M \sum_{k=0}^L b_\ell m(E_k \cap F_\ell)$$

If $m(E_k \cap F_\ell) > 0$, then $E_k \cap F_\ell \neq \emptyset$ so $a_k = b_\ell$. □

Proposition 2.5.3

Linearity

If φ and ψ are two simple functions, then for $a, b \in \mathbb{R}$, $a\varphi + b\psi$ is again a simple function, and, by definition:

$$\int_{\mathbb{R}} (a\varphi + b\psi) = a \int_{\mathbb{R}} \varphi + b \int_{\mathbb{R}} \psi$$

Proposition 2.5.4

Finite additivity

If φ is a simple function, $A, B \in \mathcal{M}$ s.t. $A \cap B = \emptyset$, then $\int_{A \cup B} \varphi = \int_A \varphi + \int_B \varphi$.

Proposition 2.5.5

Absolute value

If φ is a simple function, then $|\varphi|$ is again a simple function, and $|\int_{\mathbb{R}} \varphi| \leq \int_{\mathbb{R}} |\varphi|$.

Step 2.

Definition 2.5.6: Integral of a Non-Negative Function

If f is a non-negative measurable function, then the Lebesgue integral of f is defined as:

$$\text{Def 2} \quad \int_{\mathbb{R}} f(x) dx = \sup \left\{ \int_{\mathbb{R}} \varphi(x) dx : \varphi \text{ is a simple function, } \varphi \leq f \right\}$$

Recall Def 1:

Proposition 2.5.7

(Def 1) = (Def 2) on non-negative simple functions.

Proof. Assume $\varphi = \sum_{k=1}^L a_k 1_{E_k}$ is non-negative. Since φ is an element in the collection in (Def 2), (Def 2) $\geq$ (Def 1).

On the other hand, to show (Def 2) $\leq$ (Def 1) it suffices to show that $\forall \psi$ simple and $\psi \leq \varphi$, (Def 1)(ψ) $\leq$ (Def 1)(φ). But this is exactly the **monotonicity**. $\square$

Remark.

Given $f \geq 0$ measurable, (Def 2) $\implies \exists$ a sequence $\{\varphi_n\}$ simple such that $\varphi_n \leq f$ and $\lim \int_{\mathbb{R}} \varphi_n = \int_{\mathbb{R}} f$.

Theorem 2.5.8: Løgner's Theorem

Suppose $f \geq 0$ and measurable. If $\{\varphi_n\}$ is a sequence of simple functions such that $\varphi_n \uparrow$ and $\lim \varphi_n = f$ pointwise, then $\lim \int_{\mathbb{R}} \varphi_n = \int_{\mathbb{R}} f$
(A particular version of monotone convergence theorem.)

Proof. LHS = $\lim \int_{\mathbb{R}} \varphi_n$ (Def 1), RHS = $\int_{\mathbb{R}} f$ (Def 2). Since $\varphi_n \leq f \forall n \geq 1$, $\implies \int_{\mathbb{R}} \varphi_n \leq \int_{\mathbb{R}} f \implies \text{LHS} \leq \text{RHS}$.

To show LHS $\geq$ RHS, it's sufficient to show that $\forall \psi$ simple with $\psi \leq f$, $\int_{\mathbb{R}} \psi \leq \lim \int_{\mathbb{R}} \varphi_n$. Assume

$$\psi = \sum_{k=1}^L a_k 1_{E_k} = \sum_{k=0}^L a_k 1_{E_k}$$

Where $\{E_0, E_1, \dots, E_k\}$ is a partition of $\mathbb{R}$.

Then by additivity:

$$\int_{\mathbb{R}} = \sum_{k=0}^L a_k m(E_k) \text{ and } \int_{\mathbb{R}} \varphi_n = \sum_{k=0}^L \int_{E_k} \varphi_n$$

So it's sufficient to show that $\forall k = 0, 1, \dots, L, a_k m(E_k) \leq \lim \int_{E_k} \varphi_n$ (Δ).

If $a_k = 0$ or $m(E_k) = 0$, (Δ) holds trivially, so assume $a_k > 0, m(E_k) > 0$.

Fix k . Then $\lim \varphi_n = f \geq \psi \implies \forall x \in E_k, \lim \varphi_n(x) \geq a_k$. $\forall \varepsilon > 0$, set $C_n^\varepsilon = \{x \in E_k : \varphi_n(x) \geq (1 - \varepsilon)a_k\}$.

Since $\varphi_n \leq \varphi_{n+1}, C_n^\varepsilon \subseteq C_{n+1}^\varepsilon, \bigcup_{n=1}^\infty C_n^\varepsilon = E_k$

$$\implies \lim \int_{E_k} \varphi_n = \lim \int_{\mathbb{R}} 1_{E_k} \varphi_n \geq \lim \int_{\mathbb{R}} 1_{C_n^\varepsilon} \varphi_n \geq \lim (1 - \varepsilon)a_k, (C_n^\varepsilon) = (1 - \varepsilon)a_k m(E_k)$$

Where the inequality derives from the fact that $1_{E_k} \varphi_n \geq 1_{C_n^\varepsilon} \varphi_n \geq (1 - \varepsilon)a_k 1_{C_n^\varepsilon}$

Since ε arbitrary, $\lim \int_{E_k} \varphi_n \geq a_k m(E_k)$. □

Corollary 2.5.9

For any $f > 0$ measurable, if $\forall n \geq 1, \forall k = 0, 1, \dots, n2^n$, then

$$A_{n,k} = \left\{ \frac{k}{2^n} \leq f \leq \frac{k+1}{2^n} \right\} \text{ then } \int_{\mathbb{R}} f = \lim \sum_{k=0}^{n2^n} \frac{k}{2^n} m(A_{n,k})$$

Properties of the integral of non-negative functions.

Proposition 2.5.10

Well-definedness

If $f \geq 0, g \geq 0$ are measurable and are such that $f = g$ a.e, then $\int_{\mathbb{R}} f = \int_{\mathbb{R}} g$.

Proof. Let $\{\varphi_n\}$ and $\{\psi_n\}$ be sequences of simple functions such that $\varphi_n \uparrow$ and $\lim \varphi_n = f$; $\psi_n \uparrow$ and $\lim \psi_n = g$.

Define $h_n = \varphi_n 1_{\{f=g\}} + \psi_n 1_{\{f \neq g\}}$. Then h_n is a simple function, $h_n \uparrow$, and note that this 'blended function' has $\lim h_n = g$.

$$\implies \int_{\mathbb{R}} g = \lim \int_{\mathbb{R}} h_n = \lim \left(\int_{\{f=g\}} \varphi_n + \int_{\{f \neq g\}} \psi_n \right) = \lim \left(\int_{\{f=g\}} \varphi_n + 0 \right) = \lim \int_{\{f=g\}} \varphi_n$$

Since $\{f \neq g\}$ is a set of measure 0. Meanwhile:

$$\int_{\mathbb{R}} f = \lim \int_{\mathbb{R}} \varphi_n = \lim \left(\int_{\{f=g\}} \varphi_n + \int_{\{f \neq g\}} \psi_n \right) = \lim \int_{\{f=g\}} \varphi_n$$

□

"Subtraction is always a tricky business..." - Linan Chen

Proposition 2.5.11

Linearity

Let $f, g \geq 0$ be measurable, and let $a, b \geq 0$.

Then $\int_{\mathbb{R}} (af + bg) = a \int_{\mathbb{R}} f + b \int_{\mathbb{R}} g$.

Proof. Take $\{\varphi_n\}, \{\psi_n\}$ as in the previous proof, and $\{h_n = a\varphi_n + b\psi_n\}$ simple, $h_n \uparrow$, $\lim h_n = af + bg$. By the linearity of integrals of simple functions:

$$\int_{\mathbb{R}} (af + bg) = \lim \int_{\mathbb{R}} h_n = \lim \int_{\mathbb{R}} (a\varphi_n + b\psi_n) = \lim (a \int_{\mathbb{R}} \varphi_n + b \int_{\mathbb{R}} \psi_n) = a \int_{\mathbb{R}} f + b \int_{\mathbb{R}} g$$

□

Proposition 2.5.12

Monotonicity

If $f, g \geq 0$ are measurable and $f \leq g$ a.e, then $\int_{\mathbb{R}} f \leq \int_{\mathbb{R}} g$.

Proof. Wlog, assume that $f \leq g$ pointwise (otherwise replace f with $1_{\{f \leq g\}}$).

Then $\{\varphi : \text{simple and } \varphi \leq f\} \subseteq \{\varphi : \text{simple and } \varphi \leq g\}$; by definition, $\implies \int_{\mathbb{R}} f \leq \int_{\mathbb{R}} g$.

□

Proposition 2.5.13

Interaction with 0

If $F \geq 0$ measurable, then $\int_{\mathbb{R}} f = 0 \iff f = 0$ a.e.

Proof. $\Leftarrow$) trivial. For $\Rightarrow$): We want to prove that if $A = \{f > \infty\}$ then $m(A) = 0$.

For each $n \geq 1$, $A_n = \{f \geq \frac{1}{n}\}$; $A_n \uparrow$ and $\bigcup_{n=1}^{\infty} A_n = A$. Then $m(A) = \lim m(A_n)$. Assume $m(A) = \delta \geq 0$.

$\implies \exists N \geq 1$ s.t $m(A_N) > \delta/2$. Since $f \geq f1_{A_N} \geq \frac{1}{N}1_{A_N}$. Conveniently, we have just established monotonicity, so by monotonicity, $\int_{\mathbb{R}} f \geq \int_{\mathbb{R}} \frac{1}{N}1_{A_N} = \frac{1}{N}m(A_N) \geq \frac{1}{N}\frac{\delta}{2} > 0$. Contradiction, so $\delta = 0$.

□

Proposition 2.5.14

Interactions with 0, part ii

Let $f \geq 0$ measurable, and $A \in \mathcal{M}$. Then $\int_A f = 0 \iff$ either $f = 0$ a.e on A , or $m(A) = 0$.

Proof. By the previous, $\int_A f = 0 \iff 1_A f = 0$ a.e (on $\mathbb{R}$).

1) if $m(A) = 0$, then $1_A = 0$ a.e $\implies 1_A f = 0$ a.e

2) if $m(A) > 0$ then $f = 0$ a.e on A . □

Proposition 2.5.15

Interactions with infinity

$f \geq 0$ measurable. If $\int_{\mathbb{R}} f < \infty$, then f is finite-valued a.e.

Proof. Let $A = \{f = \infty\}$. Assume for contradiction that $m(A) = \delta > 0$.

$$\forall n \geq 1, f \geq f 1_A \geq n 1_A \implies \int_{\mathbb{R}} f \geq \int_{\mathbb{R}} n 1_A = n m(A) = n \delta \implies \int_{\mathbb{R}} f = \infty$$

For every n ! So a contradiction. □

Proposition 2.5.16

Markov's Inequality

Let $f \geq 0$ measurable, $0 < a < \infty$.

$$\text{Then } m(\{f > a\}) \leq \frac{1}{a} \int_{\mathbb{R}} f$$

(In particular,, if $\int_{\mathbb{R}} f < \infty$ then $\lim_{a \rightarrow \infty} m(\{f > a\}) = 0$)

Proof.

$$A_a = \{f > a\}. f \geq f 1_{A_a} > a 1_{A_a} \implies \int_{\mathbb{R}} f \geq a m(A_a)$$

□

We now desire to define an integral for a general measurable/integrable function.

Fact 2.5.17

Assume f is measurable. Then $\int_{\mathbb{R}} f = \int_{\mathbb{R}} f^+ - \int_{\mathbb{R}} f^-$ provided that at least one of $\int_{\mathbb{R}} f^+, \int_{\mathbb{R}} f^-$ is finite. If both are infinite, then $\int_{\mathbb{R}} f$ is undefined.

Remark.

Only having $\int_{\mathbb{R}} f$ defined is not sufficient for the desirable properties (linearity, monotonicity) to hold.

Definition 2.5.18: Integrable Function

A measurable function f is called integrable and is denoted $f \in L^1(\mathbb{R})$ if $\int_{\mathbb{R}} f^+ < \infty$, $\int_{\mathbb{R}} f^- < \infty$.

Equivalently, $f \in L^1(\mathbb{R}) \iff \int_{\mathbb{R}} |f| < \infty \iff \int_{\mathbb{R}} f$ is finite-valued.

Properties of the integral of an $f \in L^1(\mathbb{R})$.

Proposition 2.5.19

"Triangle inequality":

$$|\int_{\mathbb{R}} f(x) dx| \leq \int_{\mathbb{R}} |f(x)| dx$$

(because $-\int_{\mathbb{R}} f^- \leq \int_{\mathbb{R}} f \leq \int_{\mathbb{R}} f^+$ and $\int_{\mathbb{R}} f^{\pm} \leq \int_{\mathbb{R}} |f|$.)

Proposition 2.5.20

Finite-valuedness:

$f \in L^1(\mathbb{R}) \implies f$ is finite-valued a.e.

(because $\int_{\mathbb{R}} |f| < \infty \implies |f| < \infty$ a.e.)

Proposition 2.5.21

Linearity:

Let $f, g \in L^1(\mathbb{R})$. Then for $a, b \in \mathbb{R}$, $af + bg \in L^1(\mathbb{R})$ and $\int_{\mathbb{R}} (af + bg) = a \int_{\mathbb{R}} f + b \int_{\mathbb{R}} g$.

Proof. First we show that $af \in L^1(\mathbb{R})$ and that $\int_{\mathbb{R}} af = a \int_{\mathbb{R}} f$.

$|af| \leq |a||f|$. By monotonicity:

$$\int_{\mathbb{R}} |af| \leq |a| \int_{\mathbb{R}} |f| < \infty \implies af \in L^1(\mathbb{R})$$

.

Consider $(af)^+, (af)^-$. Then $\int_{\mathbb{R}} (af) = \int_{\mathbb{R}} (af)^+ - \int_{\mathbb{R}} (af)^-$. By linearity of " ≥ 0 " functions, $= a \int_{\mathbb{R}} f$. It is similar for $bg \in L^1(\mathbb{R})$, $\int_{\mathbb{R}} (bg) = b \int_{\mathbb{R}} g$.

Now assume wlog. that $a = b = 1$. We want to show that $f + g \in L^1(\mathbb{R})$, i.e. that $\int_{\mathbb{R}} (f + g) = \int_{\mathbb{R}} f + \int_{\mathbb{R}} g$.

$$|f + g| \leq |f| + |g| \implies \int_{\mathbb{R}} |f + g| < \infty \implies f + g \in L^1(\mathbb{R})$$

Set $h = f + g$. Then, $|f| < \infty$, $|g| < \infty$, and $|h| < \infty$ a.e. Moreover, $\int_{\mathbb{R}} |f|, \int_{\mathbb{R}} |g|, \int_{\mathbb{R}} |h|$ all $< \infty$.

Then $h^+ - h^- = f^+ - f^- + g^+ - g^- \implies h^+ + f^- + g^- = f^+ + g^+ + h^-$ (everything finite a.e).

By linearity of " ≥ 0 " functions:

$$\begin{aligned} \int h^+ + \int f^- + \int g^- &= \int f^+ + \int_g^+ + \int h^- \\ \implies \int h^+ - \int h^- &= \int f^+ - \int f^- + \int g^+ - \int g^- \implies \int h = \int f + \int g \end{aligned}$$

□

2.6 Convergence Theorems of Integrals

Q: Under what condition is $\int_{\mathbb{R}} \lim f_n = \lim \int_{\mathbb{R}} f_n$?

Theorem 2.6.1: Monotone Convergence Theorem (MON)

Assume $\{f_n\}, f$ are non-negative and measurable. If $f_n \uparrow$ and $\lim f_n = f$, then $\int_{\mathbb{R}} = \lim \int_{\mathbb{R}} f_n$

Remark.

In this and in the later convergence theorems, $\lim f_n = f$ can be replaced by $\lim f_n = f$ a.e.

Proof. By monotonicity (of ≥ 0 functions), $\lim \int_{\mathbb{R}} f_n$ exists and since $f_n \leq f$, $\lim \int_{\mathbb{R}} f_n \leq \int_{\mathbb{R}} f$. So all that remains to show is the reverse inequality.

$\forall n \geq 1$, select $\{\varphi_{n,k} : k \geq 1\}$, a sequence of simple functions such that $\varphi_{n,k} \leq \varphi_{n,k+1}$ and $\lim_{k \rightarrow \infty} \varphi_{n,k} = f_n$.

Now $f_1 \leq f_2 \leq f_k \leq \dots \leq f$ and (imagine the matrix of $\varphi_{n,k}$ s). Define $\forall k \geq 1$ set $g_k = \max\{\varphi_{1,k}, \varphi_{2,k}, \dots, \varphi_{k,k}\}$. g_k is again a simple function and $g_k \leq g_{k+1}, g_k \leq f$ so $\lim_{k \rightarrow \infty} g_k \leq f$.

But for all fixed $n \geq 1$:

$$\lim_{k \rightarrow \infty} g_k \geq \lim_{k \rightarrow \infty} \varphi_{n,k} = f_n \implies \lim_{k \rightarrow \infty} g_k \geq \lim f_n = f$$

Thus, $\lim_{k \rightarrow \infty} g_k = f$. By the theorem from the previous lecture, $\lim_{k \rightarrow \infty} \int_{\mathbb{R}} g_k = \int_{\mathbb{R}} f$. Since $\forall k \geq 1, \varphi_{1,k}, \varphi_{2,k}, \dots, \varphi_{k,k} \leq f_k \implies g_k \leq f_k$.

By the monotonicity of ≥ 0 functions:

$$\int_{\mathbb{R}} g_k \leq \int_{\mathbb{R}} f_k \implies \int_{\mathbb{R}} f = \lim_k \int_{\mathbb{R}} g_k \leq \lim_k \int_{\mathbb{R}} f_k$$

□

Corollary 2.6.2

If $\{f_n\}$ are measurable s.t $f_n \uparrow$ and $\lim f_n = f$ AND $\int_{\mathbb{R}} f_1^- < \infty$ then $\int_{\mathbb{R}} f = \lim \int_{\mathbb{R}} f_n$.

Proof. Since $f_n \uparrow$, $f_n \geq f_1 \implies f_n^- \leq f_1^-, f^- \leq f_1^-$ finite valued a.e. Then $\int_{\mathbb{R}} f_n^- \leq \int_{\mathbb{R}} f_1^- < \infty$, $\int_{\mathbb{R}} f^- \leq \int_{\mathbb{R}} f_1^- < \infty$.

For all $n \geq 1$, set $\tilde{f}_n = f_n + f_1^- = f_n^+ - f_n^- + f_1^- \geq 0$. Then $\tilde{f}_n \uparrow$ and $\lim \tilde{f}_n = f + f_1^- (\geq 0)$.

By MON:

$$\int_{\mathbb{R}} \tilde{f} = \lim \int_{\mathbb{R}} \tilde{f}_n \implies \int_{\mathbb{R}} (f + f_1^-) = \lim \int_{\mathbb{R}} (f_n + f_1^-)$$

If we could now apply linearity, then we would be done here. But we don't have general linearity.

$$\tilde{f}_n = f_n + f_1^- = f_n^+ - f_n^- + f_1^- \implies \tilde{f}_n + f_n^- = f_n^+ + f_1^-$$

(Because $f_n^- < \infty$ a.e). Linearity:

$$\int_{\mathbb{R}} \tilde{f}_n + \int_{\mathbb{R}} f_n^- = \int_{\mathbb{R}} f_n^+ + \int_{\mathbb{R}} f_1^- \implies \int_{\mathbb{R}} \tilde{f}_n = \int_{\mathbb{R}} f_n^+ - \int_{\mathbb{R}} f_n^- + \int_{\mathbb{R}} f_1^-$$

(Because $\int_{\mathbb{R}} f_n^- < \infty$). $\implies \int_{\mathbb{R}} \tilde{f}_n = \int_{\mathbb{R}} f_n + \int_{\mathbb{R}} f_1^-$.

Similarly, $\int_{\mathbb{R}} \tilde{f} = \int_{\mathbb{R}} f + \int_{\mathbb{R}} f_1^-$, which completes the proof. □

Theorem 2.6.3: Reverse MON

Assume $\{f_n\}, f$ are measurable such that $f_n \downarrow$ and $\lim f_n = f$. If $\int_{\mathbb{R}} f_1^+ < \infty$, then $\int_{\mathbb{R}} f = \lim \int_{\mathbb{R}} f_n$.

Proof: exercise, consider $\tilde{f}_n = f_n^+ = f_1^+ - f_n$.

Theorem 2.6.4: Fatou's Lemma

Assume $\{f_n\}$ is non-negative and measurable. Then:

$$\int_{\mathbb{R}} \left(\liminf_{n \rightarrow \infty} f_n \right) \leq \liminf_{n \rightarrow \infty} \left(\int_{\mathbb{R}} f_n \right)$$

Proof. Define $g_m = \inf_{n \geq m} f_n$. Then g_m is measurable, non-negative, and $g_m \uparrow \liminf_{n \rightarrow \infty} f_n$.

$\forall n \geq m, g_m \leq f_n$. By the monotonicity of non-negative functions, $\int_{\mathbb{R}} g_m \leq \int_{\mathbb{R}} f_n \forall n \geq m$, which implies that $\int_{\mathbb{R}} g_m \leq \inf_{n \geq m} \int_{\mathbb{R}} f_n$

$$\implies \int_{\mathbb{R}} (\liminf_n f_n) = \lim_m \int_{\mathbb{R}} g_m \leq \lim_m \inf_{n \geq m} \int_{\mathbb{R}} f_n = \liminf_n \left(\int_{\mathbb{R}} f_n \right)$$

□

Corollary 2.6.5

Assume $\{f_n\}$ is measurable and $\exists$ measurable g s.t. $\int_{\mathbb{R}} g^- < \infty$ and $f_n \geq g \forall n \geq 1$. Then $\int_{\mathbb{R}} (\liminf_n f_n) \leq \liminf_n (\int_{\mathbb{R}} f_n)$

Theorem 2.6.6: Reverse Fatou

Assume $\{f_n\}$ is measurable and $\exists g$ measurable s.t. $\int_{\mathbb{R}} g^+ < \infty$ and $f_n \leq g \forall n \geq 1$. Then $\int_{\mathbb{R}} (\limsup_n f_n) \geq \limsup_n (\int_{\mathbb{R}} f_n)$.

"You need an integrable floor!" - Linan Chen

Remark.

The 'floor' $\int_{\mathbb{R}} f_1^- < \infty$ in the corollary of the MON is necessary. For example, if

$$f_n(x) = \begin{cases} -1 & \text{if } x \geq n \\ 0 & \text{if } x \leq n \end{cases}$$

, then $f_n \uparrow, \lim_n f_n = 0$, but $\int_{\mathbb{R}} f_n = -\infty$ and MON does not apply.

Theorem 2.6.7: Dominated Convergence Theorem (DOM)

Assume $\{f_n\}, f$ are measurable s.t. $f_n \rightarrow f$ a.e. If $\exists g \in L^1(\mathbb{R})$ s.t. $|f_n| \leq g \forall n$, then $f_n \rightarrow f$ in $L^1(\mathbb{R})$, i.e., $\lim_{n \rightarrow \infty} \int_{\mathbb{R}} |f_n - f| = 0$ (convergence in $L^1(\mathbb{R})$). In particular:

$$\int_{\mathbb{R}} f = \lim_n \int_{\mathbb{R}} f_n$$

Proof. Since $|f_n| \leq |g|$ and $f = \lim f_n$ a.e., $|f| \leq |g|$ a.e. Then $\int_{\mathbb{R}} |f_n| \leq \int_{\mathbb{R}} |g| < \infty, \int_{\mathbb{R}} |f| \leq \int_{\mathbb{R}} |g| < \infty$ and so $f_n, f \in L^1(\mathbb{R})$.

Observe that $|f_n - f| \leq 2|g|$ and $\int_{\mathbb{R}} (2|g|) < \infty$.

Apply Reverse Fatou to $\{|f_n - f|\}$:

$$\begin{aligned} &\implies \int_{\mathbb{R}} (\limsup (|f_n - f|)) \geq \limsup \int_{\mathbb{R}} |f_n - f| \\ &\implies \lim \int_{\mathbb{R}} |f_n - f| = 0, \left| \int_{\mathbb{R}} f_n - \int_{\mathbb{R}} f \right| = \left| \int_{\mathbb{R}} (f_n - f) \right| \leq \int_{\mathbb{R}} |f_n - f| \end{aligned}$$

□

Remark.

To use DOM we must find $g \in L^1(\mathbb{R})$ such that $|g| \geq |f_n|$ for ALL n . If, say, $f_n = 1_{[n, 2n]}$, $\lim_{n \rightarrow \infty} f_n = 0$, but $\int_{\mathbb{R}} f_n = n \forall n \geq 1$. Then DOM does not apply.

Proposition 2.6.8

Assume $f \in L^1(\mathbb{R})$. $\{h_n\}$ is a sequence of measurable functions that are uniformly bounded, i.e., $\exists M \geq 0$ s.t $|h_n| \leq M$ a.e $\forall n$. If $h_n \rightarrow h$ a.e for some measurable h , then $\lim \int_{\mathbb{R}} (fh_n) = \int_{\mathbb{R}} (fh)$.

Corollary 2.6.9

If $f \in L^1(\mathbb{R})$ then:

- i) $\int_{\mathbb{R} \setminus [-n, n]} |f| \rightarrow 0$ as $n \rightarrow \infty$
- ii) $\int_{\{|f| > n\}} |f| \rightarrow 0$ as $n \rightarrow \infty$
- iii) If $\{A_n\} \subseteq \mathcal{M}$ s.t $A_n \uparrow$, then $\int_{\bigcup_{n=1}^{\infty} A_n} f = \lim_{n \rightarrow \infty} \int_{A_n} f$
- iv) If $\{B_n\} \subseteq \mathcal{M}$ are disjoint, then (countable additivity) $\int_{\bigcup_{n=1}^{\infty} B_n} f = \sum_{n=1}^{\infty} \int_{B_n} f$
- v) If $\{A_n\} \subseteq \mathcal{M}$ s.t $A_n \downarrow$, then $\int_{\bigcap_{n=1}^{\infty} A_n} f = \lim_{n \rightarrow \infty} \int_{A_n} f$

Proposition 2.6.10

Assume f is non-negative, measurable and finite valued a.e.

$\forall k \in \mathbb{Z}$, set $A_k = \{x \in \mathbb{R} : 2^k \leq f(x) < 2^{k+1}\}$, $A_k \in \mathcal{M}$. Then:

$$\int_{\mathbb{R}} f < \infty \iff \sum_{k \in \mathbb{Z}} 2^k m(A_k) < \infty$$

Proof. $\implies$) Assume $\int_{\mathbb{R}} f < \infty$. Note that A_k s are disjoint and that $\bigcup_{k \in \mathbb{Z}} A_k = \{0 < f < \infty\}$. Then:

$$\int_{\mathbb{R}} f = \int_{f=0} f + \int_{0 < f < \infty} f = \int_{f=0} f + \int_{f=\infty} f = \int_{0 < f < \infty} f = \sum_{k \in \mathbb{Z}} \int_{A_k} f$$

For each $k \in \mathbb{Z}$, $\forall x \in A_k$, $2^k \leq f(x) < 2^{k+1}$;

$$\implies 2^k m(A_k) \leq \int_{A_k} f(x) dx < 2^{k+1} m(A_k) \implies \sum_{k \in \mathbb{Z}} 2^k m(A_k) \leq \sum_{k \in \mathbb{Z}} \int_{A_k} f(x) dx = \int_{\mathbb{R}} f < \infty$$

$\Longleftarrow$) Assume $\sum_{k \in \mathbb{Z}} 2^k m(A_k) < \infty$.

$$\int_{\mathbb{R}} f = \int_{0 < f < \infty} f = (\text{by MON}) = \sum_{k \in \mathbb{Z}} \int_{A_k} f < \sum_{k \in \mathbb{Z}} 2^{k+1} m(A_k) = 2 \sum_{k \in \mathbb{Z}} 2^k m(A_k) < \infty \implies \int_{\mathbb{R}} f < \infty$$

□

Remark.

Not all tools from Riemann integrals apply to Lebesgue integrals.

Example.

Consider $f(x) = |x|^{-\alpha} 1_{[-1,1]}(x)$, $f(0) = \infty$. Then f is finite-valued a.e. What can we say about $\int_{\mathbb{R}} f$?

$$\forall k \in \mathbb{Z}, A_k = \{2^k \leq f < 2^{k+1}\} = \{x \in [-1, 1] : 2^k \leq |x|^{-\alpha} < 2^{k+1}\}, |f| \geq 1$$

$$A_k = [-2^{-\frac{k}{\alpha}}, -2^{-\frac{k+1}{\alpha}}) \cup (2^{-\frac{k+1}{\alpha}}, 2^{-\frac{k}{\alpha}}] \text{ if } k \geq 0 \quad \& \quad A_k = \emptyset \text{ if } k < 0$$

$$\implies \sum_{k \in \mathbb{Z}} 2^k m(A_k) = \sum_{k=0}^{\infty} 2^k 2(1 - 2^{-\frac{1}{\alpha}}) 2^{-\frac{k}{\alpha}} = 2(1 - 2^{-\frac{1}{\alpha}}) \sum_{k=0}^{\infty} 2^{k(1-\frac{1}{\alpha})}$$

$$\implies \sum_{k \in \mathbb{Z}} 2^k m(A_k) < \infty \iff 0 < \alpha < 1$$

$$\implies \int_{[-1,1]} |x|^{-\alpha} dx < \infty \iff \alpha < 1$$

(Lebesgue!)

Example.

Consider $g(x) = |x|^{-\beta} 1_{\mathbb{R} \setminus [-1,1]}(x)$, $(\beta > 0)$. What can we say about $\int_{\mathbb{R}} g$?

$$A_k = \{2^k \leq g < 2^{k+1}\} = \begin{cases} [-2^{-\frac{k}{\beta}}, -2^{-\frac{k+1}{\beta}}) \cup (2^{-\frac{k+1}{\beta}}, 2^{-\frac{k}{\beta}}] & \text{if } k < 0 \\ \emptyset & \text{if } k \geq 0 \end{cases}$$

$$\int_{\mathbb{R} \setminus [-1,1]} |x|^{-\beta} dx < \infty \iff \sum_{k \in \mathbb{Z}} 2^k m(A_k) < \infty \iff \beta > 1$$

(Lebesgue!)

Example.

Consider $f_n(x) = (1 + \frac{x}{n})^{-n} \sin(\frac{x}{n})$. What is $\lim_{n \rightarrow \infty} \int_{(0,\infty)} f_n(x) dx$?

$$\lim f_n(x) = e^x(0) = 0 \quad \forall x > 0; \quad |f_n(x)| < (1 + \frac{x}{n})^{-n} \leq (1 + \frac{x}{2})^{-2} = g(x) \quad \forall x > 0, n \geq 2$$

Claim: $g \in L^1((0, \infty))$, i.e. $\int_{(0,\infty)} |g| < \infty$

$$\int_{(0,\infty)} g = \int_{(0,1]} g + \int_{(1,\infty)} g \leq \int_{(0,1)} 1 + \int_{(1,\infty)} \frac{4}{x^2} dx < \infty$$

$$\text{by DOM } \lim \int_{(0,\infty)} f_n = \int_{(0,\infty)} \lim f_n = 0$$

Example.

Let $c > 0$. $f_n(x) = x^{-c} (\cosh x)^{-\frac{1}{n}}$, $\lim \int_{(1,\infty)} = ?$

$$\forall x > 1, \cosh x > 1, (\cosh x)^{-\frac{1}{n}} \uparrow, \lim (\cosh x)^{-\frac{1}{n}} = 1$$

$$\lim f_n(x) = x^{-c} \quad \forall x > 1, g(x) = x^{-c}$$

When $c > 1$, $g \in L^1((1, \infty))$; by DOM $\lim \int_{(1,\infty)} f_n(x) dx = \int_{(1,\infty)} (\lim f_n) = \int_{(1,\infty)} x^{-c} dx < \infty$

When $0 < c \leq 1$, by Fatou:

$$\liminf \int_{(1,\infty)} f_n \geq \int_{(1,\infty)} (\liminf f_n) = \int_{(1,\infty)} x^{-c} dx = \infty$$

There are lots of useful examples.

Example.

Let $c \geq 0$. $f_n(x) = \frac{n}{1+n^2x^2}$ for $x \geq 0$. $\lim \int_{[c,\infty)} f_n(x)dx = ?$

$$\lim f_n(x) = \begin{cases} 0 & \text{if } x > 0 \\ \infty & \text{if } x = 0 \end{cases}$$

$$\int_{[c,\infty)} f_n(x)dx = \int_c^\infty \frac{n}{1+n^2x^2}dx = [\arctan(nx)]_c^\infty = \frac{\pi}{2} - \arctan(nc)$$

$$\text{When } c > 0 : \lim_{n \rightarrow \infty} \left(\frac{\pi}{2} - \arctan(nc) \right) = \frac{\pi}{2} - \frac{\pi}{2} = 0 = \int_{[c,\infty)} (\lim f_n)$$

$$\text{When } c = 0 : \lim_{n \rightarrow \infty} \left(\frac{\pi}{2} - \arctan(0) \right) = \frac{\pi}{2} \neq 0 = \int_{[0,\infty)} (\lim f_n)$$

This shows the limit and integral do not always commute, even when the limit exists a.e.

2.7 Riemann vs Lebesgue integral

If f is bounded on $[a, b]$, then f is Riemann integrable on $[a, b]$ if:

- 1) f is continuous on $[a, b]$
- 2) f is monotone on $[a, b]$
- 3) f is continuous except at possibly finitely many points in $[a, b]$

Recall that $f = 1_{\mathbb{Q} \cap [0,1]}$ is not Riemann integrable but f is Lebesgue integrable because $|f| \leq 1_{[0,1]} \in L^1$.

Remark.

- 1) $\exists$ a bounded function on $[a, b]$ that is not Riemann integrable
- 2) In general, g being Riemann integrable and $|f| \leq |g| \not\Rightarrow f$ is Riemann integrable
- 3) In general, DOM and MON do not apply to Riemann integrals.

Example.

Take $\{q_n : n \geq 1\}$ to be an enumeration of $\mathbb{Q} \cap [0, 1]$. Define:

$$f_n(x) \begin{cases} 1 & \text{if } x \in \{q_1, q_2, \dots, q_n\} \\ 0 & \text{otherwise} \end{cases} \quad f_n \uparrow, \lim f_n = f = 1_{\mathbb{Q} \cap [0, 1]}$$

$\forall n \geq 1$ f_n is only discontinuous at finitely many points, so f_n is Riemann integrable on $[0, 1]$ and $(R) \int_0^1 f_n(x) dx = 0$. But $f = \lim f_n$ is NOT Riemann integrable.

Theorem 2.7.1: Riemann Giving Lebesgue

Assume f is Riemann integrable on $[a, b]$. Then f is Lebesgue integrable on $[a, b]$ (i.e. $f \in L^1([a, b])$) and $(R) \int_a^b f(x) dx = (L) \int_{[a, b]} f(x) dx$

Proof. f is Riemann integrable on $[a, b] \implies \exists M > 0$ s.t. $|f| \leq M$ on $[a, b]$.

$\exists$ step functions φ_n, ψ_n such that $\varphi_n \leq f \leq \psi_n$ on $[a, b]$, and $|\varphi_n|, |\psi_n| \leq M \forall n \geq 1$ and:

$$\lim_{n \rightarrow \infty} (R) \int_a^b \varphi_n = (R) \int_a^b f = \lim_{n \rightarrow \infty} (R) \int_a^b \psi_n$$

Denote $\varphi = \lim \varphi_n, \psi = \lim \psi_n$. Since φ_n, ψ_n are step functions. φ_n, ψ_n are measurable $\implies \varphi, \psi$ are measurable and $\varphi \leq f \leq \psi$

Observe that Lebesgue and Riemann integrals coincide on step functions:

$$(R) \int_a^b \varphi_n = \int_{[a, b]} \varphi_n, (R) \int_a^b \psi_n = \int_{[a, b]} \psi_n$$

$$\text{By DOM } \int_{[a, b]} \varphi = \lim \int_{[a, b]} \varphi_n = \lim (R) \int_a^b \varphi_n = (R) \int_a^b f = \lim (R) \int_a^b \psi_n = \lim \int_{[a, b]} \psi_n = \int_{[a, b]} \psi$$

However, $\varphi \leq \psi$ and $\int_{[a, b]} (\psi - \varphi) = 0 \implies \psi = \varphi = f$ a.e on $[a, b] \implies f$ is measurable.

Since $|f| \leq M$ on $[a, b]$, $f \in L^1([a, b])$, and finally:

$$\int_{[a, b]} f = \int_{[a, b]} \varphi = (R) \int_a^b f(x) dx$$

□

2.8 L^p spaces

Definition 2.8.1

Let $(X, \mathcal{A}, \mu)$ be a measure space and $1 \leq p < \infty$. For a measurable function $f : X \rightarrow \overline{\mathbb{R}}$, define

$$\|f\|_p = \left(\int_X |f|^p d\mu \right)^{\frac{1}{p}}$$

The L^p space is defined as

$$L^p(X, \mathcal{A}, \mu) = \{f : X \rightarrow \overline{\mathbb{R}} \text{ measurable} : \|f\|_p < \infty\}$$

When the measure space is clear from context, we write $L^p(\mu)$ or simply L^p .

Remark.

Technically, L^p is the space of equivalence classes of functions that agree μ -a.e., since $\|f\|_p = 0$ iff $f = 0$ μ -a.e. We identify functions that are equal a.e.

Remark.

For $p = \infty$, define

$$\|f\|_\infty = \inf\{M \geq 0 : |f| \leq M \text{ } \mu\text{-a.e.}\}$$

and $L^\infty = \{f \text{ measurable} : \|f\|_\infty < \infty\}$.

Proposition 2.8.2

For $1 \leq p \leq \infty$, $\|\cdot\|_p$ is a norm on L^p , and $(L^p, \|\cdot\|_p)$ is a complete normed space (i.e., a Banach space).

Lemma 2.8.3

Bounded and compactly supported functions are dense in $L^p(\mathbb{R})$ $\forall 1 \leq p < \infty$.

Given $f \in L^p(\mathbb{R})$, $f_n(x) = 1_{[-n,n]}(x)1_{\{|f|<n\}}(x)f(x)$, $f_n \rightarrow f$ in $L^p(\mathbb{R})$.

Lemma 2.8.4

Simple functions are dense in $L^p(\mathbb{R})$.

Proof. Let f_n be the same as above. For each $n \geq 1, k = 0, 1, \dots, n2^n - 1$, set:

$$A_{n,k} = \{x \in [-n, n] : \left\{ \frac{k}{2^n} \leq f_n^+ < \frac{k+1}{2^n} \right\}, \quad \varphi_n^+ = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} 1_{A_{n,k}}$$

$$B_{n,k} = \{x \in [-n, n] : \left\{ \frac{k}{2^n} \leq f_n^- < \frac{k+1}{2^n} \right\}, \quad \varphi_n^- = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} 1_{B_{n,k}}$$

Then $\varphi_n = \varphi_n^+ - \varphi_n^-$ is a simple function, and $|\varphi_n| \leq n \implies \varphi_n \in L^p(\mathbb{R})$ (since φ_n is supported on $[-n, n]$).

$\forall x \in \mathbb{R}, \lim_n \varphi_n(x) = f(x)$:

$$|f_n(x) - \varphi_n(x)| \leq |f_n^+(x) - \varphi_n^+(x)| + |f_n^-(x) - \varphi_n^-(x)| \leq 2 \cdot 2^{-n}$$

$$\|f - \varphi_n\|_p \leq \|f - f_n\|_p + \|f_n - \varphi_n\|_p$$

Then the first term goes to 0 as $n \rightarrow \infty$, and the second term gives:

$$\|f - \varphi_n\|_p \leq \left(\int_{[-n,n]} |f_n - \varphi_n|^p \right)^{\frac{1}{p}} \leq ((2 \cdot 2^{-n})^p m([-n, n]))^{\frac{1}{p}} = (2^p 2^{-np} 2^n)^{\frac{1}{p}}$$

Which $\rightarrow 0$ as $n \rightarrow \infty$. □

Theorem 2.8.5

Denote by $Cc(\mathbb{R})$ the space of continuous and compactly supported functions. $Cc(\mathbb{R})$ is dense in $L^p(\mathbb{R})$ for $1 \leq p < \infty$.

Proof. Given $f \in L^p(\mathbb{R})$, choose $\{\varphi_n\}$ the same as above. Recall that $\forall n \geq 1, \exists$ a step function θ_n such that θ_n is bounded by $\sup |\varphi_n(x)| \leq n$ and θ_n is supported on $[n-1, n+1]$ and $\{\theta_n \neq \varphi_n\}$ has arbitrarily small measure.

In particular, $m(\{\theta_n \neq \varphi_n\}) \leq 2^{-n-1}$.

Recall that given θ_n a step function, $\exists \tilde{\theta}_n$ continuous on $\mathbb{R}$ where $\tilde{\theta}_n$ is supported on $[-n-2, n+2]$ and $m(\{\tilde{\theta}_n \neq \theta_n\}) \leq 2^{-n-1}$.

Then $\{\tilde{\theta}_n\} \subseteq Cc(\mathbb{R})$ and:

$$m(\{\tilde{\theta}_n \neq \varphi_n\}) \leq m(\{\tilde{\theta}_n \neq \theta_n\}) + m(\{\theta_n \neq \varphi_n\}) \leq 2^{-n}.$$

Now we turn to the integral.

$$\|f - \tilde{\theta}_n\|_p \leq \|f - \varphi_n\|_p + \|\varphi_n - \tilde{\theta}_n\|_p$$

Where as $n \rightarrow \infty$ the first term goes to 0 and the second term becomes:

$$\|f - \tilde{\theta}_n\|_p \leq \left(\int_{\mathbb{R}} |\varphi_n - \tilde{\theta}_n|^p \right)^{\frac{1}{p}} = \left(\int_{\{\tilde{\theta}_n \neq \varphi_n\}} |\varphi_n - \tilde{\theta}_n|^p \right)^{\frac{1}{p}} \leq ((2n)^p 2^{-n})^{\frac{1}{p}} \rightarrow 0.$$

□

Remark.

The density of $Cc(\mathbb{R})$ in $L^p(\mathbb{R})$ is useful in the study of properties of general L^p functions. (i.e. if $f \in L^p(\mathbb{R})$, then $\lim_{n \rightarrow \infty} \int_{\mathbb{R}} |f(x + \frac{1}{n}) - f(x)|^p dx = 0$, i.e. $f(\cdot + \frac{1}{n}) \rightarrow f \in L^p(\mathbb{R})$).

Remark.

$Cc(\mathbb{R})$ is NOT dense in $L^\infty(\mathbb{R})$.

Proof: find some $f \in L^\infty(\mathbb{R})$ that cannot be approximated by some sequence in $Cc(\mathbb{R})$.

2.9 Convergence modes, uniform integrability

Given $\{f_n\}$, f measurable and finite-valued a.e:

- 1) $f_n \rightarrow f$ in measure $\implies \exists$ subseq $\{n_k\}$ such that $f_{n_k} \rightarrow f$ a.e.
- 2) $f_n \rightarrow f$ a.e on $A \in \mathcal{M}$ with $m(A) < \infty \implies f_n \rightarrow f$ in measure on A .
- 3) $f_n \rightarrow f$ in $L^p(\mathbb{R}) \implies f_n \rightarrow f$ in measure.

Proposition 2.9.1

If $\{f_n\}$, $f \in L^p(\mathbb{R})$ and $f_n \rightarrow f$ in $L^p(\mathbb{R})$, then $f_n \rightarrow f$ in measure.

Proof.

$$\forall \delta > 0 m(\{|f_n - f| > \delta\}) = \int_{\{|f_n - f| > \delta\}} 1 dx \leq \int_{\{|f_n - f| > \delta\}} \frac{|f_n - f|^p}{\delta^p} dx \leq \frac{1}{\delta^p} \|f_n - f\|_p^p \rightarrow 0$$

(Since $1 \leq \frac{|f_n - f|^p}{\delta^p}$ and by *Markov's inequality.)

□

Remark.

In general, convergence in $L^p \not\implies$ convergence a.e.

We also do not have in general that convergence in measure $\implies$ convergence in L^p unless one of the integral convergence theorems holds.

Proposition 2.9.2

(mMON) If f_n is non-negative, $f_n \uparrow$, and $f_n \rightarrow f$ in measure.

Then $\int_{\mathbb{R}} f = \lim_n \int_{\mathbb{R}} f_n$.

Proof. $f_n \rightarrow f$ in measure $\implies f_{n_k} \rightarrow f$ a.e for $\{n_k\} \implies f$ is non-negative.

Assume the proposition doesn't hold. Then $\exists \{n_q\}$ such that $\int_{\mathbb{R}} f_{n_q} \not\rightarrow \int_{\mathbb{R}} f$.

However, $f_{n_q} \rightarrow f$ in measure $\implies \exists \{n_{qp}\}$ such that $f_{n_{qp}} \rightarrow f$ a.e. By (MON) applied to $\{f_{n_{qp}}\}$, $\int_{\mathbb{R}} f_{n_{qp}} \rightarrow \int_{\mathbb{R}} f$ as $p \rightarrow \infty$. Contradiction. $\square$

Proposition 2.9.3

(mDOM) Let $f_n \in L^1(\mathbb{R})$ and $f_n \rightarrow f$ in measure, and $\exists g \in L^1(\mathbb{R})$ such that $|f_n| \leq |g| \forall n \geq 1$. Then $f_n \rightarrow f \in L^1(\mathbb{R})$.

(Proof is same as above.)

Fact 2.9.4

$f \in L^1(\mathbb{R}) \implies \int_{\{|f|>n\}} |f| \rightarrow 0$ as $n \rightarrow \infty$, but $\Leftarrow$ is not true.

Notation: For $A \in \mathcal{M}$, $f \in L^1(A)$ if $\int_A |f| < \infty$.

Proposition 2.9.5

Given $A \in \mathcal{M}$ with $m(A) < \infty$, $f \in L^1(A) \iff \lim_n \int_{A \cap \{|f|>n\}} |f| = 0$.

Proof. $\implies$) is trivial.

$\Leftarrow$) Write:

$$\int_A |f| = \int_{A \cap \{|f| \leq N\}} |f| + \int_{A \cap \{|f| > N\}} |f| \leq Nm(A) + 1 < \infty$$

$\square$

Definition 2.9.6: Uniformly Integrable

Given $\{f_n\}$ measurable and $A \in \mathcal{M}$, we say that $\{f_n\}$ is uniformly integrable on A if:

$$\lim_{M \rightarrow \infty} \sup_{n \geq 1} \int_{A \cap \{|f_n| > M\}} |f_n| = 0$$

Note: this is a property that is "between" boundedness in L^1 and boundedness in L^p .

Proposition 2.9.7

Let $\{f_n\}$ measurable and $A \in \mathcal{M}$.

- 1) if $m(A) < \infty$ and $\{f_n\}$ is uniformly integrable on A , then $\{f_n\}$ is bounded in $L^1(A)$, i.e., $\sup_{n \geq 1} \int_A |f_n| < \infty$.
- 2) if $\{f_n\}$ is bounded in $L^p(A)$ for any $1 < p < \infty$, then $\{f_n\}$ is uniformly integrable on A .

Proof. 1) Choose M such that $\sup_{n \geq 1} \int_{A \cap \{|f_n| > M\}} |f_n| \leq 1$. Then:

$$\sup_{n \geq 1} \int_A |f_n| = \sup_{n \geq 1} \left(\int_{A \cap \{|f_n| \leq M\}} |f_n| + \int_{A \cap \{|f_n| > M\}} |f_n| \right) \leq Mm(A) + 1 < \infty$$

2) For any $M > 0$:

$$\sup_n \int_{A \cap \{|f_n| > M\}} |f_n| \leq \sup_n \int_{A \cap \{|f_n| > M\}} |f_n| \frac{|f_n|^{p-1}}{M^{p-1}} \leq \frac{1}{M^{p-1}} \sup_n \int_A |f_n|^p \rightarrow 0$$

(Using Markov's inequality since $1 \leq \left(\frac{|f_n|}{M}\right)^{p-1}$). Then we get the result as $M \rightarrow \infty$. $\square$

Proposition 2.9.8

Given $\{f_n\}$ measurable, $A \in \mathcal{M}$, with $m(A) < \infty$. TFAE:

- i) $f_n \in L^1(A) \forall n \geq 1, f \in L^1(A)$ and $f_n \rightarrow f$ in $L^1(A)$
- ii) $\{f_n\}$ is uniformly integrable on A and $f_n \rightarrow f$ in measure on A .

Proof. NOTE: This proof is similar in strategy to one you need to do for homework. ($\Rightarrow$)

Assume $f_n \rightarrow f$ in $L^1(A) \implies \int_A |f_n| \rightarrow \int_A |f|$. So $\{f_n\}$ is bounded in $L^1(A)$.

$$\forall M > 0, \int_{A \cap \{|f_n| > M\}} |f_n| \leq \int_{A \cap \{|f_n| > M\}} |f_n - f| + \int_{A \cap \{|f_n| > M\}} |f|$$

The first of those terms $\leq \int_A |f_n - f| \rightarrow 0$. The second term:

$$\begin{aligned}
 &= \int_{A \cap \{|f_n| > M\} \cap \{|f| \leq \sqrt{M}\}} |f| + \int_{A \cap \{|f_n| > M\} \cap \{|f| > \sqrt{M}\}} |f| \\
 &\leq \sqrt{M} \cdot m(A \cap \{|f_n| > M\}) + \int_{\{A \cap \{|f| > \sqrt{M}\}\}} |f| \\
 (\text{Markov}) \quad &\leq \sqrt{M} \frac{\sup_n \int_A |f_n|}{M} + \int_{A \cap \{|f| > \sqrt{M}\}} |f| \rightarrow 0 \text{ as } M \rightarrow \infty
 \end{aligned}$$

. This shows uniform integrability. For convergence in measure: given $\varepsilon > 0$,

$$m(A \cap \{|f_n - f| > \varepsilon\}) \leq \frac{1}{\varepsilon} \int_A |f_n - f| \rightarrow 0$$

by Markov's inequality.

($\Leftarrow$) Assume $\{f_n\}$ is uniformly integrable and $f_n \rightarrow f$ in measure on A .

Given $\varepsilon > 0$, by uniform integrability, $\exists M > 0$ such that $\sup_n \int_{A \cap \{|f_n| > M\}} |f_n| < \frac{\varepsilon}{3}$ and also $\int_{A \cap \{|f| > M\}} |f| < \frac{\varepsilon}{3}$.

Then

$$\int_A |f_n - f| = \int_{A \cap \{|f_n| \leq M, |f| \leq M\}} |f_n - f| + \int_{A \cap \{|f_n| > M \text{ or } |f| > M\}} |f_n - f|$$

The second term is $\leq \int_{A \cap \{|f_n| > M\}} |f_n| + |f| + \int_{A \cap \{|f| > M\}} |f_n| + |f| < \frac{2\varepsilon}{3}$ for n large.

For the first term, since $f_n \rightarrow f$ in measure and $m(A) < \infty$, Egorov's theorem gives $\exists B \subseteq A$ with $m(A \setminus B) < \frac{\varepsilon}{6M}$ and $f_n \rightarrow f$ uniformly on B .

$$\int_{A \cap \{|f_n| \leq M, |f| \leq M\}} |f_n - f| \leq \int_B |f_n - f| + \int_{A \setminus B} |f_n| + |f| \leq \varepsilon/3 + 2M \cdot \frac{\varepsilon}{6M} = \frac{2\varepsilon}{3}$$

for n sufficiently large. Thus $\int_A |f_n - f| < \varepsilon$ for n large, so $f_n \rightarrow f$ in $L^1(A)$. $\square$

Chapter 3: Product Spaces

Proposition 3.0.1

Given a measure $(X, \mathcal{F}, \mu)$, with X nonempty, $\mathcal{F}$ a σ -algebra of subsets of X , and μ a σ -finite measure ($\exists$ a sequence $\{X_n\} \subseteq \mathcal{F}$ s.t. $X_n \uparrow$ and $\bigcup_{n=1}^{\infty} X_n = x$ and $\mu(X_n) < \infty \forall n \geq 1$).

Definition 3.0.2: $\mathcal{F}$ -measurable

We say that $f : X \rightarrow \overline{\mathbb{R}}$ is $\mathcal{F}$ -measurable if $\forall a \in \mathbb{R}, f^{-1}([-\infty, a)) \in \mathcal{F}$.

We have similar properties as in the Lebesgue setting:

- i) if f is $\mathcal{F}$ -measurable, $cf, f^k, |f|, f \wedge a, f^+,$ etc are too.
- ii) if f, g $\mathcal{F}$ -measurable, $f + g, fg, f \wedge g,$ etc are too.
- iii) if $\{f_n\}$ are $\mathcal{F}$ -measurable, $\sup f_n, \liminf f_n, \lim f_n,$ etc are too.

Again, we can dissect f in the same way as before, and construct integration in the exact same way. Nothing changes at all; it is just that f is $\mathcal{F}$ -measurable instead of Lebesgue measurable.

$$\left(\text{i.e. } \int_X \varphi_n^{+/-} d\mu = \sum_{k=0}^{n2^n} \frac{k}{2^n} \mu(A_{n,k}^{+/-}) \right)$$

$$\left(\text{and } \int_X f d\mu = \int_X f^+ d\mu - \int_X f^- d\mu \right)$$

3.1 Product σ -algebra

We will only demonstrate the construction of product space with $\mathbb{R}^2$ but the same construction can be generalized to $\mathbb{R}^d$.

Definition 3.1.1: Product σ -algebra of $\mathbb{R}^2$

Consider $\mathbb{R}^2$, a product space $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ with elements (x, y) . The product σ -algebra (of subsets of $\mathbb{R}^2$) denoted by $\mathcal{M} \otimes \mathcal{M} = \mathcal{M}^{2\otimes} = \mathcal{M}^2$ is defined as:

$$\mathcal{M}^2 = \sigma(\{A \times B : A, B \in \mathcal{M}\})$$

(Then $A \times B$ is a product set: $A \times B = \{(x, y) : x \in A, y \in B\}$)

$\mathcal{M}^2$ contains:

i) rectangles $I_1 \times I_2$ with I_1, I_2 intervals

"What do I mean by rectangles?" - Linan Chen

ii) singletons $\{(x, y)\}$

iii) open sets, closed sets: $\mathcal{B}_{\mathbb{R}^2} = \sigma(\{\text{open sets in } \mathbb{R}^2\}) \subseteq \mathcal{M}$.

Given G open, $\forall x \in G, \exists$ a disc centered at x that is entirely in G . Moreover, there exist (a_1, a_2) and (b_1, b_2) , $a_i, b_i \in \mathbb{Q}$ such that $x \in (a_1, a_2) \times (b_1, b_2) \subseteq G$, and $G = \bigcup_{x \in G} (a_1, a_2) \times (b_1, b_2)$.

Definition 3.1.2: Slice

Given $E \subseteq \mathbb{R}^2, \forall x \in \mathbb{R}$:

$$E_x = \{y \in \mathbb{R}, (x, y) \in E\} \subseteq \mathbb{R}$$

is called the slice of E at x (likewise $E^y = \{x \in \mathbb{R}, (x, y) \in E\} \subseteq \mathbb{R}$ is the slice of E at y)

Proposition 3.1.3

If $E \subseteq \mathcal{M}^2$, then $\forall x \in \mathbb{R}, E_x \in \mathcal{M}$ and $\forall y \in \mathbb{R}, E^y \in \mathcal{M}$.

Proof. We prove by the good set strategy. Define $\mathcal{A} = \{E \subseteq \mathbb{R}^2 : \forall x \in \mathbb{R}, E_x \in \mathcal{M}\}$.

Claim: $\mathcal{A}$ is a σ -algebra of subsets of $\mathbb{R}^2$.

i) $\mathbb{R}^2 \in \mathcal{A} : \forall x \in \mathbb{R}, (\mathbb{R}^2)_x = \mathbb{R} \in \mathcal{M} \checkmark$

ii) $E \in \mathcal{A} \implies \forall x \in \mathbb{R}, E_x \in \mathcal{M} \implies (E^c)_x = (E_x)^c \in \mathcal{M} \implies E^c \in \mathcal{A}$

iii) $\{E_n\} \subseteq \mathcal{A} \implies \forall x \in \mathbb{R}, (E_n)_x \in \mathcal{M} \implies (\bigcup_{n=1}^{\infty} E_n)_x = \bigcup_{n=1}^{\infty} (E_n)_x \in \mathcal{M} \implies \bigcup_{n=1}^{\infty} E_n \in \mathcal{A}$.

Then, $\mathcal{A}$ is a σ -algebra. We claim that for all $A, B \in \mathcal{M}$, we claim that $A \times B \in \mathcal{A}$.

$$\forall x \in \mathbb{R}, (A \times B)_x = \emptyset \text{ if } x \notin A, \text{ but } B \text{ if } x \in A$$

$$\implies \mathcal{A} \supseteq \sigma(\{A \times B : A, B \in M\}) = M^2$$

(”Everything in M^2 is a good set.”)

(It is easy to show the same for E^y .) □

Remark.

In general, marginal measurability (the above) $\not\Rightarrow$ product measurability.

Fact 3.1.4

In fact, $\exists E \subseteq \mathbb{R}^2$ s.t. $E_x \in \mathcal{M}$ but $E \notin M^2$. As an example, consider $E = N \times A$ for $A \in M$, N the Vitali set.

Then, for all $x \in \mathbb{R}$, $E_x = A$ if $x \in \mathcal{N}$, $\emptyset$ otherwise. And so $E_x \in M \forall x \in \mathbb{R}$, but $E \notin M^2$ (as $E^y = N$ if $y \in A$, $\emptyset$ otherwise is not $\in \mathcal{M}$)

Remark.

In fact, there exists an $E \subseteq \mathbb{R}^2$ such that $E_x, E^y \in \mathcal{M}$, but $E \notin M^2$ (Sierpiński set).

However, if $E \subseteq \mathbb{R}^2$ is a product set, i.e. $E = A \times B$, then if $A, B \in \mathcal{M}$, $E \in M^2$.

Definition 3.1.5

Assume $f : \mathbb{R}^2 \rightarrow \overline{\mathbb{R}}$ is a function.

Then $\forall x \in \mathbb{R}$, $f_x : \mathbb{R} \rightarrow \overline{\mathbb{R}}$ is such that $f_x(y) = f(x, y) \forall y \in \mathbb{R}$.

Also, $\forall y \in \mathbb{R}$, $f^y : \mathbb{R} \rightarrow \overline{\mathbb{R}}$ is such that $f^y(x) = f(x, y) \forall x \in \mathbb{R}$

Proposition 3.1.6

If $f : \mathbb{R}^2 \rightarrow \overline{\mathbb{R}}$ is M^2 -measurable, then $\forall x \in \mathbb{R}$, f_x is M -measurable, and $\forall y \in \mathbb{R}$, f^y is M -measurable as well.

Proof. Fix $x \in \mathbb{R}$. We show f_x is M -measurable. Let $\alpha \in \mathbb{R}$. Then

$$\begin{aligned} \{y \in \mathbb{R} : f_x(y) > \alpha\} &= \{y \in \mathbb{R} : f(x, y) > \alpha\} \\ &= \{y \in \mathbb{R} : (x, y) \in f^{-1}((\alpha, \infty])\}. \end{aligned}$$

Define $\phi : \mathbb{R} \rightarrow \mathbb{R}^2$ by $\phi(y) = (x, y)$. Note that ϕ is continuous, hence M -measurable. Since

f is $\mathcal{M}^2$ -measurable, we have $f^{-1}((\alpha, \infty]) \in \mathcal{M}^2$. Therefore

$$\{y \in \mathbb{R} : f_x(y) > \alpha\} = \phi^{-1}(f^{-1}((\alpha, \infty])) \in \mathcal{M}$$

as the preimage of a measurable set under a measurable function. Thus f_x is $\mathcal{M}$ -measurable. The proof for f^y is analogous, using $\psi : \mathbb{R} \rightarrow \mathbb{R}^2$ defined by $\psi(x) = (x, y)$. $\square$

Fact 3.1.7

Some remarks:

- i) If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is continuous then f is $\mathcal{M}^2$ -measurable.
- ii) If $f = 1_E$ for some $E \subseteq \mathbb{R}^2$ then f $\mathcal{M}^2$ -measurable $\iff E \in \mathcal{M}^2$.
- iii) In general, $\exists f : \mathbb{R}^2 \rightarrow \overline{\mathbb{R}}$ s.t that f_x is $\mathcal{M}$ -measurable $\forall x \in \mathbb{R}$ but f is NOT $\mathcal{M}^2$ -measurable.

Proposition 3.1.8

If $f(x, y) = h(x)g(y)$ for some $h, g, \mathbb{R} \rightarrow \overline{\mathbb{R}}$ that are non-trivial, then f is $\mathcal{M}^2$ measurable $\iff$ both h, g are $\mathcal{M}$ -measurable.

3.2 Product Measure

Definition 3.2.1

Given $E \in \mathcal{M}^2$, define functions:

$$I_E^1 : \mathbb{R} \rightarrow \overline{\mathbb{R}} \quad (\forall x \in \mathbb{R}) \quad I_E^1(x) = m(E_x)$$

$$I_E^2 : \mathbb{R} \rightarrow \overline{\mathbb{R}} \quad (\forall y \in \mathbb{R}) \quad I_E^2(y) = m(E^y)$$

(Returns the measure of the slices)

Theorem 3.2.2: Equality of Slices

Given $E \in \mathcal{M}^2$, I_E^1 and I_E^2 are $\mathcal{M}$ -measurable functions, and:

$$\int_{\mathbb{R}} I_E^1(x) dx = \int_{\mathbb{R}} I_E^2(y) dy$$

Proof. If indeed I_E^1 and I_E^2 are $\mathcal{M}$ -measurable, then the integrals in the theorem are well-defined as I_E^1 and I_E^2 are non-negative.

The proof is once again by the 'good set strategy'.

Set $\Sigma = \{E \in \mathcal{M}^2 : \text{all of the above hold for } E_N = E \cap [-N, N]^2 \forall N \geq 1\}$

i.e. $\Sigma = \{E \in \mathcal{M}^2 : I_{E_N}^1, I_{E_N}^2 \text{ are } \mathcal{M}\text{-measurable and (Equality of Slices) holds for } E_N\}$

Note that $\forall E \in \mathcal{M}^2, \forall N > 0 :$

$$I_{E_N}^1(x) = \begin{cases} m((E_N)_x) & \text{if } x \in [-N, N] \\ 0 & \text{if } x \notin [-N, N] \end{cases} \quad I_{E_N}^2(y) = \begin{cases} m((E_N)_y) & \text{if } y \in [-N, N] \\ 0 & \text{if } y \notin [-N, N] \end{cases}$$

$$\implies I_{E_N}^1 = 1_{[-N, N]} I_{E_N}^1, I_{E_N}^2 = 1_{[-N, N]} I_{E_N}^2$$

Set $\mathcal{C} = \{A \times B : A, B \in \mathcal{M}\}. \quad (\mathcal{M}^2 = \sigma(\mathcal{C})).$

i) Claim: $\mathcal{C} \subseteq \Sigma. \quad \forall E \in \mathcal{C}, \exists A, B \in \mathcal{M} \text{ s.t. } E = A \times B.$

$\forall N > 0, E_n = (A \times B) \cap [-N, N]^2 = A_N \times B_N \quad (A_N = A \cap [-N, N], \text{ etc, of course}).$

$$\implies I_{E_N}^1(x) = I_{A_N \times B_N}^1(x) = \begin{cases} m(B_N) & \text{if } x \in A_N \\ 0 & \text{if } x \notin A_N \end{cases} \quad I_{E_N}^2(y) = \begin{cases} m(A_N) & \text{if } y \in B_N \\ 0 & \text{if } y \notin B_N \end{cases}$$

So it's clear that $I_{A_N \times B_N}^1$ and $I_{A_N \times B_N}^2$ are M -measurable and $\int_{\mathbb{R}} I_{A_N \times B_N}^1(x) dx = m(B_N)m(A_N) = \int_{\mathbb{R}} I_{A_N \times B_N}^2(y) dy$

Then $E = A \times B \in \Sigma$, i.e., $\mathcal{C} \subseteq \Sigma$.

ii) Now we claim that $\mathbb{R}^2 \in \Sigma. \quad \forall N > 0 :$

$$I_{[-N, N]^2}^1(x) = \begin{cases} 2N & \text{if } x \in [-N, N] \\ 0 & \text{if } x \notin [-N, N] \end{cases} \quad I_{[-N, N]^2}^2(y) = \begin{cases} 2N & \text{if } y \in [-N, N] \\ 0 & \text{if } y \notin [-N, N] \end{cases}$$

Clearly $I_{[-N, N]^2}^1, I_{[-N, N]^2}^2$ are both $\mathcal{M}$ -measurable, and both integrals are equal to $4N^2$.

iii) Claim: if $E \in \Sigma$, then $E^c \in \Sigma$.

For each $N > 0$, denote by $F_N = E^c \cap [-N, N]^2$. Obviously $I_{F_N}^1(x) = 0$ if $x \notin [-N, N]$, and if $x \in [-N, N]$:

$$(F_N)_x = \{y : (x, y) \in E^c \cap [-N, N]^2\} = [-N, N] \setminus E_x = [-N, N] \setminus (E_N)_x \implies I_{F_N}^1(x) = 2N - I_{E_N}^1(x)$$

$$\text{Similarly } I_{F_N}^2(y) = \begin{cases} 2N - I_{E_N}^2(y) & \text{if } y \in [-N, N] \\ 0 & \text{if } y \notin [-N, N] \end{cases}$$

Then $I_{F_N}^1, I_{F_N}^2$ are M -measurable, and:

$$\int_{\mathbb{R}} I_{F_N}^1(x) dx = \int_{[-N, N]} (2N - I_{E_N}^1(x)) dx = 4N^2 - \int_{\mathbb{R}} I_{E_N}^1(x) dx$$

And likewise $\int_{\mathbb{R}} I_{F_N}^2(y)dy = 4N^2 - \int_{\mathbb{R}} I_{E_N}^2(y)dy$. And so $E^c \in \Sigma$.

iv) Claim: $\{E_k : k \geq 1\} \subseteq \Sigma \implies E = \bigcup_{k=1}^{\infty} E_k \in \Sigma$.

Wlog., assume that the E_k s are disjoint (otherwise replace E_k with $E_k \setminus \left(\bigcup_{j=1}^{k-1} E_j\right)$). For each $N > 0$, set $E_N = \bigcup_{k=1}^{\infty} E_{k,N}$ where $E_{k,N} = E_k \cap [-N, N]^2$. Then:

$$I_{E_N}^1(x) = 1_{[-N, N]}(x) m\left(\bigcup_{k=1}^{\infty} (E_{k,N})_x\right) = 1_{[-N, N]}(x) \sum_{k=1}^{\infty} m((E_{k,N})_x) = \sum_{k=1}^{\infty} I_{E_{k,N}}^1(x)$$

(Where we used additivity in the equality.) Similarly, $I_{E_N}^2(y) = \sum_{k=1}^{\infty} I_{E_{k,N}}^2(y)$ And so $I_{E_N}^1, I_{E_N}^2$ are $\mathcal{M}$ -measurable, and by (MON) and using that $E_k \in \Sigma$:

$$\int_{\mathbb{R}} I_{E_N}^1(x)dx = \sum_{k=1}^{\infty} \int_{\mathbb{R}} I_{E_{k,N}}^1(x)dx \quad \int_{\mathbb{R}} I_{E_N}^2(y)dy = \sum_{k=1}^{\infty} \int_{\mathbb{R}} I_{E_{k,N}}^2(y)dy$$

$$\implies E = \bigcup_{k=1}^{\infty} E_k \subseteq \Sigma \implies \Sigma \text{ is a } \sigma\text{-algebra of subsets of } \mathbb{R}^2, \text{ and } \mathcal{C} \subseteq \Sigma \implies \Sigma = M^2.$$

Thus $\forall E \in \mathcal{M}^2, E \in \Sigma \implies$ all the statements are true for $E_N \forall N > 0$.

$$I_E^1(x) = \lim_{N \rightarrow \infty} 1_{[-N, N]}(x) m((E_N)_x) = \lim_N I_{E_N}^1(x)$$

(Using the fact that $m(E_x) = \lim_N m((E_N)_x)$.)

$$\text{Moreover, } I_{E_N}^1 \leq I_{E_{N+1}}^1 \implies I_E^1 \text{ is } \mathcal{M}\text{-measurable and } \int_{\mathbb{R}} I_E^1(x)dx = \lim_N \int_{\mathbb{R}} \int_{\mathbb{R}} I_{E_N}^1(x)dx$$

(Where we get the last equality by (MON).) Similarly, I_E^2 is $\mathcal{M}$ -measurable and $\int_{\mathbb{R}} I_E^2(y)dy = \lim_N \int_{\mathbb{R}} \int_{\mathbb{R}} I_{E_N}^2(y)dy$. $\square$

Definition 3.2.3: Lebesgue Measure on $\mathbb{R}^2$

Define a non-negative set function on the measurable space $(\mathbb{R}^2, \mathcal{M}^2)$ by:

$$\forall E \in \mathcal{M}^2, m(E) = \int_{\mathbb{R}} I_E^1(x) dx = \int_{\mathbb{R}} I_E^2(y) dy$$

This m is called the Lebesgue measure on $\mathbb{R}^2$.

"Being a measure is not free, right?" - Linan Chen

Indeed it is a measure on $(\mathbb{R}^2, \mathcal{M}^2)$: i) Quite trivially $m(\emptyset) = 0$

ii) If $\{E_k : k \geq 1\} \subseteq \mathcal{M}^2$ is a sequence of disjoint sets and $E = \bigcup_{k=1}^{\infty} E_k$, then $m(E) = \sum_{k=1}^{\infty} m(E_k)$ because $\forall x \in \mathbb{R}, E_x = \bigcup_{k=1}^{\infty} (E_k)_x$ disjoint $\implies \int_{\mathbb{R}} I_E^1(x) dx = \sum_{k=1}^{\infty} \int_{\mathbb{R}} I_{E_k}^1(x) dx$.

Remark.

For any rectangle $E = I_1 \times I_2$, $m(E) = \text{Area}(E) = \ell(I_1) \cdot \ell(I_2)$

$\implies$ any singleton, any countable set, any line on $\mathbb{R}^2$ is a null set.

Remark.

If $A \subseteq \mathbb{R}$ is a null set under m on $\mathbb{R}$, then $A \times \mathbb{R}, \mathbb{R} \times A$ are null sets under m on $\mathbb{R}^2$.

Remark.

$\mathcal{M}^2$ is NOT complete under m on $\mathbb{R}^2$ (because e.g. with N the 1-D Vitali set, and $a \in \mathbb{R}$, $\{a\} \times \mathcal{N} \subseteq \{a\} \times \mathbb{R} \implies \{a\} \times \mathcal{N}$ is a subset of a null set under m on $\mathbb{R}^2$, but $\{a\} \times \mathcal{N} \notin \mathcal{M}^2$ (not measurable)).

Fact 3.2.4

It is possible to construct m on $\mathbb{R}^2$ through the "outer measure approach", i.e:

$$\forall E \subseteq \mathbb{R}^2, m^*(E) = \inf \left\{ \sum_{n=1}^{\infty} \text{Area}(R_n) : R_n \text{ s are closed finite rectangles s.t. } \bigcup_{n=1}^{\infty} R_n \supseteq E \right\}$$

And m^* satisfies similar properties as in the 1D case.

Define $E \subseteq \mathbb{R}^2$ is measurable if $\forall F \subseteq \mathbb{R}^2, m^*(F) = m^*(F \cap E) + m^*(F \cap E^c)$.

$$\overline{\mathcal{M}^2} = \{E \subseteq \mathbb{R}^2 : E \text{ is measurable}\}.$$

Then $\overline{\mathcal{M}^2}$ is a σ -algebra. Let m be the restriction of m^* to $\overline{\mathcal{M}^2}$. Then m is a measure, and in fact, m matches the Lebesgue measure constructed from the product measure approach we did earlier. Moreover:

$\overline{\mathcal{M}^2}$ is the completion of $\mathcal{M}^2$ under m . Also, $\overline{\mathcal{M}^2} = \overline{\mathcal{B}(\mathbb{R}^2)}$.

Fact 3.2.5

The Lebesgue measure m on $\mathbb{R}^2$ is the ONLY measure on $\mathcal{M}^2$ (or $\overline{\mathcal{B}(\mathbb{R}^2)}$ or $\overline{\mathcal{M}^2}$) such that $\forall$ rectangles $I_1 \times I_2, m(I_1 \times I_2) = \ell(I_1)\ell(I_2)$.

(This is because $I = \{I_1 \times I_2 : I_1, I_2 \text{ finite intervals}\}$ is a π -system and $\sigma(I) = \mathcal{B}(\mathbb{R}^2)$.)

Fact 3.2.6

The Lebesgue measure m on $\mathbb{R}^2$ is translation-invariant, i.e., $\forall E \in \mathcal{M}^2, \forall z = (x, y) \in \mathbb{R}^2, m(E + z) = m(E)$.

(Because $\text{Area}(I_1 \times I_2)$ is invariant under translation.)

Define $m_z : \mathcal{M}^2 \rightarrow [0, \infty]$ s.t $m_z(E) = m(E + z) \forall E \in \mathcal{M}^2$. Verify that m_z is a measure and that $m_z = m$ on I .

Remark.

The Lebesgue measure m on $\mathbb{R}^2$ is the only measure on $\mathcal{M}^2$ and its brethren that is translation-invariant.

3.3 Fubini's Theorem

Definition 3.3.1

Let $f : \mathbb{R}^2 \rightarrow \overline{\mathbb{R}}$ be $\mathcal{M}^2$ -measurable and non-negative. Define two functions:

$$I_f^1(x) = \int_{\mathbb{R}} f(x, y) dy = \int_{\mathbb{R}} f_x(y) dy \quad \forall x \in \mathbb{R}$$

$$I_f^2(y) = \int_{\mathbb{R}} f(x, y) dx = \int_{\mathbb{R}} f^y(x) dx \quad \forall y \in \mathbb{R}$$

Note on notation: given $f : \mathbb{R}^2 \rightarrow [0, \infty]$ $\mathcal{M}^2$ -measurable and non-negative, the integral of f w.r.t m (the Lebesgue measure on $\mathbb{R}^2$) is denoted by $\int_{\mathbb{R}^2} f(x, y) dx dy = \int_{\mathbb{R}^2} f$.

Theorem 3.3.2: Tonelli's Theorem

Let $f : \mathbb{R}^2 \rightarrow [0, \infty]$ be M^2 -measurable and non-negative. Then:

$$\int_{\mathbb{R}^2} f(x, y) dx dy = \int_{\mathbb{R}} I_f^1(x) dx = \int_{\mathbb{R}} I_f^2(y) dy$$

$$\text{i.e. } \int_{\mathbb{R}^2} f(x, y) dx dy = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) dy \right) dx = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x, y) dx \right) dy$$

Proof. Since f is $\mathcal{M}^2$ -measurable and non-negative, $\exists \{\varphi_n\}$ a sequence of simple functions such that $\varphi_n \uparrow$, $\lim_{n \rightarrow \infty} \varphi_n = f$ and $\int_{\mathbb{R}^2} f = \lim_{n \rightarrow \infty} \int_{\mathbb{R}^2} \varphi_n$.

e.g. $\varphi_n = \sum_{k=0}^{n2^n} \frac{k}{2^n} 1_{A_{n,k}}$, for $A_{n,k} = \{(x, y) \in [-n, n]^2 : \frac{k}{2^n} \leq f(x, y) < \frac{k+1}{2^n}\} \in \mathcal{M}^2$.

$$\implies \int_{\mathbb{R}^2} f(x, y) dx dy = \lim_{n \rightarrow \infty} \sum_{k=0}^{n2^n} \frac{k}{2^n} m(A_{n,k})$$

$$\implies \forall x \in \mathbb{R} : I_f^1(x) = \int_{\mathbb{R}} f(x, y) dy \stackrel{\text{MON}}{=} \lim_{n \rightarrow \infty} \int_{\mathbb{R}} \varphi_n(x, y) dy = \lim_{n \rightarrow \infty} \sum_{k=0}^{n2^n} \frac{k}{2^n} I_{A_{n,k}}^1(x)$$

$\parallel$
 $m((A_{n,k})_x)$

$$\implies \int_{\mathbb{R}} I_f^1(x) dx \stackrel{\text{MON}}{=} \lim_{n \rightarrow \infty} \int_{\mathbb{R}} \left(\sum_{k=0}^{n2^n} \frac{k}{2^n} I_{A_{n,k}}^1(x) \right) dx = \lim_{n \rightarrow \infty} \sum_{k=0}^{n2^n} \frac{k}{2^n} \int_{\mathbb{R}} I_{A_{n,k}}^1(x) dx$$

$$\text{Similarly } \int_{\mathbb{R}} I_f^2(y) dy = \lim_{n \rightarrow \infty} \sum_{k=0}^{n2^n} \frac{k}{2^n} \int_{\mathbb{R}} I_{A_{n,k}}^2(y) dy.$$

So by the definition of m on $\mathbb{R}^2$, $m(A_{n,k}) = \int_{\mathbb{R}} I_{A_{n,k}}^1(x)dx = \int_{\mathbb{R}} I_{A_{n,k}}^2(y)dy$

$$\implies \int_{\mathbb{R}^2} f(x,y)dxdy = \int_{\mathbb{R}} I_f^1(x)dx = \int_{\mathbb{R}} I_f^2(y)dy$$

□

Definition 3.3.3: Integrable in $\mathbb{R}^2$

Given $f : \mathbb{R}^2 \rightarrow \overline{\mathbb{R}}$ that is $\mathcal{M}^2$ -measurable, we write $f \in L^1(\mathbb{R}^2)$ if $\int_{\mathbb{R}^2} |f(x,y)|dxdy < \infty \iff \int_{\mathbb{R}^2} f^+ < \infty$ and $\int_{\mathbb{R}^2} f^- < \infty$.

To set up for splitting, write $\int_{\mathbb{R}^2} f(x,y)dxdy = \int_{\mathbb{R}^2} f^+(x,y)dxdy - \int_{\mathbb{R}^2} f^-(x,y)dxdy$.

Assume $f \in L^1(\mathbb{R}^2)$. By Tonelli, $\int_{\mathbb{R}^2} |f(x,y)|dxdy = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(x,y)|dy \right) dx = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(x,y)|dx \right) dy$.

Theorem 3.3.4: Fubini's Theorem

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is $\mathcal{M}^2$ -measurable and $f \in L^1(\mathbb{R}^2)$, then:

(i) $I_f^1 \in L^1(\mathbb{R}), I_f^2 \in L^1(\mathbb{R})$ (product integrable implies marginally integrable)

(ii) $I_f^1(x)$ finite valued for a.e $x \in \mathbb{R} \implies f_x \in L^1(\mathbb{R})$ for a.e $x \in \mathbb{R}$

and $I_f^2(y)$ finite valued for a.e $y \in \mathbb{R} \implies f^y \in L^1(\mathbb{R})$ for a.e $y \in \mathbb{R}$

$$(iii) \int_{\mathbb{R}^2} f(x,y)dxdy = \int_{\mathbb{R}} I_f^1(x)dx = \int_{\mathbb{R}} I_f^2(y)dy$$

Proof. Assume $f \in L^1(\mathbb{R}^2)$.

$$\int_{\mathbb{R}} |I_f^1(x)|dx \leq \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(x,y)|dy \right) dx \stackrel{\text{Tonelli}}{=} \int_{\mathbb{R}^2} |f(x,y)|dxdy < \infty$$

$\implies I_f^1 \in L^1(\mathbb{R})$, and similarly $I_f^2 \in L^1(\mathbb{R})$. So (i) is proven. (i) $\implies$ (ii) is immediate.

Next: wrote $f = f^+ - f^-$. Note: $f \in L^1(\mathbb{R}^2) \implies f^+, f^- \in L^1(\mathbb{R}^2) \implies f^+, f^-$ are finite-valued. By Tonelli:

$$\int_{\mathbb{R}} I_{f^+}^1(x)dx = \int_{\mathbb{R}} I_{f^+}^2(y)dy = \int_{\mathbb{R}^2} f^+(x,y)dxdy < \infty$$

$$\int_{\mathbb{R}} I_{f-}^1(x)dx = \int_{\mathbb{R}} I_{f-}^2(y)dy = \int_{\mathbb{R}^2} f^-(x, y)dx dy < \infty$$

Thus $I_{f+}^1, I_{f+}^2, I_{f-}^1, I_{f-}^2 \in L^1(\mathbb{R})$ and hence are finite-valued a.e.

By linearity:

$$\int_{\mathbb{R}} I_{f+}^1(x)dx - \int_{\mathbb{R}} I_{f-}^1(x)dx = \int_{\mathbb{R}} (I_{f+}^1(x) - I_{f-}^1(x)) dx$$

For a.e $x \in \mathbb{R}$, $f_x^+ \in L^1(\mathbb{R})$ and $f_x^- \in L^1(\mathbb{R})$ which, by linearity:

$$\implies I_{f+}^1(x) - I_{f-}^1(x) = \int_{\mathbb{R}} f_x^+(y)dy - \int_{\mathbb{R}} f_x^-(y)dy = \int_{\mathbb{R}} (f_x^+(y) - f_x^-(y)) dy = \int_{\mathbb{R}} f_x dy = I_f^1(x)$$

Thus $\int_{\mathbb{R}} I_{f+}^1(x)dx - \int_{\mathbb{R}} I_{f-}^1(x)dx = \int_{\mathbb{R}} I_f^1(x)dx$ and similarly $\int_{\mathbb{R}} I_{f+}^2(y)dy - \int_{\mathbb{R}} I_{f-}^2(y)dy = \int_{\mathbb{R}} I_f^2(y)dy$. Then:

$$\int_{\mathbb{R}} I_f^1(x)dx = \int_{\mathbb{R}} I_f^2(y)dy = \int_{\mathbb{R}^2} (f^+ - f^-) = \int_{\mathbb{R}^2} f$$

□

Remark.

In general, $I_f^1, I_f^2 \in L^1(\mathbb{R}) \not\Rightarrow f \in L^1(\mathbb{R}^2)$.

e.g. take $f(x, y) = 1$ for $x < y < x + 1$, $f(x, y) = -1$ for $x - 1 < y < x$, and $f(x, y) = 0$ elsewhere. Then $I_f^1(x) = \int_{\mathbb{R}} f(x, y)dy = 0 = I_f^2(y)$, but $f \notin L^1(\mathbb{R}^2)$.

Remark.

If $f : \mathbb{R}^2 \rightarrow \overline{\mathbb{R}}$ is $\overline{\mathcal{M}^2}$ -measurable, then Tonelli and Fubini still apply (and we use the same notation). In fact, $\exists \tilde{f} : \mathbb{R}^2 \rightarrow \overline{\mathbb{R}}$ $\mathcal{M}^2$ -measurable s.t $\tilde{f} = f$ a.e.

Remark.

The construction can be extended to $\mathbb{R}^d$ exactly as you would expect it to. Tonelli and Fubini still hold, and m on $\mathbb{R}^d$ has similar properties as in $\mathbb{R}$; translation invariant, scaling property, and regularity property.

Chapter 4: Differentiation

In the "Riemann" setting, the notion of differentiation is closely related to integration, i.e. integrable often implies differentiable and vice versa via the fundamental theorem of calculus. How much of this is true for Lebesgue?

4.1 Hardy-Littlewood maximal function

Definition 4.1.1: H-L max function

Assume $f \in L^1(\mathbb{R})$; then the H-L max function of f , denoted by f^* , is defined as:

$$\forall x \in \mathbb{R}, f^*(x) = \sup_{I \in \mathcal{I}(x)} \frac{1}{m(I)} \int_I |f(t)| dt$$

Where $\mathcal{I}(x) = \{I : I \text{ is an open interval and } x \in I\}$.

Proposition 4.1.2

Given $f \in L^1(\mathbb{R})$, let f^* be the H-L max function of f . Then f^* is measurable.
(Note: it is not immediately obvious that f^* is measurable!)

Proof. It's sufficient to show that $\forall a > 0, \{f^* > a\} \in \mathcal{M}$.

If $x \in \{f^* > a\}$, then $a < f^*(x) = \sup_{I \in \mathcal{I}(x)} \frac{1}{m(I)} \int_I |f(t)| dt \implies \exists I \in \mathcal{I}(x)$ s.t. $\frac{1}{m(I)} \int_I |f| > a$.

Note: I is an open interval and $x \in I \implies \exists \delta$ s.t. $(x - \delta, x + \delta) \subseteq I$.

$\forall y \in (x - \delta, x + \delta), y \in I \implies I \in \mathcal{I}(y)$

$$\implies f^*(y) = \sup_{J \in \mathcal{I}(y)} \frac{1}{m(J)} \int_J |f| \geq \frac{1}{m(I)}$$

$$\implies y \in \{f^* > a\} \implies (x - \delta, x + \delta) \subseteq \{f^* > a\} \implies \{f^* > a\} \text{ is open} \in \mathcal{M}$$

So f^* is $\mathcal{M}$ -measurable. □

Recall. Assume $f \in L^1([a, b])$. Define $F(x) = \int_{[a, x]} f(t)dt$ for $x \in [a, b]$. Since $\int_{[a, x]} = \int_{(a, x)} = \int_{(a, x)}$, we write these all as $\int_a^x$, and so $F(x) = \int_a^x f(t)dt$ for $x \in [a, b]$. If we are interested in " $F'(x)$ ", then we should investigate:

$$\lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t)dt = \lim_{h \rightarrow 0} \frac{1}{m([x, x+h])} \int_{[x, x+h]} f(t)dt$$

Lemma 4.1.3: Vitali's Covering Lemma

Assume that $\mathcal{I} = \{I_1, I_2, \dots, I_N\}$ is a finite collection of open intervals. Then $\exists$ a subcollection $\{I_{k_1}, I_{k_2}, \dots, I_{k_M}\} \subseteq \mathcal{I}$

such that $I_{k_i} \cap I_{k_j} = \emptyset \forall i \neq j$, and:

$$m\left(\bigcup_{n=1}^N I_n\right) \leq 3 \sum_{j=1}^M m(I_{k_j})$$

Proof. Wlog. assume that $m(I_n) < \infty \forall n$. (If otherwise, $\exists n \in \{1, \dots, N\}$ s.t $m(I_n) = \infty$, then we take the subcollection to be $I_{k_1} = I_n$, in which case the lemma is trivially true with infinity on both sides.)

Set $\mathcal{I}_{k_1} = \{I \in \mathcal{I} : I \cap I_{k_1} \neq \emptyset\}$. For any $I \in \mathcal{I}_{k_1}$, with $I \cap I_{k_1} \neq \emptyset$ and $m(I) \leq m(I_{k_1}) \implies I \subseteq 3I_{k_1} \implies \bigcup_{I \in \mathcal{I}_{k_1}} I \subseteq 3I_{k_1}$.

Consider $\mathcal{I} \setminus \mathcal{I}_{k_1}$. Choose the biggest interval in $\mathcal{I} \setminus \mathcal{I}_{k_1}$, and call it I_{k_2} ; set $\mathcal{I}_{k_2} = \{I \in \mathcal{I} \setminus \mathcal{I}_{k_1} : I \cap I_{k_2} \neq \emptyset\}$. Following the same argument, $\bigcup_{I \in \mathcal{I}_{k_2}} I \subseteq 3I_{k_2}$.

By choice, $I_{k_2} \cap I_{k_1} = \emptyset$. Repeat this process until you exhaust $\mathcal{I}$ (which is finite). We get $I_{k_1}, I_{k_2}, \dots, I_{k_M}$ disjoint with $\mathcal{I}_{k_1}, \mathcal{I}_{k_2}, \dots, \mathcal{I}_{k_M}$ a partition of $\mathcal{I}$.

$$\implies m\left(\bigcup_{n=1}^N I_n\right) = \sum_{j=1}^M m\left(\bigcup_{I \in \mathcal{I}_{k_j}} I\right) \leq 3 \sum_{j=1}^M m(I_{k_j})$$

□

Proposition 4.1.4

Suppose $f \in L^1(\mathbb{R})$, and let f^* be the H-L max function of f . Then:

$$\forall \varepsilon > 0, m(\{x \in \mathbb{R} : f^*(x) > \varepsilon\}) \leq \frac{3}{\varepsilon} \|f\|_1 \quad \left(= \frac{3}{\varepsilon} \int_{\mathbb{R}} |f| \right)$$

(Markov-type inequality but f^* is not necessarily integrable.)

Proof. Fix $\varepsilon > 0$. Write $B = \{x \in \mathbb{R} : f^*(x) > \varepsilon\}$. Recall that $m(B) = \sup\{m(K) : K \text{ compact, } K \subseteq B\}$ (inner regularity).

Therefore it suffices to show that $m(K) \leq \frac{3}{\varepsilon} \|f\|_1 \forall$ compact $K \subseteq B$. For every $x \in K \subseteq B$, $f^*(x) > \varepsilon \implies \exists$ open interval I_x s.t $x \in I_x$ and $\frac{1}{m(I_x)} \int_{I_x} |f| > \varepsilon$.

$$\implies K \subseteq \bigcup_{x \in K} I_x \implies \exists \mathcal{I} = \{I_1, \dots, I_N\} \text{ s.t } K \subseteq \bigcup_{n=1}^N I_n$$

By the covering lemma, $m(K) \leq m\left(\bigcup_{n=1}^N I_n\right) \leq 3 \sum_{j=1}^M m(I_{k_j})$ for some disjoint $I_{k_1}, \dots, I_{k_M}$. Meanwhile, $\forall 1 \leq j \leq M, m(I_{k_j}) \leq \frac{1}{\varepsilon} \int_{I_{k_j}} |f|$

$$\implies m(K) \leq 3 \sum_{j=1}^M m(I_{k_j}) \leq \frac{3}{\varepsilon} \sum_{j=1}^M \int_{I_{k_j}} |f| = \frac{3}{\varepsilon} \int_{\bigcup_{j=1}^M I_{k_j}} |f| \leq \frac{3}{\varepsilon} \|f\|_1$$

□

Corollary 4.1.5

Given $f \in L^1(\mathbb{R})$, f^* is finite-valued a.e.

Proof. $\forall N > 0, m(\{f^* > N\}) \leq \frac{3}{N} \|f\|_1 \xrightarrow{N \rightarrow \infty} 0$.

Observe that $m(\{f^* = \infty\}) \leq m(\{f^* > N\}) \implies m(\{f^* = \infty\}) = 0$.

□

Remark.

$f \in L^1(\mathbb{R}) \not\Rightarrow f^* \in L^1(\mathbb{R})$.

e.g.: $f = 1_{[-1,1]} \in L^1(\mathbb{R})$.

$$\text{Since } f^* = \frac{1}{b-a} \int_a^b f(t) dt = \begin{cases} 0 & \text{if } (a, b) \cap [-1, 1] = \emptyset \\ \frac{\min\{b, 1\} - \max\{a, -1\}}{b-a} & \text{if } (a, b) \cap [-1, 1] \neq \emptyset \end{cases}$$

$$\implies f^*(x) = 1 \forall x \in (-1, 1), f^*(x) = \frac{2}{|x|+1} \forall x \notin (-1, 1) \implies f^* \notin L^1(\mathbb{R})$$

4.2 Lebesgue Differentiation Theorem

Theorem 4.2.1: Lebesgue's Theorem

Given $f \in L^1(\mathbb{R})$, for a.e $x \in \mathbb{R}$, if $\{I_n\}$ is a sequence of open intervals s.t $x \in I_n \forall n$, and $\lim_{n \rightarrow \infty} m(I_n) = 0$ (" $\{I_n\}$ is a seq of intervals shrinking to x "), then:

$$\lim_{n \rightarrow \infty} \frac{1}{m(I_n)} \int_{I_n} |f(t) - f(x)| dt = 0; \quad \text{in particular, } \lim_{n \rightarrow \infty} \frac{1}{m(I_n)} \int_{I_n} f(t) dt = f(x)$$

Proof. Note: the last statement is due to the common fact that (assume x is s.t that $f(x)$ is finite-valued.

$$\left| \frac{1}{m(I_n)} \int_{I_n} f(t) dt - f(x) \right| = \left| \frac{1}{m(I_n)} \int_{I_n} (f(t) - f(x)) dt \right| \leq \frac{1}{m(I_n)} \int_{I_n} |f(t) - f(x)| dt$$

Wlog. assume f is finite-valued everywhere and so we will only consider finite intervals I_n . For every $k \geq 1$, consider $B_k = \{x \in \mathbb{R} : \exists \{I_n\} \subseteq \mathcal{I}(x) \text{ with } \lim_{n \rightarrow \infty} m(I_n) = 0 \text{ s.t } \limsup_{n \rightarrow \infty} \frac{m(I_n)}{f} \int_{I_n} |f(t) - f(x)| dt \geq \frac{1}{k}\}$.

It's sufficient to show that $m(B_k) = 0 \forall k \geq 1$. Fix an arbitrary $\varepsilon > 0$, Since $Cc(\mathbb{R})$ is dense in $L^1(\mathbb{R})$, $\exists g \in Cc(\mathbb{R})$ s.t $\|f - g\|_1 \leq \varepsilon$. Since g is continuous and supported on a compact set, it implies that $\forall x \in \mathbb{R}, \forall k \geq 1, \exists \alpha > 0$ s.t if $|t - x| \leq \alpha$, then $|g(t) - g(x)| \leq \frac{1}{3k}$.

Given any $x \in \mathbb{R}$, any sequence $\{I_n\} \subseteq \mathcal{I}(x)$ s.t $\lim_{n \rightarrow \infty} m(I_n) = 0$:

$$\frac{1}{m(I_n)} \int_{I_n} |f(t) - f(x)| dt \leq \frac{1}{m(I_n)} \int_{I_n} (|f(t) - g(t)| + |g(t) - g(x)| + |g(x) - f(x)|) dt$$

For n large enough that $I_n \subseteq (x - \alpha, x + \alpha)$, we have

$$\leq \frac{1}{m(I_n)} \int_{I_n} |f(t) - g(t)| dt + \frac{1}{3k} + |g(x) - f(x)|$$

By the Lebesgue Differentiation Theorem for $|f - g|$, for a.e. $x \in \mathbb{R}$,

$$\limsup_{n \rightarrow \infty} \frac{1}{m(I_n)} \int_{I_n} |f(t) - g(t)| dt = |f(x) - g(x)|$$

Therefore, for a.e. $x \in \mathbb{R}$,

$$\limsup_{n \rightarrow \infty} \frac{1}{m(I_n)} \int_{I_n} |f(t) - f(x)| dt \leq |f(x) - g(x)| + \frac{1}{3k} + |g(x) - f(x)| = 2|f(x) - g(x)| + \frac{1}{3k}$$

Let $E = \{x : |f(x) - g(x)| > \frac{1}{6k}\}$. By Markov's inequality,

$$m(E) \leq 6k \int_{\mathbb{R}} |f - g| \leq 6k\varepsilon$$

For $x \notin E$, we have $\limsup_{n \rightarrow \infty} \frac{1}{m(I_n)} \int_{I_n} |f(t) - f(x)| dt \leq 2 \cdot \frac{1}{6k} + \frac{1}{3k} = \frac{2}{3k} < \frac{1}{k}$. Thus $B_k \subseteq E$, so $m(B_k) \leq 6k\varepsilon$. Since $\varepsilon > 0$ was arbitrary, $m(B_k) = 0$. $\square$

Definition 4.2.2: Lebesgue Point

x is called a Lebesgue point of f if the conclusion of the Lebesgue Differentiation Theorem holds at x .

Remark.

Note that in the Lebesgue differentiation theorem, "a.e $x \in \mathbb{R}$ " cannot be replaced by "pointwise".

Corollary 4.2.3

Given $f \in L^1(\mathbb{R})$, for a.e $x \in \mathbb{R}$:

$$\lim_{h \rightarrow 0} \frac{1}{2h} \int_{x-h}^{x+h} |f(t) - f(x)| dt = 0 \quad \left(\implies \lim_{h \rightarrow 0} \frac{1}{2h} \int_{x-h}^{x+h} f(t) dt = f(x) \right)$$

Or, equivalently, for a.e $x \in \mathbb{R}$:

$$\lim_{h \rightarrow 0} \frac{1}{2h} \int_{x-h}^{x+h} |f(x+y) - f(x)| dy = 0 \quad \left(\implies \lim_{h \rightarrow 0} \frac{1}{2h} \int_{x-h}^{x+h} f(x+y) dy = f(x) \right)$$

Proof: in lecture 22.

Corollary 4.2.4

Given $f \in L^1(\mathbb{R})$, let f^* be the H-L max function of f . Then $f^* \geq |f|$ a.e.

Proof: in lecture 22.

Corollary 4.2.5

Given $f \in L^1(\mathbb{R})$ and $a \in \mathbb{R}$, $F(x) = \int_a^x f(t)dt$ for $x \geq a$, then F is uniformly continuous on $[a, \infty)$ and for a.e $x \in \mathbb{R}$, $F'(x)$ exists and $F'(x) = f(x)$.

Proof: uses the following lemma:

Lemma 4.2.6

If $f \in L^1(\mathbb{R})$, then $\forall \varepsilon > 0 \exists \delta > 0$ s.t $\forall$ open intervals I with $m(I) \leq \delta$, $\int_I |f| \leq \varepsilon$.

Proof: in lecture 22.

Remark.

x is a Lebesgue point of $f \implies F'$ exists at x . The reverse is not true in general.

Remark.

If $f \in L^1_\alpha(\mathbb{R})$, i.e. $f \in L^1([-n, n]) \forall n \geq 1$, then the conclusion of the Lebesgue Diff Thm still holds.

4.3 Monotonic (Increasing) functions

We take on the second question: what kind of F satisfies $F(b) - F(a) = \int_a^b F'(t)dt$?

In the Lebesgue setting, many things about this equation can fail. We will start with functions that are easier to tackle: increasing functions. First, notation, for a general F and $x \in \mathbb{R}$:

$$\begin{aligned} \overline{D}_r F(x) &= \limsup_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h}; & \underline{D}_r F(x) &= \liminf_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\ \overline{D}_\ell F(x) &= \limsup_{h \rightarrow 0} \frac{F(x) - F(x-h)}{h}; & \underline{D}_\ell F(x) &= \liminf_{h \rightarrow 0} \frac{F(x) - F(x-h)}{h} \end{aligned}$$

We restrict to finite-valued functions.

Proposition 4.3.1

Assume that F is an increasing function on $[a, b]$. Then F is continuous on $[a, b]$ except at countably many points. (If needed, extend F to $\mathbb{R}$ by setting $F(x) = F(a) \forall x < a$, $F(x) = F(b) \forall x > b$).

Proposition 4.3.2

Assume that F is an increasing function on $[a, b]$. Then F is differentiable a.e on $[a, b]$.

Theorem 4.3.3: Vitali's Covering Theorem

Given a set $E \subseteq \mathbb{R}$, a collection J of intervals is called a Vitali covering of E if $\forall x \in E \forall \varepsilon > 0, \exists I \in J$ s.t $x \in I$ and $m(I) \leq \varepsilon$. If $E \in \mathcal{M}$ with $m(E) < \infty$, and J is a Vitali covering of E then $\forall \varepsilon > 0 \exists I_1, I_2, \dots, I_N \in J$ s.t $I_i \cap I_j = \emptyset \forall i \neq j$ and $m(E \setminus \bigcup_{i=1}^N I_i) \leq \varepsilon$.

When do we have FTC for Lebesgue integrals?

Proposition 4.3.4

If $F : [a, b] \rightarrow \mathbb{R}$ is increasing, then F' exists and is ≥ 0 a.e on $[a, b]$; also, $F' \in L^1([a, b])$ and $\int_a^b F' \leq F(b) - F(a)$.

Remark.

The inequality $\int_a^b F' \leq F(b) - F(a)$ can be strict in some cases. (Cantor function; see lecture 23.)

4.4 Functions of Bounded Variation

Definition 4.4.1: of Bounded Variation

A function $F : [a, b] \rightarrow \mathbb{R}$ is of bounded variation, i.e. $F \in BV([a, b])$ if $T_F(a, b) = \sup\{\sum_{k=1}^N |F(x_k) - F(x_{k-1})| : N \geq 1, a = x_0 < x_1 < \dots < x_N = b\} < \infty$. $T_F(a, b)$ is called the total variation of F over $[a, b]$.

For the properties of $F \in BV([a, b])$ and $T_F(a, b)$, look at lecture 23.

Theorem 4.4.2

A function $F : [a, b] \rightarrow \mathbb{R}$ is in $BV([a, b])$ if and only if $\exists H, G : [a, b] \rightarrow \mathbb{R}$ two increasing functions on $[a, b]$ s.t $F = H - G$.

Proposition 4.4.3

If $F : [a, b] \rightarrow \mathbb{R} \in BV([a, b])$, then F is continuous on $[a, b]$ except at most countably many points in $[a, b]$, and F' exists a.e on $[a, b]$ and $F' \in L^1([a, b])$.

Summary: We have proven functions in $BV([a, b])$ have similar "differentiability" properties as increasing functions on $[a, b]$. But we haven't found the conditions for $\int_a^b F'(t)dt = F(b) - F(a)$.

4.5 Absolutely Continuous Functions**Definition 4.5.1: Absolutely Continuous**

A function $F : [a, b] \rightarrow \mathbb{R}$ is called absolutely continuous on $[a, b]$, denoted by $F \in AC([a, b])$ if $\forall \varepsilon > 0 \exists \delta > 0$ s.t for any disjoint intervals $(a_k, b_k) \subseteq (a, b)$, $1 \leq k \leq N$, with $\sum_{k=1}^N (b_k - a_k) \leq \delta$, it holds that $\sum_{k=1}^N |F(b_k) - F(a_k)| \leq \varepsilon$.
 (a_k, b_k) s are not necessarily a partition of (a, b) .

Fact 4.5.2

Some properties of $F \in \text{AC}([a, b])$:

1. $F \in \text{AC}([a, b]) \implies F$ is uniformly continuous on $[a, b]$

Idea: $\forall \varepsilon > 0$, let δ be the constant in the def of AC. $\forall x, y \in [a, b], x < y$, if $y - x < \delta$, then $|F(y) - F(x)| \leq \varepsilon$

2. If $f \in L^1([a, b])$ and $F(x) = \int_a^x f(t)dt$ for $x \in [a, b]$, then $F \in \text{AC}([a, b])$.

Proof: Recall that $\forall \varepsilon > 0, \exists M > 0$ s.t $\int_{\{f^* > M\}} |f| < \frac{\varepsilon}{2}$. Set $\delta = \frac{\varepsilon}{2M}$. $\forall (a_k, b_k) \subseteq (a, b)$ disjoint s.t $\sum_{k=1}^N (b_k - a_k) \leq \delta$ we have:

$$\begin{aligned} \sum_{k=1}^N |F(b_k) - F(a_k)| &\leq \sum_{k=1}^N \int_{a_k}^{b_k} |f| = \sum_{k=1}^N \int_{(a_k, b_k) \cap \{f^* > M\}} |f| + \sum_{k=1}^N \int_{(a_k, b_k) \cap \{f^* \leq M\}} |f| \\ &\leq \int_{\bigcup_{k=1}^N (a_k, b_k) \cap \{f^* > M\}} |f| + M \sum_{k=1}^N (b_k - a_k) = \int_{\bigcup_{k=1}^N (a_k, b_k) \cap \{f^* > M\}} |f| + M\delta \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

3. If $F, G \in \text{AC}([a, b])$, then $F + G, F - G$ also $\in \text{AC}([a, b])$.

4. If $F \in \text{AC}([a, b])$, then $F \in \text{BV}([a, b])$.

Proof: Set $\varepsilon = 1$. Take δ to be the constant in the definition of AC. Consider a partition of $[a, b] : a = t_0 < t_1 < \dots < t_L = b$ s.t $t_{i+1} = t_i + \frac{b-a}{L}$, and L is s.t $\frac{b-a}{L} \leq \delta$. For each $0 \leq i \leq L$, for **any** partition of $[t_i, t_{i+1}] : t_i = x_0 < x_1 < \dots < x_N = t_{i+1}$; then (x_k, x_{k+1}) s are disjoint and $\sum_{k=0}^{N-1} (x_{k+1} - x_k) = t_{i+1} - t_i \leq \delta$

$$\implies \sum_{k=0}^{N-1} |F(x_{k+1}) - F(x_k)| \leq 1 \implies T_F(t_i, t_{i+1}) \leq 1$$

Then $T_F(a, b) = \sum_{i=0}^{L-1} T_F(t_i, t_{i+1}) \leq L < \infty$.

5. If F is continuous on $[a, b]$ and (truly) differentiable everywhere on (a, b) and $\exists M > 0$ s.t $|F'(x)| \leq M \forall x \in (a, b)$, then $F \in \text{AC}([a, b])$.

Proof: Recall that we have used the mean value theorem to show that $\forall x, y \in [a, b]$, $|F(x) - F(y)| \leq M|x - y|$.

Theorem 4.5.3

If $F \in \text{AC}([a, b])$, then F' exists a.e on $[a, b]$ and $F' \in L'([a, b])$.

Follows immediately from the fact that $\text{AC}([a, b]) \subseteq \text{BV}([a, b])$.

Theorem 4.5.4

Given $F \in AC([a, b])$, F is constant on $[a, b]$ if and only if $F' = 0$ a.e on $[a, b]$.

Proof. For $\implies$: if $F \equiv c$ then $F' \equiv 0$ on (a, b) trivially.

For $\impliedby$: Assume $F \in AC([a, b])$, and $F' = 0$ a.e on $[a, b]$. We want to show that $\forall c \in (a, b), F(c) = F(a)$. Fix a $c \in (a, b)$. Set $E = \{x \in [a, c] : F'(x) = 0\} \implies m([a, c] \setminus E) = 0$. Fix an arbitrary $\varepsilon > 0$. Take δ as the constant in the defn. of AC.

Define $\mathcal{J} = \{I = [x, x+h] \subseteq (a, c) : x \in E, h > 0 \text{ and } |F(x+h) - F(x)| \leq \varepsilon h\}$.

$\forall x \in E : x \in (a, c)$ open and $F'(x) = 0 \implies \exists I = [x, x+h] \in \mathcal{J}$ with $x \in I$ and h arbitrarily small. Then this implies that $\mathcal{J}$ is a Vitali covering of $E \implies \exists$ disjoint $I_1, \dots, I_n \in \mathcal{J}$ s.t $m(E \setminus \bigcup_{i=1}^N I_i) \leq \delta \implies m([a, c] \setminus \bigcup_{i=1}^N I_i) \leq \delta$.

Assume $I_i = [x_i, x_{i+1}]$ for $i = 1, \dots, N$ and ordered as $x_1 < x_1 + h_1 < x_2 < \dots < x_N < x_N + h_N$; then for all i , $|F(x_i + h_i) - F(x_i)| \leq \varepsilon h_i$.

$$|F(a) - F(c)| \leq |F(a) - F(x_1)| + \sum_{i=1}^N |F(x_{i+1}) - F(x_i)| + \sum_{i=1}^{N-1} |F(x_{i+1}) - F(x_i + h_i)| + |F(c) - F(x_N + h_N)|$$

And the specific term:

$$\sum_{i=1}^N |F(x_i + h_i) - F(x_i)| \leq \varepsilon \sum_{i=1}^N h_i \leq \varepsilon(c - a)$$

Since $(a, x_1), (x_1 + h_1, x_2), (x_N + h_N, c)$ are disjoint and the union of all of them $= [a, c] \setminus \bigcup_{i=1}^N I_i$, then the sum of the lengths $\leq \delta$.

We see that:

$$(x_1 - a) + \sum_{i=1}^{N-1} (x_{i+1} - (x_i + h_i)) + (c - (x_N + h_N)) = m\left([a, c] \setminus \bigcup_{i=1}^N I_i\right) \leq \delta$$

By the defn. of AC:

$$\begin{aligned} |F(x_1) - F(a)| + \sum_{i=1}^N |F(x_{i+1}) - F(x_i + h_i)| + |F(c) - F(x_N + h_N)| &\leq \varepsilon \\ \implies |F(c) - F(a)| &\leq \varepsilon(c - a) + \varepsilon = (c - a + 1)\varepsilon \end{aligned}$$

Since ε is arbitrary, $F(c) = F(a) \implies F \equiv F(a)$ on $[a, b]$. □

Remark.

The condition $F \in AC([a, b])$ cannot be dropped. For example, if F is the Cantor-Lebesgue function on $[0, 1]$, then $F' \equiv 0$ a.e on $[0, 1]$, but F is not constant. Note that this implies that the Cantor-Lebesgue function is not AC.

However, it is BV.

Theorem 4.5.5: Fundamental Theorem of Calculus

If $F \in AC([a, b])$, then F' exists a.e on $[a, b]$, $F' \in L^1([a, b])$, and $\forall x \in [a, b]$, then $F(x) - F(a) = \int_a^x F'(t)dt \forall x \in [a, b]$.

$$\text{In particular: } F(b) - F(a) = \int_a^b F'(t)dt$$

Proof. Assume $F \in AC([a, b])$. Define $G(x) = F(a) + \int_a^x F'(t)dt \forall x \in [a, b]$. According to the second property, $G \in AC([a, b])$. Moreover, by the previous theorem, we know $G' = F'$ a.e on $[a, b]$.

Thus, $H = F - G \in AC([a, b])$, and $H' = F' - G' = 0$ a.e on $[a, b]$. By the previous theorem, $H(x) = H(a) = 0 \forall x \in [a, b]$.

$$\implies F(x) = G(x) = F(a) + \int_a^x F'(t)dt \forall x \in [a, b]$$

□

4.6 A Sprinkling of Probability

Assume μ is a probability measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ (i.e. μ is defined on $\mathcal{B}_{\mathbb{R}}$ and $\mu(\mathbb{R}) = 1$).

Definition 4.6.1: Distribution Function

Define a function F_{μ} on $\mathbb{R}$ by $F_{\mu}(x) = \mu((-\infty, x]) \forall x \in \mathbb{R}$.

Then F_{μ} is the distribution function of μ .

(If X is a random variable on $(\Omega, \mathcal{F}, P)$ then $\mu((-\infty, x]) = P(X \leq x)$.)

Proposition 4.6.2

'Intrinsic' properties of F_μ :

1. F_μ is increasing: $\forall x < y, F_\mu(x) = \mu((-\infty, x]) \leq \mu((-\infty, y]) = F_\mu(y)$.
2. $\lim_{x \rightarrow \infty} F_\mu = \mu(\mathbb{R}) = 1$ and $\lim_{x \rightarrow -\infty} F_\mu(x) = 0$.
(e.g. $\lim_{n \rightarrow \infty} F_\mu(n) = \lim_{n \rightarrow \infty} \mu((-\infty, n]) = \mu(\bigcup_{n=1}^{\infty} (-\infty, n]) = \mu(\mathbb{R}) = 1$ and vice versa.)
3. F_μ has at most countable discontinuities, and F_μ is càdlàg.
(i.e. $\lim_{n \rightarrow \infty} F_\mu(x + \frac{1}{n}) = \lim_{n \rightarrow \infty} \mu((-\infty, x + \frac{1}{n}]) = \mu(\bigcap_{n=1}^{\infty} (-\infty, x + \frac{1}{n}]) = \mu((-\infty, x]) = F_\mu(x)$)
BUT $\lim_{n \rightarrow \infty} F_\mu(x - \frac{1}{n}) = \mu(\bigcup_{n=1}^{\infty} (-\infty, x - \frac{1}{n}]) = \mu((-\infty, x)) = F_\mu(-x)$

Proposition 4.6.3

Derived properties of F_μ :

4. F_μ is continuous at x iff. $\mu(\{x\}) = 0$.
More specifically, $\forall x \in \mathbb{R}, F_\mu(x) - F_\mu(x-) = \mu(\{x\})$.
"Cannot assign mass to any one specific point"
5. $\forall a, b \in \mathbb{R}, a < b, F_\mu(b) - F_\mu(a) = \mu((a, b])$ "closed left, open right", but there are other results:
 $F_\mu(b-) - F_\mu(a) = \mu((a, b)); \quad F_\mu(b) - F_\mu(a-) = \mu([a, b]); \quad F_\mu(b-) - F_\mu(a-) = \mu([a, b)).$

Proposition 4.6.4

Unique determination

6. F_μ uniquely determines μ , i.e., if μ, ν are both probability measures on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$, and $F_\mu = F_\nu$, then $\mu = \nu$.

Proof: Set $\mathcal{I} = \{(a, b) : a, b \in \mathbb{R}\}$. $\forall a, b \in \mathcal{I}, \mu((a, b)) = F_\mu(b-) - F_\mu(a) = F_\nu(b-) - F_\nu(a) = \nu((a, b))$

$\implies \mu = \nu$ on $\mathcal{I}$; $\mathcal{I}$ is a π -system and $\sigma(\mathcal{I}) = \mathcal{B}_{\mathbb{R}} \implies \mu = \nu$ on $\mathcal{B}_{\mathbb{R}}$.

7. In fact, any $F : \mathbb{R} \rightarrow [0, 1]$ satisfying the intrinsic properties 1, 2, and 3 is the distribution function of some probability measure μ on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$.

(By following the same steps as the construction of the Lebesgue measure.)

Corollary 4.6.5

Assume $\mu \ll m$ with density f . Then if $g : \mathbb{R} \rightarrow \mathbb{R}$ is $\mathcal{B}_{\mathbb{R}}$ -measurable then $\int_{\mathbb{R}} |g| d\mu = \int_{\mathbb{R}} g(x)f(x)dx$.

Theorem 4.6.6: Radon-Nikodym Theorem

Let μ, ν be two σ -finite measures on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. If $\forall A \in \mathcal{B}_{\mathbb{R}}$, we have that $\nu(A) = 0 \implies \mu(A) = 0$, i.e. if it's a null set under ν then it is null under μ , then $\mu \ll \nu$ and hence $\exists f \in L^1(\nu)$ s.t. $\forall B \in \mathcal{B}_{\mathbb{R}}, \mu(B) = \int_B f d\nu$.

We call f the Radon-Nikodym derivative of μ wrt. ν and denote it $f = \frac{d\mu}{d\nu}$.

Note: this works in the other direction too; it is an iff.

Theorem 4.6.7: Lebesgue Decomposition Theorem

Given μ , any probability measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$, μ admits a unique decomposition that we write $\mu = \mu_a + \mu_s$ s.t:

- 1) μ_a is a finite measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ and $\mu_a \ll m$
- 2) μ_s is a finite measure on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ s.t. $\mu_s \perp m$ " μ_s is singular to m " (i.e. $\exists E \in \mathcal{B}_{\mathbb{R}}$ s.t. $m(E) = 0, \mu_s(E^c) = 0$).

(Heuristically: $(\text{support}[\mu_s]) \cap (\text{support}[m]) = \emptyset$; " μ_s and m never see each other".)